ARTICLE

Learning across diseases and outcomes for risk prediction in lymphoid cancers

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TUNING REHEARSAL REVIEW: numerical contrasts are exploratory; clinical refresh, attribution and national results remain pending.

Abstract

Background Prognosis shapes treatment, supportive care and patients' decisions about their future, yet most prognostic tools learn from one disease and one outcome at a time. We asked whether, what and how to borrow information across diseases, outcomes and patient histories to predict risk in lymphoid cancers, a specialty of many uncommon diseases.

Methods We studied 37,531 Danish adults starting first-line treatment for 24 lymphoid cancers, predicting 69 outcomes at six horizons from 30 days to two years from routinely recorded histories. Otherwise identical sequence models learned from one disease–outcome pair, one outcome across all diseases, all outcomes within one disease, or every disease and outcome together. OPERA, a single model for all tasks, adds corrections specific to each patient's disease and outcome and lets each prediction borrow those of other tasks. Encoders were pretrained on the cohort or the national population. Comparators were per-task and pooled XGBoost and the prognostic indices in clinical use, in patients treated from 2023.

Findings Borrowing from other diseases increased the area under the ROC curve (AUC) by 0.120 (95% CI 0.107 to 0.133) over single-task models. Adding all other outcomes to the disease-pooled model changed AUC by −0.001 (95% CI −0.008 to 0.005) ([claim label]), so one model could predict every outcome without loss. OPERA improved on the joint model by 0.009 (95% CI 0.003 to 0.015), with borrowed corrections adding 0.005 (95% CI 0.000 to 0.009), and differed from per-task XGBoost by −0.028 (95% CI −0.036 to −0.019). **[National pretraining.] [Against the N prognostic indices in routine use for these diseases, for each index's own endpoint and patients: the number in which OPERA was better, with the NCCN-IPI result in DLBCL.]**

Interpretation **[From the test results: borrowing across diseases improved prognosis most where data were scarce, and one model served every disease and outcome at no cost. How this compares with tuned tabular models and clinical indices, without claiming clinical benefit.]**

Funding [Funders, to be confirmed by the authors.]

Research in context

Evidence before this study

[We searched PubMed, Embase and arXiv/medRxiv from inception to [date] for studies combining (multi-task OR transfer OR pretrain* OR foundation model OR shared representation) AND (prognos* OR risk prediction) AND (electronic health record OR registry), and for prognostic indices in lymphoid cancers. Report numbers screened and included.] Clinical prediction models are almost always built for one disease and one outcome. Multi-task and transfer learning on health records have helped some tasks and harmed others, often against weak baselines. Models pretrained on population records predict many outcomes, but whether they improve specialised prognosis over models trained within the specialty, or over tuned gradient-boosted trees, is unknown. Lymphoid cancers rely on disease-specific indices built from a few variables, and many uncommon diseases have none.

Added value of this study

To our knowledge, this is the first study to measure, within one specialty and on the same patients, what prognosis gains from borrowing across diseases, across outcomes and from the national population. Matched models separate each source across 24 diseases, 69 outcomes and six horizons. One model learns how much each prediction should borrow, and every task is compared with tuned tabular models and every prognostic index in routine use. **[Principal test findings with effect sizes.]**

Implications of all the available evidence

Biological boundaries between diseases need not limit prognostic learning. Other diseases can supply the examples a rare disease lacks, and one model can serve a specialty. **[Implications for disease-specific indices and task-specific models, and what the national population adds.]** Better prediction is not evidence that acting on it improves care.

Introduction

Prognosis is central to patient care. In haematology, it guides the choice of treatment, determines eligibility for clinical trials and shapes what patients and their families can expect. For the lymphoid cancers, a group of blood cancers including lymphomas, lymphocytic leukaemias and myeloma, prognostic tools such as the IPI, NCCN-IPI and CLL-IPI each predict a single outcome in a single disease from a few variables [1–4]. This specificity comes at a price: it restricts the signals, patients and outcomes such tools can learn from. Learning opportunities are scarce, because many patient groups and outcomes are rare in isolation and rarer still in combination [5]. Adding more patients with the same disease is seldom possible when the disease itself is rare. To predict better, information must be borrowed from elsewhere.

Even for a single disease and outcome, other diseases, outcomes and patient histories can be informative. Haematological cancers are biologically distinct yet share diagnostic work-up, treatments and complications, and frailty and prior illness affect outcomes globally. Learning such patterns across patients and tasks can improve prediction from few labelled examples [7–9]. Borrowing can also harm. In clinical prediction, training tasks jointly frequently degraded individual tasks, even where multi-task pretraining helped [10]. The question is therefore not only whether to borrow, but what to borrow and how, without losing clinically important differences.

The simplest way for a model to borrow is to learn the same outcome across several diseases, or several outcomes within the same disease, so that each task gains examples from the others. Such pooling can cost the idiosyncratic differences between subgroups. Ideally, a model relies on borrowed information where it must, as for rare diseases and outcomes, and on its own where that is sufficient. Statistics strikes this balance by partial pooling [11], and neural models can learn it from the data [12, 13].

A more recent way to borrow is to pretrain on patient histories, so that a model learns general patterns of how illness, investigation and treatment unfold before it learns any outcome [14–16]. A model pretrained on a seemingly unrelated cohort can then pass much of this learned structure to a downstream prognostic model. Pretrained on the records of a whole population, it borrows from people far outside the specialty [17, 18]. Whether these approaches improve on strong

conventional alternatives, particularly when labels are scarce, remains uncertain [19, 20].

Lymphoid cancers are a natural proving ground for borrowing information across diseases and outcomes. Their many entities are individually uncommon [21, 22] yet share investigation and care. At the start of treatment, clinicians weigh disease control alongside infection, cytopenia, organ toxicity and admission, which existing indices address only in part [23, 24]. Testing borrowing requires all of these diseases and outcomes under one roof, recorded from the same sources in the same formats. The Danish setting makes this possible: DALY-CARE links nationwide registers, health records and laboratory data for all Danish adults with lymphoid cancers [25]. Few, if any, other resources do so.

We studied 69 outcomes in 37,531 patients with 24 lymphoid cancers and asked three questions, answered in the Results sections named (Fig. 1).

1. *Whether to borrow.* Do sequence models improve when they learn across diseases, outcomes or both, compared with one model per disease and outcome? How do they compare with XGBoost and the clinical indices? (*Whether to borrow*, *Practical alternatives* and *Prognostic indices*.)
2. *What to borrow.* What do other diseases, other outcomes and the national population each add? (*What to borrow*.)
3. *How to borrow.* Does letting each prediction balance its own corrections against borrowed ones improve on borrowing uniformly, and where does any gain come from? (*How to borrow*.)

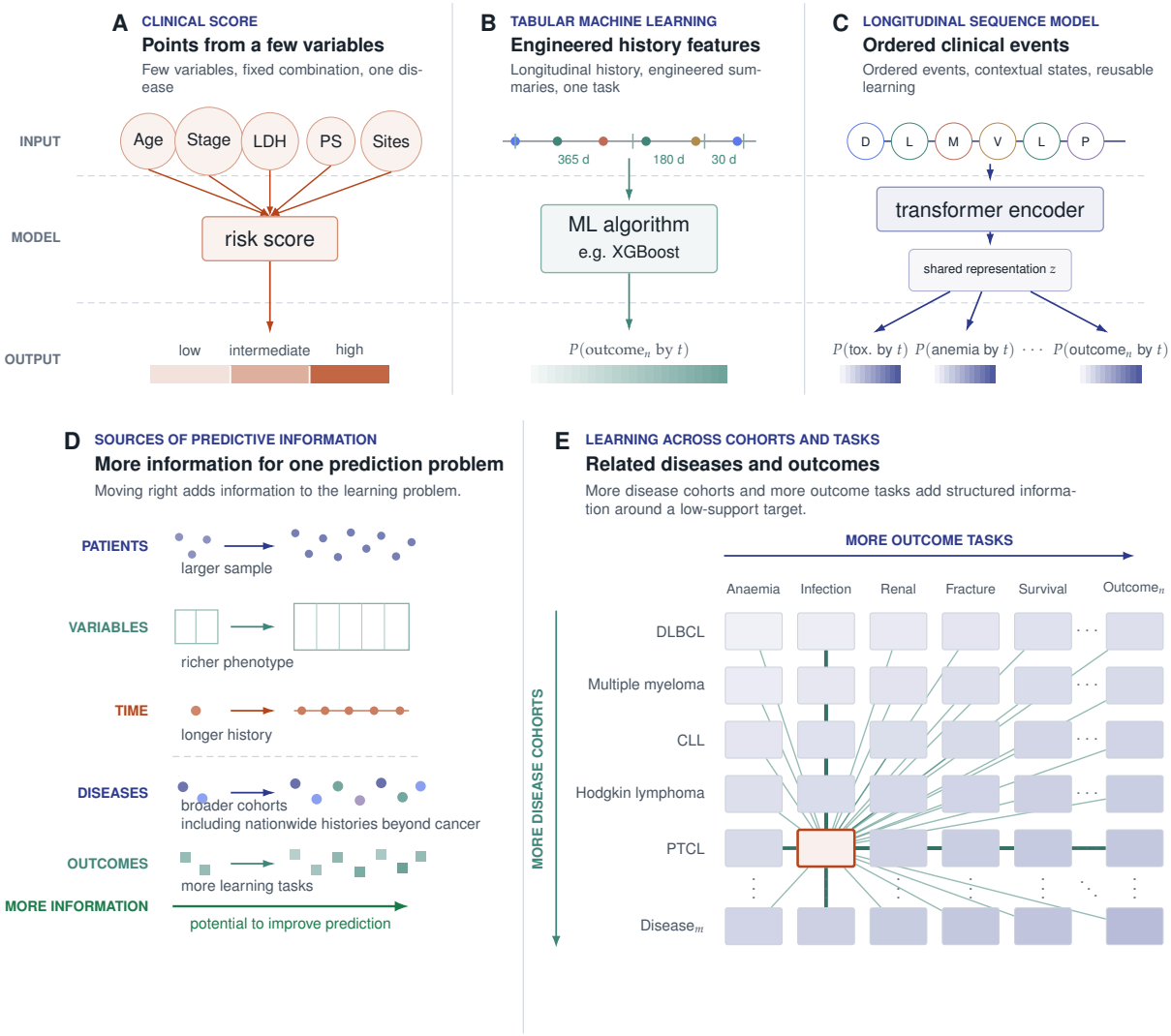


Figure 1. Approaches to risk prediction and the information each can use. Top, three approaches to risk prediction in lymphoid cancers. **A**, clinical risk scores combine a small diagnostic snapshot with a fixed rule into broad risk strata. **B**, classical machine learning uses a wider history, aggregated over time into a task-specific feature table. **C**, a longitudinal sequence model reads the ordered clinical events and can share one patient representation across outcomes. Its encoder may be pretrained on the cohort or on the whole population. Bottom, sources of additional signal. **D**, conventional scaling adds patients, variables or observations to one isolated prediction problem. **E**, shared learning also draws on related diseases (rows) and outcomes (columns).

Methods

Population, prediction time and outcomes

We included all patients in Denmark diagnosed with a lymphoid cancer (lymphoma, lymphocytic leukaemia or plasma-cell neoplasm) from 2002 who were aged 18 or older at diagnosis and started first-line treatment, as recorded in DALY-CARE [25]. Every prediction was made at the start of treatment, based only on records from before it. These comprised diagnoses, procedures, pathology, medication, laboratory results and hospital contacts with their region and specialty of care, together with age and sex (Fig. S2). The study was conducted under the DALY-CARE approvals from the Danish Health Data Authority (P-2020-561) and the National Ethics Committee (1804410).

We predicted 69 outcomes at six horizons from 30 days to two years, spanning disease course, infection, cytopenia, organ toxicity, cardiovascular events and acute care, with death as a competing event for non-fatal outcomes. Laboratory outcomes followed the Common Terminology Criteria for Adverse Events (CTCAE) [26] as closely as routine data permitted. These are graded abnormal laboratory values and not necessarily toxicities caused by treatment. Definitions, adaptations and ascertainment, including which patients were eligible for each laboratory outcome, are given in the Supplementary Methods.

We used a pseudo-prospective design, splitting patients by the date treatment started. Models were trained on patients treated from 2002 to 2021, tuned on those treated in 2022 (the tuning set) and tested on those treated from 2023, a set left unopened until every choice was fixed. Pretraining used only records from before 2022. The current figures and numbers are tuning analyses. Held-out and national-pretraining results remain pending.

Borrowing across diseases and outcomes

To investigate the effect of borrowing across diseases and outcomes, we organised all 1,656 prediction tasks (24 diseases by 69 outcomes) in a grid (Fig. S1). Each cell is a prediction question, such as the risk of infection in a patient with DLBCL, at six horizons. Four matched types of model differed only in what they could borrow. A specialist learned from one cell. A disease-pooled model learned one outcome across all diseases. An outcome-pooled model learned all outcomes within one disease. A joint model learned across both diseases and outcomes. All shared a causal transformer encoder pretrained to predict the next recorded event, building on CORE-BEHRT [27], and were fine-tuned to estimate cumulative incidence at each horizon, with death as a competing event [28, 29]. Every outcome was predicted and evaluated separately (Supplementary Methods).

Learning how much to borrow

Borrowing uniformly can blur differences that matter, by assuming that diseases and outcomes are similar when they are not. Outcome Prediction with Expert Routing Across diseases and outcomes (OPERA) instead lets each prediction balance shared and task-specific information (Fig. S14). We added small trainable modules, called adapters, that correct the shared patient representation: one for each disease, one for each outcome, and one shared by all tasks. A router weighs the disease and outcome corrections for each patient and target outcome, so that a prediction can rely on its own where they are informative and borrow where its own data are scarce [30, 31]. The idea is that of partial pooling [11], with the balance learned from the data rather than set by a prior. Model selection is described in the Supplementary Methods.

To understand how these corrections helped predictions, we constructed three separately trained

controls. The first keeps only the shared correction, the second adds the patient's own disease and outcome corrections, and the third, the full model, also borrows those of other tasks. Replacing other diseases' corrections by their averages showed how much the fitted model depended on each (Supplementary Methods).

Borrowing from the population

To test the limits of borrowing, we pretrained a causal transformer, PHAIR, not only on patients with lymphoid cancers but on the nationwide Danish registers, covering around **[8–9 million]** people, using records from before 1 January 2022. We fine-tuned the joint model and OPERA from it directly and after continued pretraining on the cohort's records [32]. The encoder used by OPERA in all other comparisons was chosen among these on the tuning set (Supplementary Methods). The encoders share registers, architecture and training schedule, but the national vocabulary keeps rarer codes and finer laboratory bins, so the comparison concerns pretraining strategies rather than the number of patients alone (Supplementary Methods). Because pretraining should matter most when labels are scarce, joint models were also fine-tuned on 10%, 25% and 50% of the training patients, mimicking smaller cohorts.

Comparators

We compared the sequence models with XGBoost [33] and ridge logistic regression, which remain hard to outperform on routine records [34]. Features summarised the same histories over **[7-, 30-, 90- and 365-day]** windows, using code counts and the latest, minimum, maximum, variance and slope of laboratory values [35]. This aggregation already improved on the clinical prognostic indices in DLBCL [36]. A separate model was tuned for each disease, outcome and horizon. We also made versions of the XGBoost models that pooled across diseases.

Each clinical index was compared with the model on the same patients. Ordered by predicted risk, the model placed patients into as many groups as the index has categories, each the size of the corresponding category, and survival was compared between the two sets of groups. No outcome information set these boundaries. These comparisons evaluate the model as a whole, including its use of the full record, rather than borrowing alone.

Evaluation

We assessed discrimination with the time-dependent area under the ROC curve (AUC) at each horizon and probability accuracy with the Brier score and index of prediction accuracy (IPA), with censoring weights [37–39]. Calibration was assessed at every horizon, without recalibrating any model. Comparisons were paired on the same patients and restricted to disease–outcome–horizon cells with at least five cases and five controls, the minimum needed to estimate an AUC. Cells meeting it can still be imprecise. Because diseases differ greatly in size, and larger diseases contribute more cells, every summary is reported two ways: with each supported cell weighing equally, and with each disease weighing equally (endpoint families, outcomes and horizons balanced within it). Intervals came from 1,000 patient bootstrap resamples within disease, shared across models (Supplementary Methods). Integrated gradients, and evaluations with recent weeks or entire sources of the record hidden, described what the predictions depend on. Reporting follows TRIPOD+AI [40].

Role of the funding source

[The funders' roles in study design, data collection, analysis, interpretation, writing and the decision to submit.]

Results

Patients and the tasks that could be evaluated

Of 78,668 adults in DALY-CARE, whose records before 2022 pretrained the cohort encoder, 37,531 started first-line treatment with an assigned disease (Fig. S3 and Table 1): 29,105 were treated through 2021, 1,980 in 2022 and 6,446 from 2023. The median age was 69 years, and diffuse large B-cell lymphoma and multiple myeloma made up half of the cohort. Laboratory records and the third version of the National Patient Register (LPR3, from 2019) came online late in the training period, so later patients had longer, more laboratory-rich records. They were also older and more often lacked a recorded clinical score, so the test evaluated performance under a shift in both patients and records. 3,004 disease–outcome–horizon cells, 600 of them at one year, had enough events for the specialist and joint models to be compared (Figs. S4 and S5).

Table 1. Baseline characteristics at first-line treatment. One record per patient, by temporal split; *n* (%) or median (IQR).

Characteristic	Overall *All years* *n* = 37,531	Training *Through 2021* *n* = 29,105	Tuning *2022* *n* = 1,980	Test *From 2023* *n* = 6,446
Demographics				
Age at treatment, years	69 (60–77)	69 (59–77)	71 (61–78)	72 (62–79)
Age ≥70 years	18,108 (48.2%)	13,470 (46.3%)	1,065 (53.8%)	3,573 (55.4%)
Female	15,665 (41.7%)	12,195 (41.9%)	808 (40.8%)	2,662 (41.3%)
Disease				
Large B-cell lymphomas	10,388 (27.7%)	8,356 (28.7%)	500 (25.3%)	1,532 (23.8%)
Diffuse large B-cell (DLBCL)	9,935 (26.5%)	7,996 (27.5%)	476 (24.0%)	1,463 (22.7%)
Transformed follicular (tFL)†	264 (0.7%)	210 (0.7%)	10 (0.5%)	44 (0.7%)
Richter transformation (RT)†	189 (0.5%)	150 (0.5%)	14 (0.7%)	25 (0.4%)
Burkitt and lymphoblastic lymphomas	1,365 (3.6%)	1,074 (3.7%)	86 (4.3%)	205 (3.2%)
Lymphoblastic (LBL)†	1,073 (2.9%)	838 (2.9%)	71 (3.6%)	164 (2.5%)
Burkitt (BL)	292 (0.8%)	236 (0.8%)	15 (0.8%)	41 (0.6%)
Mantle-cell lymphoma (MCL)	1,467 (3.9%)	1,117 (3.8%)	86 (4.3%)	264 (4.1%)
Indolent B-cell lymphomas	5,414 (14.4%)	3,891 (13.4%)	308 (15.6%)	1,215 (18.8%)
Follicular (FL)	2,929 (7.8%)	2,242 (7.7%)	152 (7.7%)	535 (8.3%)
Lymphoplasmacytic (LPL)	1,580 (4.2%)	1,168 (4.0%)	101 (5.1%)	311 (4.8%)
Extranodal marginal-zone (EMZL)	500 (1.3%)	318 (1.1%)	32 (1.6%)	150 (2.3%)
Unspecified B-cell (BCL)†	234 (0.6%)	87 (0.3%)	21 (1.1%)	126 (2.0%)
Splenic marginal-zone (SMZL)	144 (0.4%)	71 (0.2%)	2 (0.1%)	71 (1.1%)
Nodal marginal-zone (NMZL)†	27 (0.1%)	5 (0.0%)	0 (0.0%)	22 (0.3%)
CLL and SLL	5,183 (13.8%)	4,102 (14.1%)	254 (12.8%)	827 (12.8%)
Chronic lymphocytic leukaemia (CLL)	4,091 (10.9%)	3,230 (11.1%)	195 (9.8%)	666 (10.3%)
Small lymphocytic lymphoma (SLL)	1,092 (2.9%)	872 (3.0%)	59 (3.0%)	161 (2.5%)
Hairy-cell leukaemia (HCL)†	424 (1.1%)	295 (1.0%)	29 (1.5%)	100 (1.6%)
Hodgkin lymphoma (HL)	2,636 (7.0%)	2,072 (7.1%)	141 (7.1%)	423 (6.6%)
T-cell lymphomas	1,646 (4.4%)	1,287 (4.4%)	70 (3.5%)	289 (4.5%)
Peripheral T-cell, NOS (PTCL)	521 (1.4%)	443 (1.5%)	18 (0.9%)	60 (0.9%)
Angioimmunoblastic (AITL)	484 (1.3%)	352 (1.2%)	26 (1.3%)	106 (1.6%)
Anaplastic large-cell (ALCL)†	417 (1.1%)	343 (1.2%)	11 (0.6%)	63 (1.0%)
Unspecified T-cell (TCL)†	224 (0.6%)	149 (0.5%)	15 (0.8%)	60 (0.9%)
Plasma-cell neoplasms	8,962 (23.9%)	6,889 (23.7%)	499 (25.2%)	1,574 (24.4%)
Multiple myeloma (MM)	8,630 (23.0%)	6,656 (22.9%)	470 (23.7%)	1,504 (23.3%)
Plasma-cell leukaemia (PCL)†	201 (0.5%)	143 (0.5%)	20 (1.0%)	38 (0.6%)
Solitary plasmacytoma†	131 (0.3%)	90 (0.3%)	9 (0.5%)	32 (0.5%)
Amyloidosis	46 (0.1%)	22 (0.1%)	7 (0.4%)	17 (0.3%)
Performance status (ECOG)				
0–1	22,795 (60.7%)	18,539 (63.7%)	1,301 (65.7%)	2,955 (45.8%)
2–4	4,303 (11.5%)	3,572 (12.3%)	289 (14.6%)	442 (6.9%)
Missing	10,433 (27.8%)	6,994 (24.0%)	390 (19.7%)	3,049 (47.3%)
Risk group by disease-specific index‡				
Low	3,938 (10.5%)	3,187 (11.0%)	249 (12.6%)	502 (7.8%)
Intermediate	13,020 (34.7%)	10,797 (37.1%)	744 (37.6%)	1,479 (22.9%)
High	5,809 (15.5%)	4,656 (16.0%)	398 (20.1%)	755 (11.7%)
Missing	14,764 (39.3%)	10,465 (36.0%)	589 (29.7%)	3,710 (57.6%)
Pre-index history				
Recorded history, years	13.9 (8.2–18.9)	11.6 (6.9–15.8)	20.2 (19.5–20.6)	22.0 (21.0–23.0)
Recorded events	886 (416–1,819)	695 (341–1,353)	2,213 (1,322–3,575)	1,922 (1,095–3,318)
Distinct diagnoses	21 (13–32)	20 (12–30)	24 (15–37)	25 (15–38)
Distinct medications	21 (12–34)	19 (10–30)	32 (21–46)	34 (22–49)
Laboratory results	197 (0–487)	119 (0–349)	484 (276–801)	524 (295–881)
Days with hospital contact	19 (10–35)	16 (9–27)	37 (22–61)	42 (24–70)
Distinct contact specialties	8 (5–11)	7 (5–9)	10 (8–13)	11 (8–14)
Days from last record to treatment	2 (1–5)	2 (1–5)	2 (1–4)	2 (1–4)
Region (most contacts)				
North Denmark	3,977 (11.4%)	3,081 (11.6%)	207 (10.6%)	689 (10.8%)
Central Denmark	6,189 (17.7%)	4,291 (16.2%)	455 (23.2%)	1,443 (22.5%)
Southern Denmark	8,908 (25.5%)	7,072 (26.7%)	442 (22.5%)	1,394 (21.8%)
Capital Region	11,046 (31.7%)	8,615 (32.5%)	560 (28.5%)	1,871 (29.2%)
Zealand	4,776 (13.7%)	3,470 (13.1%)	298 (15.2%)	1,008 (15.7%)
Follow-up				
Follow-up (reverse Kaplan–Meier), years	9.1 (4.4–14.5)	11.4 (7.7–16.1)	3.7 (3.4–4.0)	1.6 (0.8–2.4)
Died during follow-up	18,847 (50.2%)	17,378 (59.7%)	539 (27.2%)	930 (14.4%)

History summaries use records before the index date. †No dedicated, externally validated prognostic index (Supplementary Table S5). ‡Category recorded with each disease's registry index (e.g. FLIPI, CLL-IPI, ISS), collapsed to low, intermediate and high; for DLBCL the NCCN-IPI computed from registry components (low 0–1; intermediate 2–5, its low- and high-intermediate groups; high ≥6), or the recorded index where it could not be computed. Missing includes conflicting records.

Whether to borrow: borrowing against task-specific models

Borrowing from other tasks improved the sequence models. Compared with specialists, AUC changed by 0.120 (95% CI 0.107 to 0.133) when models borrowed from other diseases (disease-balanced **[X]**), by 0.027 (95% CI 0.015 to 0.040) when they borrowed from other outcomes (**[Y]**) and by 0.119 (95% CI 0.104 to 0.133) when they borrowed from both (**[Z]**). IPA changed by 0.065 (95% CI 0.040 to 0.083), −0.005 (95% CI −0.011 to 0.001) and 0.063 (95% CI 0.035 to 0.083) (Fig. 2, Table 2). The joint model had the higher AUC in 523 of 675 cells (77%). **[The domains and horizons in which the gain was largest and smallest]** (Fig. 2D, Fig. S8).

The gain from borrowing was largest where a task had least data of its own **[(confirm on test)]** (Fig. 2C). In diseases with fewer than 500 training patients, the joint model differed from the specialist by 0.173 (95% CI 0.134 to 0.211). Borrowing also produced predictions where no specialist could be trained. At one year, **[102]** disease–outcome cells had no training events and **[19]** specialist fits failed, but the joint model predicted all of them, and **[N]** had enough events to be evaluated (Figs. S4 and S7).

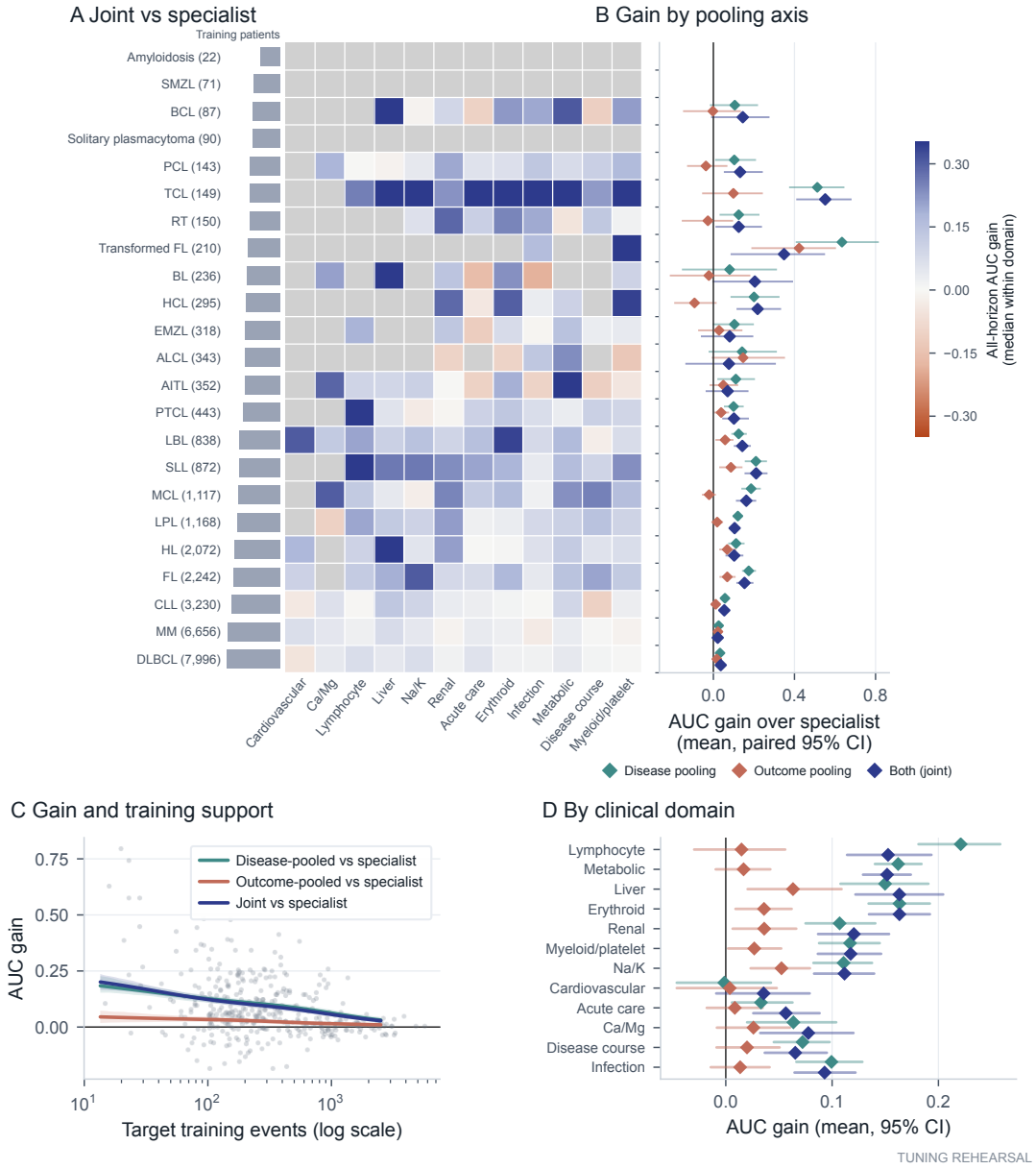


Figure 2. Effect of pooling across diseases and outcomes on discrimination. **A**, AUC gain of the joint model over the specialist, pooled over the six horizons, by disease (rows) and clinical domain (columns), median over the domain's outcomes. Diseases run from fewest to most training patients, given beside each name with a log-scale bar, and domains from fewest to most training events. **B**, For each disease, the mean gain from pooling over diseases, pooling over outcomes and both (joint), on the same rows. **C**, Gain against the number of training events of the target cell (at one year), with the hierarchical support curve and 95% credible band. Points are cells with at least 10 training events and a 95% interval narrower than 0.4 (all cells enter the curve). Tuning preview: the support curve is a kernel smoother over cells with a 95% bootstrap band. The hierarchical model is fitted for the test-set analysis. **D**, Gains by clinical domain (paired mean and 95% bootstrap interval where available, with descriptive preview fallbacks labelled in the plot). Positive values favour the pooled model, and grey cells could not be estimated. Cell-level maps for each pooling axis, their interaction and all horizons are in Figs. S7 and S8.

Table 2. What each source of signal adds, across all horizons. Paired differences on the test set (first model minus second), pooled over all six horizons with equal weight per disease–outcome–horizon cell, with 95% paired bootstrap intervals. The same table at each horizon is in Supplementary Table [S]. Cells: cells estimable for both models. Rare diseases: fewer than 500 training patients. Cells better: share of cells with a higher AUC point estimate, irrespective of uncertainty. Bold: the interval lies above zero. Red: below zero. Claim: the prespecified label from the paired bootstrap of the pooled difference: *better* (*worse*) if the interval lies above (below) zero, marked *above margin* when the estimate also exceeds the practical margin of 0.01. *Equivalent* if the interval lies within ±0.01, otherwise *inconclusive*. Paired bootstrap *p*-values and the hierarchical estimates, with their separately specified weighting, are in Supplementary Table [S].

Comparison	Cells	ΔAUC (95% CI)	ΔIPA (95% CI)	ΔAUC, rare diseases	Cells better	Claim
Learning across diseases and outcomes (against the specialist)						
Other diseases (disease-pooled)	3,004	**0.120 (0.107 to 0.133)**	**0.065 (0.040 to 0.083)**	**0.162 (0.125 to 0.199)**	77%	better, above margin
Other outcomes (outcome-pooled)	3,004	**0.027 (0.015 to 0.040)**	−0.005 (−0.011 to 0.001)	0.016 (−0.018 to 0.054)	59%	better, above margin
Both (joint)	3,004	**0.119 (0.104 to 0.133)**	**0.063 (0.035 to 0.083)**	**0.173 (0.134 to 0.211)**	77%	better, above margin
Learning how much to borrow						
Routed vs joint	3,004	**0.009 (0.003 to 0.015)**	**0.031 (0.024 to 0.050)**	0.009 (−0.003 to 0.023)	57%	better
Population pretraining (joint model)						
National vs cohort encoder	–	–	–	–	–	–
Adapted national vs cohort encoder	–	–	–	–	–	–
Population pretraining (OPERA)						
OPERA (national) vs OPERA (local)	–	–	–	–	–	–
OPERA (national + adaptation) vs OPERA (local)	–	–	–	–	–	–
Sequence models against tabular models						
Specialist vs per-task XGBoost	3,004	−0.155 (−0.168 to −0.143)	−0.130 (−0.144 to −0.123)	−0.185 (−0.221 to −0.151)	10%	worse, beyond margin
Pooled vs per-task XGBoost	3,004	**0.012 (0.006 to 0.017)**	0.009 (−0.007 to 0.020)	**0.028 (0.012 to 0.043)**	55%	better, above margin
OPERA (local) vs per-task XGBoost	3,004	−0.028 (−0.036 to −0.019)	−0.036 (−0.053 to −0.025)	−0.003 (−0.024 to 0.018)	32%	worse, beyond margin
OPERA (national + adaptation) vs per-task XGBoost	–	–	–	–	–	–
Joint vs per-task XGBoost	3,004	−0.037 (−0.045 to −0.028)	−0.067 (−0.095 to −0.056)	−0.012 (−0.036 to 0.009)	28%	worse, beyond margin
Joint vs pooled XGBoost	3,004	−0.048 (−0.056 to −0.041)	−0.076 (−0.097 to −0.067)	−0.040 (−0.058 to −0.022)	21%	worse, beyond margin
Joint vs XGBoost with registry variables	–	–	–	–	–	–

What to borrow: diseases, outcomes and the population

Other diseases supplied most of what was worth borrowing. Once outcomes were shared, adding other diseases changed AUC by 0.091 (95% CI 0.081 to 0.101). Once diseases were shared, adding all other outcomes changed it by −0.001 (95% CI −0.008 to 0.005), **[within the prespecified equivalence margin of 0.01 (claim label)]**. The joint model thus predicted all 69 outcomes as well as the disease-pooled models did one at a time. Borrowing from other outcomes alone, without other diseases, changed IPA by −0.005 (95% CI −0.011 to 0.001) relative to the specialist.

Borrowing from the national population changed the joint model's AUC by [X] with direct transfer and by [Y] after adaptation to the cohort. For OPERA, the corresponding differences were [X] and [Y] (Fig. 3A). With 10%, 25% and 50% of the training patients, the adapted national encoder differed from the cohort encoder by [A], [B] and [C] (Fig. 3B). **[Where the national encoder helped, by training events and disease]** (Fig. 3C–D).

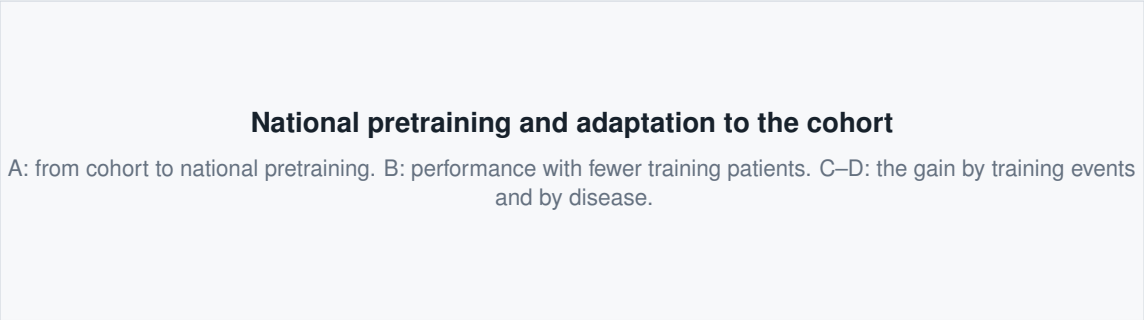


Figure 3. National pretraining, adaptation to the cohort and training-set size. A, One-year AUC of the joint model (filled) and the routed model (outline) initialised from the encoder pretrained on the cohort alone, from the national encoder (pretrained on the whole population, the cohort included) used directly, and from the national encoder adapted to the cohort by continued pretraining on its records (95% CI, averaged over the cells evaluated for all six models). Arrows give the paired effect of each step, and the bracket the two together. **B**, Median one-year AUC over the same cells of the joint model from the cohort encoder and from the adapted national encoder, trained on 10%, 25%, 50% or all of the training patients (nested random subsets with the same patients for every outcome). Numbers give the median paired difference at each share. The dashed line marks the cohort-encoder model trained on all patients, and the open circle the share at which the national model reaches it (interpolated on the log scale). **C**, Paired effect of the adapted national versus the cohort encoder for each cell, by the number of training events in the cell (line and band: median and interquartile range within bins). **D**, The same effects by disease (diamond and bar: median and interquartile range). Tuning and test patients were never subsampled.

How to borrow: own corrections and borrowed corrections

Letting each prediction weigh its own corrections against borrowed ones improved modestly on borrowing uniformly. OPERA changed AUC by 0.009 (95% CI 0.003 to 0.015) and IPA by 0.031 (95% CI 0.024 to 0.050) relative to the joint model, with the higher AUC in 390 of 675 cells (58%).

In the separately trained controls (Fig. 4), own disease and outcome corrections changed AUC by 0.007 (95% CI 0.001 to 0.014) (disease-balanced −0.005 (95% CI −0.019 to 0.009)), and borrowed corrections by a further 0.005 (95% CI 0.000 to 0.009) (0.014 (95% CI 0.004 to 0.026)). Own corrections thus helped mainly in the large diseases, which dominate the summary over cells, and borrowed corrections in the smaller ones, which the disease-balanced summary weights equally **[(confirm on test)]**. **[Borrowed corrections gained most in tasks with few training events]** (Fig. 4B–C).

Borrowing was broad rather than targeted. **[The share of routing on the patient's own disease and outcome, and the effective number of sources.]** Replacing every other disease's correction by its average changed AUC by **[X (95% CI)]**, and **[no single source was indispensable]** (Figs. S18 and S37).

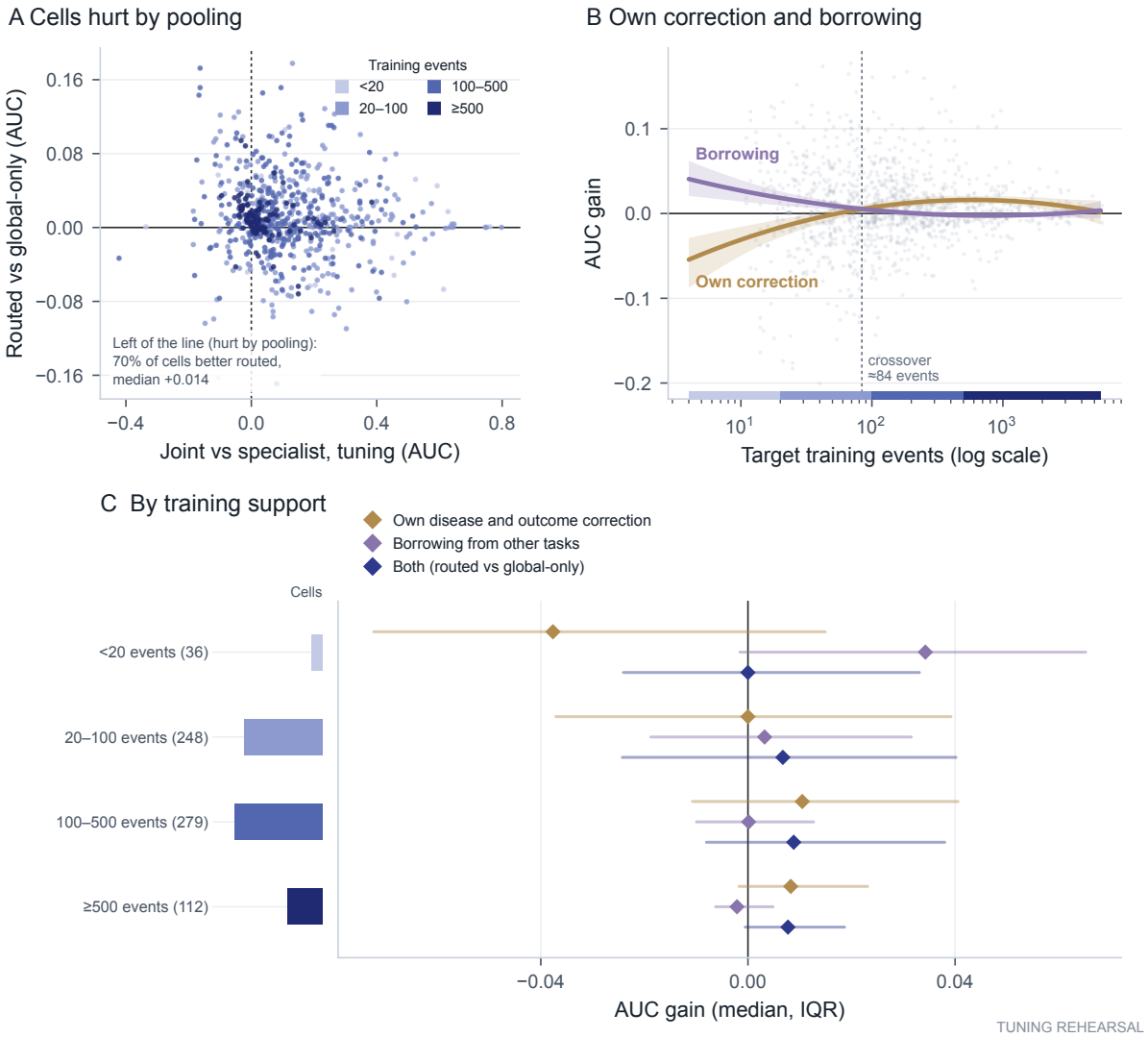


Figure 4. Own corrections and borrowed corrections, by training support. Three separately trained versions of the routed model are compared: shared correction only, with the patient's own disease and outcome corrections added (own), and with the corrections of other tasks also available (borrowed). Own: own-correction model minus shared-only. Borrowed: full model minus own-correction model. Both: full model minus shared-only. These are differences between trained models, not an additive split of one model's AUC. Gains are AUC pooled over the six horizons within each disease–outcome cell. **A**, Each cell by the joint model's gain over the specialist on the tuning set (left of the dashed line, borrowing had hurt) against the routed model's gain, coloured by the cell's training events. **B**, Own and borrowed corrections against the target's training events at one year (log scale), with a quadratic in log events and a bootstrap band over cells, both descriptive. Points are cells with at least 10 training events and a 95% interval narrower than 0.4 (all cells enter the fits). **C**, Median and interquartile range across cells within each band of training events, with the number of cells. The spread is between cells, not a confidence interval.

Borrowing against practical alternatives

Borrowing within the cohort did not make the sequence models the strongest predictors. OPERA differed from per-task XGBoost by −0.028 (95% CI −0.036 to −0.019) in AUC and −0.036 (95% CI −0.053 to −0.025) in IPA, with the higher AUC in 212 of 675 cells (31%, Figs. 5 and 6). XGBoost also gained from borrowing: pooled across diseases, it differed from per-task XGBoost by 0.012 (95% CI 0.006 to 0.017), and the joint model differed from pooled XGBoost by −0.048 (95% CI −0.056 to −0.041). **[Where the sequence models came closest to or exceeded XGBoost, by**

training events and recorded history.] [Calibration slopes and intercepts by model] (Fig. S9). One OPERA model covered the tasks of **[N]** separately trained specialists.

Borrowing from the national population **[narrowed / closed / did not change]** the gap to XGBoost. With the adapted national encoder, OPERA differed from per-task XGBoost by **[X]** in AUC and **[Y]** in IPA, with the higher AUC in **[N of M]** cells (**[X%]**). The joint model differed by **[Z]**. With the national encoder used directly, the differences were **[X]** for OPERA and **[Z]** for the joint model. **[Where the national encoder narrowed the gap most, by training events and disease]** (Fig. 3).

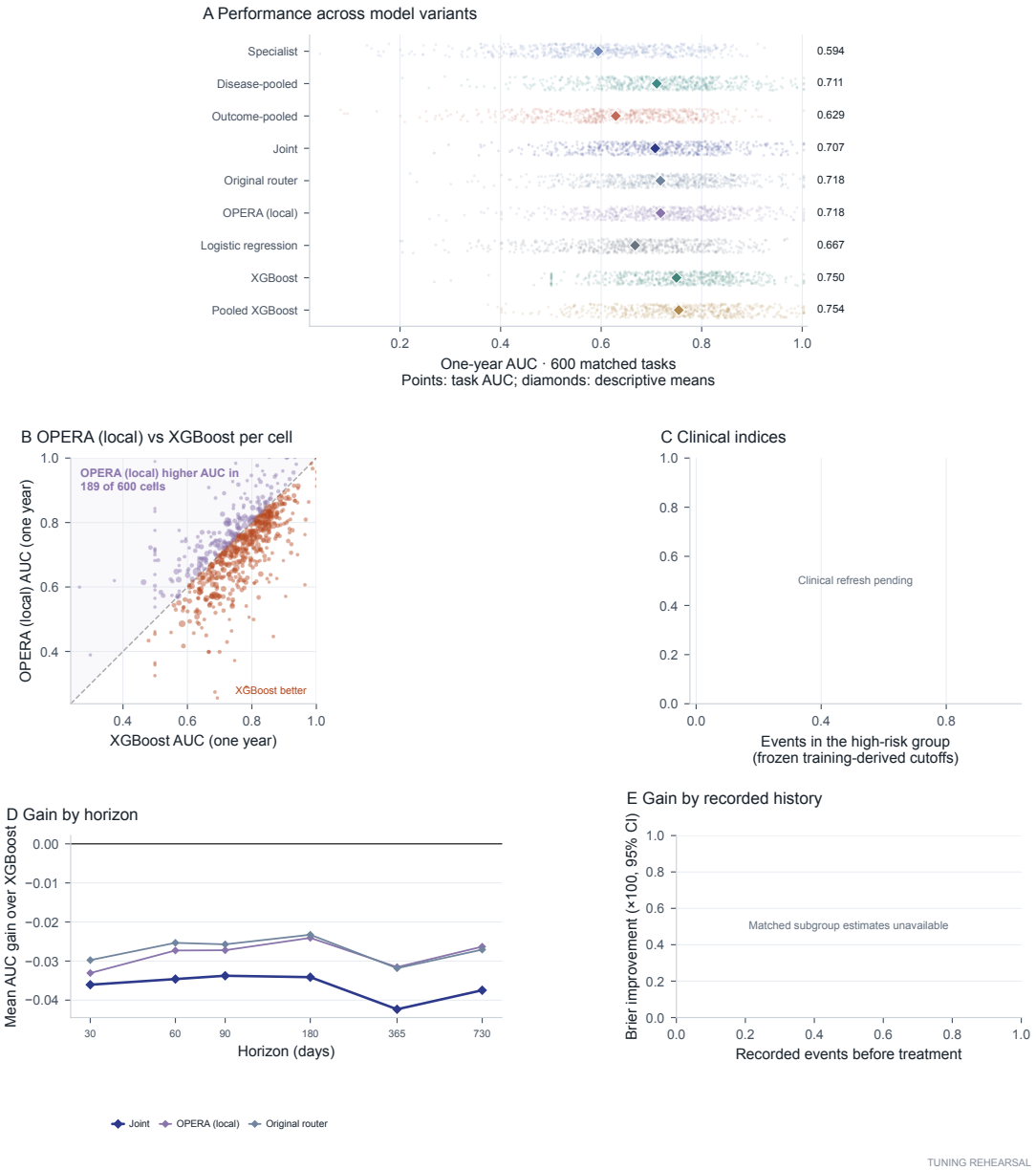


Figure 5. Performance of OPERA relative to reference methods. **A**, One-year AUC for every available sequence-model and tabular variant on the same complete-case task roster. Small points show individual task AUCs and diamonds show descriptive means. These are not population-pooled AUCs or inferential comparisons. Paired differences with confidence intervals are reported in Table 2. **B**, One-year AUC of OPERA against per-task XGBoost in each cell, with point size reflecting training events. **C**, Share of patients with the event placed in the high-risk group by each clinical index and by OPERA at frozen training-derived cutoffs. Specificity was targeted in training and may differ in evaluation. **D**, Mean AUC gain over per-task XGBoost of the joint model, primary OPERA, original-router sensitivity and available national OPERA variants at each horizon. **E**, Improvement in Brier score, pooled over horizons, over the specialist and over XGBoost by the number of events recorded before treatment (×100, 95% CI). Within each contrast, models are evaluated on the same test patients.

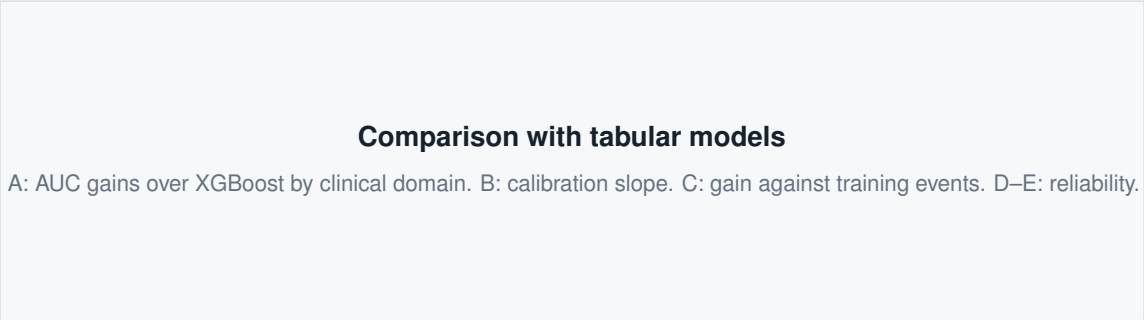


Figure 6. Comparison with tabular models: discrimination and calibration. A, AUC gain over per-task XGBoost, pooled over the six horizons, of OPERA, the joint model and the specialist, and of the joint model over XGBoost pooled across diseases and XGBoost given quality-registry variables, by clinical domain (median and interquartile range across cells). **B**, Calibration slope of OPERA, the joint, specialist and XGBoost models by domain, over every cell and horizon (ideal 1). **C**, Gain of OPERA over XGBoost against the target's training events, with the support curve. Points are cells with at least 10 training events and a 95% interval narrower than 0.4 (all cells enter the curve). Tuning preview: the support curve is a kernel smoother over cells with a 95% bootstrap band. The hierarchical model is fitted for the test-set analysis. **D–E**, Reliability of OPERA, the joint model and XGBoost at one year for the two example cells, chosen by a rule on training counts (the cell with the most training events in its domain among cells with at least 200 training events and an event share between 10% and 50%). Observed risks are censoring-weighted and point size reflects patients per bin. Bins with fewer than 20 patients are not drawn. No model was recalibrated on test data. The full calibration analysis is in Fig. S9.

Comparison with the prognostic indices in clinical use

A single model was compared with each of the **[N]** prognostic indices in clinical use, on the patients for whom that index was recorded. For **[N of N]** indices, its size-matched risk groups stratified at least as well as the index's own categories. The model's lowest group had survival as high as or higher than the index's lowest category, and its highest group survival as low as or lower than the index's highest category. **[For N of these, the difference in the lowest or highest group was beyond chance. The exceptions and the comparisons that could not be estimated]** (Fig. S13).

In diffuse large B-cell lymphoma, the largest comparison, NCCN-IPI was the main comparator. Among **[N]** patients with NCCN-IPI recorded, the model's lowest group, holding as many patients as the NCCN-IPI low-risk category, had one-year survival of **[X% (95% CI)]**, against **[Y%]** for that category. In the highest groups, survival was **[A% and B%]** (Fig. 7).

The same model also covered patients the indices do not. Ten of the 24 diseases had no dedicated, externally validated index (Table S5), and **[X%]** of patients had none recorded, whereas the model predicted every outcome for every patient.

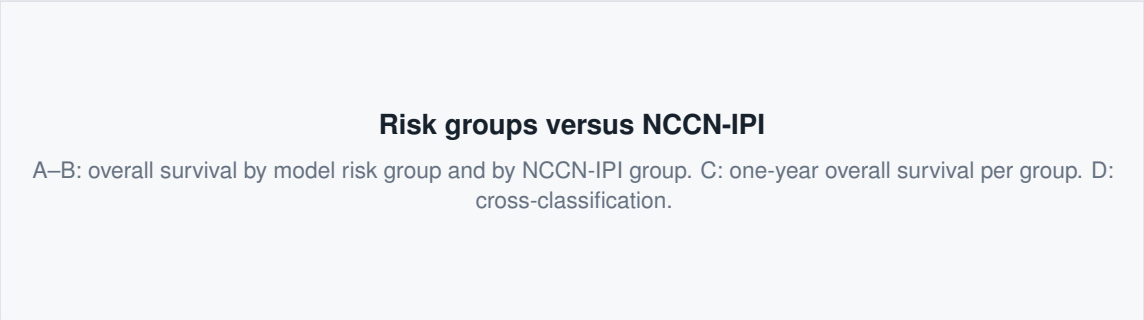


Figure 7. Risk groups defined by OPERA and by NCCN-IPI in DLBCL. Patients with diffuse large B-cell lymphoma and NCCN-IPI recorded at diagnosis, compared on overall survival, the endpoint NCCN-IPI was developed for. The model places patients into as many groups as NCCN-IPI has categories, each with the same number of patients as the corresponding NCCN-IPI category, by predicted risk. No outcome information sets the boundaries. **A**, Kaplan–Meier overall survival by model group, with numbers at risk below the curves. **B**, The same by NCCN-IPI category. Shaded bands are 95% bootstrap intervals and the dashed line marks one year. **C**, One-year overall survival per group with 95% CI, labelled with deaths by one year and patients. **D**, Cross-classification: rows are NCCN-IPI categories and columns model groups, each cell with its patients' one-year survival and interval above and their number below. Outlined cells hold patients placed in the same group by both, cells to the left are patients the model places at lower risk, and cells to the right at higher risk.

What the predictions rest on, and where they hold

Hiding the last 7, 14, 21 and 28 days before treatment changed AUC by **[A, B, C and D (95% CIs)]**, most in **[domains]** (Fig. S35), and only the last year of history retained **[X]** of the AUC. Laboratory results carried **[X]** of the attribution to recorded content (Fig. S38). **[Performance by calendar year, region and patient characteristics, and the prespecified sensitivity analyses]** (Figs. S10 and S21).

Discussion

Borrowing decided whether many prognostic tasks in lymphoid cancers could be learned at all. Trained on one disease and one outcome, the sequence models discriminated little better than chance in **[about half]** of the evaluable tasks. Borrowing from other diseases made most of them useful predictors (0.120 (95% CI 0.107 to 0.133)), adding every other outcome cost nothing measurable, and OPERA added a smaller gain (0.009 (95% CI 0.003 to 0.015)). XGBoost gained far less from pooling (0.012 (95% CI 0.006 to 0.017)), as it already learned usefully from few events, and remained the more accurate model **[(final OPERA–XGBoost difference)]**. That both model families gained suggests the shared information is real rather than an artefact of one architecture.

Classifications separate the lymphoid cancers by biology, and prognostic indices follow them [21]. Our results suggest that biological distinctness and predictive separateness are different things. An infection or a falling neutrophil count may arise through different mechanisms in different diseases, yet the records that precede it, and how it is investigated and treated, are similar enough that examples from one disease inform prediction in another. The gain was largest where a disease had least data of its own, so the limit for small diseases appears to be the number of examples rather than their biology. This is not an argument for redrawing diagnostic boundaries. It is an argument that the boundaries needed for diagnosis and treatment need not also bound what a predictive model learns from.

Borrowing could also have harmed. Joint training has frequently degraded individual clinical

tasks [10], and with 69 outcomes of very different frequency, the rare ones risked being drowned by the common. Instead, adding every other outcome to a model shared across diseases left discrimination unchanged within the prespecified margin (−0.001 (95% CI −0.008 to 0.005)). The outcomes arise during the same course of treatment, and the loss balanced endpoint families, which may both have protected against negative transfer. Other outcomes also added little on their own: models pooling outcomes within one disease improved far less than models pooling diseases. What the models borrowed was mainly patients with the same outcome, not related outcomes in the same patients, and what other outcomes could add may already have been learned through other diseases. An equivalent average can hide losses for individual tasks **[(number of cells worse, and by how much)]**. Still, it means that one model can cover a whole specialty without the price joint training has often exacted.

OPERA was designed to borrow where a task's own data are scarce and to deviate from the shared pattern where a task differs. It did both, but modestly. Own corrections helped mainly in the large diseases and borrowed corrections in the small ones, as partial pooling would predict [11], but the gain over borrowing uniformly was small **[(revisit with the national encoder)]**. Borrowing was also broad rather than along clinical lines. No single source was indispensable, and pairs that clinical knowledge suggests should matter, such as small lymphocytic lymphoma and chronic lymphocytic leukaemia, showed no specific dependence **[(inconclusive on tuning, confirm on test)]**. **[Integrated gradients: features behind each outcome clinically plausible, diagnosis-specific patterns contributing comparatively little (confirm).]** What transfers appears to be general, such as treatment exposure, frailty, prior illness and contact with care, and the shared encoder may already place related diseases together, leaving routing little to add. Pooling therefore need not be specified by hand. Corrections for disease groups, such as the indolent B-cell or T-cell lymphomas, would test this directly. These dependencies describe the fitted model, not biology.

The national arm tests whether foundation models deliver within a specialty. Population-scale models of health records have been proposed as general foundations [17, 18], but whether they improve on strong conventional models for a specific clinical question has rarely been tested [19]. Here, a model pretrained on the nationwide Danish registers was fine-tuned for one specialty and compared, on the same patients, with tuned XGBoost and with models pretrained on the cohort alone. **[Interpret the transfer and label-efficiency results, whether positive, null or harmful.]** If national pretraining closes the gap to XGBoost or reduces the labelled patients needed, it sets a precedent for how national foundation models can raise performance within specialties: in Denmark, by pairing the national registers with clinical databases such as DALY-CARE, and elsewhere wherever national registers exist. If it does not, that is equally informative, because a well-tuned tabular model on the cohort's own records would then remain the better choice.

Few settings would allow this study. DALY-CARE holds every Danish adult with a lymphoid cancer, recorded from the same sources in the same formats, and the national registers cover everyone in Denmark, so each kind of borrowing could be measured on the same patients with otherwise identical models. The test set was opened once, after every choice had been fixed, against a tuned tabular model and the indices clinicians use. A single model stratified patients at least as well as **[each of]** the clinical indices and covered the ten diseases without one **[(interpret NCCN-IPI and recent-record results)]**. These risk groups describe prognosis under the treatment received, not whether treatment could safely be changed [41], and where XGBoost predicts better, that accuracy must be weighed against the convenience of one model for every task. The main limitations concern validation, outcome measurement and precision. Evaluation was temporal within one health system, and external validation remains necessary. Test patients' records before 2022 could contribute to self-supervised pretraining, without outcome labels. Register-derived outcomes are imperfect, and laboratory outcomes depend on monitoring. Clinical comparisons are limited to patients with a recorded index. Reported intervals condition on the fitted models and exclude uncertainty from model selection, and many uncommon tasks had too few events to

be evaluated.

Prognosis in lymphoid cancers need not be learned one disease and one outcome at a time. For many tasks, borrowing across diseases is what makes learning possible, and one model can serve a whole specialty without losing accuracy to negative transfer. **[If the national encoder helps: specialties need not build prognosis from their own patients alone. They can start from a model of the whole population and borrow across their diseases and outcomes, which sets a precedent for prediction within specialties in Denmark and elsewhere. If not: the conclusion rests on borrowing within the specialty.]** The next steps are external validation and testing whether acting on the predictions improves care.

Data availability

According to Danish legislation, the health data used in this study may not be shared publicly. Within the approved DALY-CARE protocol, the data may be accessed on a collaborative basis under a data processing agreement, and all analyses must be performed within the DALY-CARE data resource at the Danish National Genome Center [25].

Code availability

All code, model configurations, outcome definitions and analysis recipes are publicly available at **[repository URL and archival DOI]**. The conversion of the registers to the event format is available at `https://github.com/FGA-DIKU/ehr2meds`. Trained model weights were fitted on patient-level data and are not released.

Patient and public involvement

[Statement.]

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Author contributions

[CRediT statement.]

Declaration of interests

[Statement.]

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Supplementary Methods: model and analysis details

Data sources and processing

All data were held and processed in the DALY-CARE environment at the Danish National Genome Center [25]. Registers were linked through the civil registration number and converted to the Medical Event Data Standard (MEDS) with ehr2meds (`https://github.com/FGA-DIKU/ehr2meds`). MEDS stores every record as an event with a patient, a time, a code and an optional value, so all sources share one format. Table S1 lists the sources, what each contributes and when its events are dated, and Fig. S2 shows how they form one sequence per patient.

Table S1. Sources of the event sequences. Register names follow the DALY-CARE resource [25]. Coverage gives the years with national data in the extract.

Source	Events	Time of the event	Coverage
Civil registration	sex, date of birth, date of death	fixed per patient	**[years]**
National Patient Register (LPR2 to 2019, LPR3 from 2019)	diagnoses, procedures, and the reason, priority, type, specialty and region of hospital contacts	diagnoses at contact end, contact attributes at contact start, procedures when performed	**[years]**
National Pathology Register	SNOMED codes, including molecular findings such as TP53 aberrations **[confirm]**; specialty and region of the requesting department	receipt of the requisition	**[years]**
Register of Pharmaceutical Sales	redeemed prescriptions (ATC)	dispensing	**[years]**
Hospital Medication Register	hospital-administered medication (ATC), with specialty and region	start of the order	**[years]**
Laboratory database	results by NPU code, as value bin or text result	sampling	**[years, varies by region]**

The registers arrive as separate tables with their own identifiers and conventions for time, so the conversion first gives them a common form. Diagnoses and procedures are recorded against a hospital contact rather than a patient, so they were joined to their contact. Contacts, pathology requisitions and hospital medication orders were assigned a region and a primary specialty from the national register of health-care organisations, using each organisation as it was registered on the date of the record, because departments merge, move and change specialty over time. The change of patient register in 2019 also changed how some contact attributes are coded; where a documented correspondence exists, older values were mapped to their newer equivalents, so that a token means the same before and after the change.

The time given to an event determines what the model can see at the start of treatment. The patient registers do not record when during a contact a diagnosis was made, so diagnoses were placed at the end of their contact, the first moment they are certain to have been known; a diagnosis made during an admission that spans the start of treatment is therefore not visible to the model. Attributes known when a contact opens were placed at its start, procedures when performed, prescriptions at dispensing, hospital medication at the start of the order and laboratory results at sampling.

Rare codes give the model too few examples to learn from. The lymphoma conversion configuration uses adaptive truncation with a default minimum of 100 events. Less frequent diagnosis, procedure, medication and pathology codes are mapped to the deepest permitted prefix that meets that count. The permitted prefix lengths depend on the source. Codes without a qualifying prefix retain their full form. Laboratory and contact codes are not truncated. A shared pathology

prefix need not retain the molecular-result specificity of the original code. Attribution concepts therefore require the appropriate result-specific token, checked against the saved mapping, rather than inferring a positive result from a mixed prefix. Laboratory values were binned separately for each test: values were clipped at the 1st and 99th percentiles and assigned to quantile bins, with more bins for tests with more distinct values, following the power law that Montgomery et al. fitted for transformers on health records ($B = \text{round}(1.14\, N_{\text{distinct}}^{0.237})$, between 2 and 100) [42]. Each result became one token combining the test with its bin, or with its text value for non-numeric results such as "<0.1". In the lymphoma configuration, bins and code counts are fitted on events before 1 January 2022 from a random 90% of patients. This split serves only these statistics and is unrelated to the study's training, tuning and test periods. **[Confirm the frozen build configuration and retain its adaptive-code mapping and numeric metadata with the model.]**

BONSAI [43] then turned each patient's events into the sequence the models read. The vocabulary was fitted on events before 1 January 2022, and codes first seen later were read as an unknown token, so no code that appears only in the test period has a learned embedding. Sex and age opened the sequence, a separator marked each change of time so that events recorded together stay grouped, and every event kept its time. For each prediction the sequence was cut strictly before the start of treatment and, if longer than 8,192 tokens, shortened to the sex and age tokens and the most recent events. The conversion can be rerun on the same registers inside the DALY-CARE environment from the configurations in the ehr2meds repository.

Model development

Encoder and pretraining. The encoder builds on CORE-BEHRT [27], adapted from a bidirectional to a causal transformer [44] that predicts each next event from the events before it, with every event's time encoded by a scaled Time2Vec representation [45] (Table S2). It was pretrained on the histories before 1 January 2022 of the pretraining population, every DALY-CARE patient: adults diagnosed with a lymphoid cancer or a precursor condition (MGUS, MBL) since 2002 (78,668), for 10 epochs, the budget of the national encoder. An encoder twice as wide was considered before the analysis was locked and not adopted: fine-tuned as the joint model, it improved tuning AUC by 0.003, within the difference between two fits of the same model, so the narrower encoder was kept. This encoder initialised every supervised model, each trained, tuned and tested on the final cohort (37,531 patients).

Sampling and loss weighting. Patients were sampled in their natural proportions, without oversampling patients with events or reweighting classes, to avoid deliberately changing class proportions. This does not guarantee calibration, which is evaluated separately. Each training patient has events for a few of the 69 outcomes and none for most, so no single weighting of patients could balance all disease–outcome–horizon tasks at once; balance between tasks instead comes from the loss, averaged so that each endpoint group weighs equally. As a sensitivity analysis fixed before the test set was opened, the routed and the joint model were refitted with importance-weighted event sampling: batches were enriched towards patients with events at the six horizons, and each patient's loss was divided by their sampling weight, so the likelihood remained that of the natural population (**[X]** and **[Y]** in AUC against natural sampling).

Memorization check. Test patients' records before 2022 were part of pretraining, so we checked whether the encoder reproduces the records it was trained on. The pretrained model was given the first X events of a patient's record (X = 16 to 4,096) and continued it greedily for Y events (Y = 1 to 64); the continuation was compared with the real record by four scores: all Y events reproduced in order, the share of the next Y events reproduced in any order, the same within 30

days of when they occurred, and the share of the patient's rare codes (held by fewer than 1 in 1,000 training patients) reproduced. Records are partly predictable, so each score was compared between up to 2,000 pretraining patients and up to 2,000 patients the encoder never saw, and memorization was read as the excess of the former, considering only combinations of X and Y with at least 50 scored patients in each group. The next-event loss could not tell the two groups apart (AUC 0.505, 95% CI 0.480 to 0.531), and **[the largest excess of any score was X (CI) at X = , Y =]** (code: `bonsai/run/memorization_check.py` in the BONSAI repository).

Table S2. Model and training specification.

Encoder	causal transformer; hidden size 256, 4 layers, 4 attention heads of 64 dimensions; **[feed-forward width, parameters]**
Input	up to 8,192 tokens; scaled Time2Vec time encoding; laboratory results as test–bin tokens
Pretraining	next-event prediction; events before 1 January 2022; 10 epochs; **[optimiser, learning rate, batch, tokens]**
Prediction head	linear; piecewise-constant hazards for each outcome and for competing death; cumulative incidence at 30, 60, 90, 180, 365 and 730 days
Supervised training	effective batch of 256 patients; up to 30 epochs, stopping after 5 without improvement in tuning loss; tuning-best checkpoint

Prediction head and loss. Every supervised model consisted of the pretrained encoder, trained further, and a linear head that produced, for each outcome, piecewise-constant hazards for the outcome and for competing death, from which the cumulative incidence at each horizon was derived [28, 29]. The loss was a binary cross-entropy at each horizon over the patients whose status was known: those with the event before the horizon were positive, those who died of another cause or were followed beyond it negative, and those censored earlier did not contribute at that horizon. Censoring weights were used in evaluation, not in training. Losses were averaged within each outcome and then across endpoints, treating the grade variants and nested definitions of one finding as one endpoint family (69 separately predicted outcomes, 52 endpoint families). Losses were averaged within each family before families received equal weight, so that findings represented by several grades or nested definitions did not receive disproportionate training weight. This grouping affected loss weighting, not the number of outputs or their separate evaluation.

Supervision scopes. The four scopes, described in Methods, differed only in which cells of the disease-by-outcome grid provided labels (Fig. S1). Every model predicted all six horizons and was trained with the same settings (Table S2) on the same target patients, labels and horizons. Each selected its checkpoint on the tuning labels within its own scope, so a contrast between scopes includes what the extra labels do to training as well as to the final prediction. A disease without tuning patients therefore cannot have a model of its own: nodal marginal-zone lymphoma had five training patients and none in 2022 (Table 1), so its specialists and its outcome-pooled model were not estimable by design and were recorded as such, not as failures; its predictions come only from the disease-pooled, joint and routed models. Because equal epochs do not mean equal computation, optimisation steps and compute are reported for each scope.

Outcome panel

We predicted 69 outcomes, each at 30, 60, 90, 180, 365 and 730 days, in 12 clinical domains (definitions supplement): disease course and later malignancy (including second-line treatment,

treatment failure, transformation, new malignancy and death); infection and antimicrobial use; erythroid findings and transfusion; neutrophil, white-cell and platelet findings; lymphocyte and eosinophil counts; renal injury and support; liver injury and coagulation; sodium and potassium; calcium and magnesium; metabolic and cell-turnover markers; cardiovascular and thrombotic events; and acute care and other events. Laboratory outcomes were defined with reference to the CTCAE laboratory criteria; for each outcome, the definitions supplement states whether it equals a CTCAE grade, follows one with a stated difference, or has no counterpart [26]. The laboratory correspondence audit identified 13 outcomes matching CTCAE grade thresholds, 13 following a grade with a documented deviation, and three without a direct CTCAE counterpart. Deviations include boundary conventions and omitted baseline-dependent criteria. Symptom- and intervention-dependent grades cannot be reproduced from laboratory measurements alone. The ascertainment supplement provides the version-specific correspondence and implementation qualifications. Infection was also captured through antimicrobial treatment, labelled as such. A patient contributed to an outcome only if it could be ascertained for them, for example if laboratory follow-up was available, and was otherwise left out rather than counted as negative. Death was a competing event for every non-fatal outcome. Definitions, code lists and risk-set sizes are given in the definitions supplement.

Tabular comparators: features and tuning

For each disease–outcome cell and horizon, we trained gradient-boosted trees (XGBoost) [33] and ridge-penalised logistic regression on the same patients, labels and competing-event handling as the neural models. Their features came from the pipeline of our earlier DLBCL study [36]: using timeseriesflattener [35], prescriptions, laboratory results, procedures and referrals, pathology codes and diagnoses were summarised over 90-day, one-year and three-year windows before the index date (counts and distinct counts for codes; latest, minimum, maximum, variance and slope for laboratory values). Variables recorded for fewer than 10% of training patients were represented by per-source counts of such rare occurrences, and comorbidity by these counts rather than a weighted index such as Charlson. Features were constructed separately within each disease, so each disease's models used a feature set suited to its own patients. The XGBoost pooled across diseases used one matrix constructed in the same way across all supervised patients and, like the neural pooled models, was not given the disease. To align the register sources of the tabular and sequence models, quality-registry variables such as stage and performance status were removed, and age and sex at the index date were added. Numeric features were median-imputed with missingness indicators and standardised, and categorical features one-hot encoded, with all preprocessing fitted on training data. Hyperparameters were chosen for each cell by log loss on the tuning set from a prespecified grid: for XGBoost, shallow trees (depth 1–3), deeper trees (depth 4 and 6) and the setting of the earlier study (depth 8, learning rate 0.01, up to 3,000 trees), with the number of trees also chosen on the tuning set; for logistic regression, seven penalty strengths. Neither model was class-reweighted or recalibrated. **[Confirm against the delivered feature export: anchoring at first-line treatment, source list and number of features.]**

As a reference outside the information-matched comparison, XGBoost was also trained for each cell with the quality-registry predictors of the earlier study kept: variables recorded at diagnosis (such as performance status, stage, lactate dehydrogenase, extranodal involvement and cytogenetic risk, where the registry for the disease records them) and registry-recorded adverse events before the index date, left missing where not recorded. The sequence models never saw these variables, so this arm estimates what the curated registry adds, not a like-for-like comparison.

National encoder: data and representation

The national (PHAIR) encoder was pretrained on the registers of the whole Danish population, converted to MEDS with the same pipeline. The registers are the same, but the national extract uses older deliveries of some of them: pathology without the requesting department's specialty and region, and an earlier research extract of the laboratory database; it also includes psychiatric contacts. Its vocabulary and laboratory bins were fitted on the national data. To use it, our patients' records were taken from the national extract and tokenised with its vocabulary, so that it read them in the form it was trained on; outcomes, cohort membership and splits were unchanged. Because the national population contains the cohort, the national encoder had seen the cohort's pre-2022 records as well as everyone else's, and continued pretraining on the cohort adds no patients: it is domain adaptation. The national and cohort encoders therefore differ in population, register deliveries and training budget, and the comparisons concern the encoders as delivered.

With the same threshold of 100 events, the national conversion keeps specific codes, such as particular molecular findings or rare diagnoses, that the cohort's conversion merged into their parents, and bins laboratory results more finely. Finer resolution is one route by which the national encoder, or any larger dataset, could improve prediction, separate from what it learns from other patients. In the national conversion, rare codes were counted over the whole period, including events after 2021. No outcomes were involved, and everything the models learned from was restricted to events before 2022, but which codes kept their own token depends on these later frequencies; we note it as a limitation.

The national encoder was pretrained for 10 epochs with a 1 January 2022 cutoff. Continued pretraining on the pretraining population (adaptation) and pretraining on it from scratch ran for the same 10 epochs; on tuning data, encoders pretrained for 10 and 20 epochs gave fine-tuned performance within the difference between two fits of the same model. As references, a randomly initialised encoder of the same architecture was trained with the same supervised protocol, and the pretrained cohort encoder was kept frozen while only the prediction head was trained. Patients, tokens and optimisation steps are reported separately, because equal epochs do not guarantee equal exposure. **[National training population, observation years, events and tokens, encoder version and PHAIR reference.]**

Data efficiency of national pretraining

To test whether population-scale pretraining is most useful when local labels are scarce, the joint model was retrained from the cohort encoder and from the national encoder adapted to the cohort on 10%, 25% and 50% of the training patients, with otherwise identical training. The subsets are nested (10% within 25% within 50%), identical for every outcome, and keep each disease's share. Tuning and test patients were never subsampled, model selection used the full tuning set, and national and cohort encoders were compared at each share with the paired bootstrap of the main analysis.

Comparison with clinical prognostic indices

The quality registries record, for each patient, the prognostic index scored at diagnosis (for example the IPI, FLIPI, MIPI, CLL-IPI or ISS) [1, 3, 46–48] and its risk category. Every index recorded for at least 50 patients of a subtype was compared with OPERA's prediction for the endpoint the index was developed for, in the patients who had that index, and every index used in several subtypes also across all of them, since the IPI, for instance, is applied to many rare lymphomas without a score of their own. For DLBCL we also computed the NCCN-IPI from its components in the lymphoma registry, following the original definition [2]: age (41–60, 61–75 and

over 75 years: 1, 2 and 3 points), LDH relative to the upper limit of normal (above 1 and above 3 times: 1 and 2 points), Ann Arbor stage III–IV, involvement of bone marrow, central nervous system, liver, gastrointestinal tract or lung, and performance status $\geq$2 (1 point each), grouped as low (0–1), low-intermediate (2–3), high-intermediate (4–5) and high ($\geq$6). Patients missing a component were not scored.

Model risk groups were developed using the fitted model's training-cohort predictions at the designated endpoint and horizon, then applied unchanged to tuning and test patients. Index event risks and matching specificity were censoring-weighted within the score-available training population. These are apparent training predictions, not out-of-fold estimates. Matching in training does not guarantee matched risks or specificity in evaluation. The low cutoff was the highest predicted risk at which the group below it, of at least 20 patients, had an observed risk no higher than the index's low-risk category. The high cutoff approximately matched the specificity of the index's high-risk category, so that both flagged the same share of patients without the event; this is also the operating point marked in the precision–recall comparison. Patients in between were divided in the proportions of the index's intermediate categories. A model that ranks patients better can then place more patients at low risk at the same observed risk, and capture more events at the same specificity. For each index we report group sizes, observed risks, and the sensitivity and specificity of classifying a patient in a group or higher, and, for the model, the observed risk in its highest-risk 5% and 10%, which the index cannot resolve within its top category. As a sensitivity analysis, model groups were given exactly the index's group sizes.

Overall survival was the endpoint for the IPI, NCCN-IPI, CLL-IPI, ISS, R-ISS, FLIPI and MIPI. The IPS, built for freedom from progression [49], was compared on treatment failure, a proxy that is reported separately as endpoint-approximate. The horizon was two years where at least a third of the index's test patients could have been followed for two years (judged from index dates and the data cutoff alone, without reference to outcomes), and one year otherwise; one year is reported alongside. Censoring before the horizon is handled by inverse probability of censoring weighting, so the criterion concerns precision, not bias. The original indices were developed for longer horizons, usually five years, so these are shorter-horizon evaluations, not reproductions of the original analyses. Discrimination used the index category (or NCCN-IPI points) as an ordinal score.

Routing formulation and selection

To test whether a joint model benefits from conditioning its predictions on disease beyond ordinary shared learning, we compared it with a routed model trained on the same panel. In the routed model, the patient representation h_i is adjusted for each target outcome k by a weighted combination of small residual adapters, one per source s:

$$\begin{aligned} h_{ik} &= h_i + \lambda \left[G(h_i) + \sum_s \pi_{iks}\, c_s\, A_s(h_i) \right], \\ A_s(h) &= U_s\, \mathrm{GELU}(V_s h), \qquad c_s = \frac{e_s}{e_s + \tau}. \end{aligned} \tag{1}$$

Here A_s is a low-rank adapter, $\lambda = 0.7$ a fixed residual scale and c_s shrinks sources with few training events e_s ($\tau = 10$ events). In the primary local model, G is a global adapter trained across the panel and applied outside the routed mixture. The weights π_{iks} come from a softmax over the sources and one further slot that applies no adapter (its logit starting at 0). That slot adds no source correction; the base path, global adapter and own-source floor remain active. The weights depend on the patient, the patient's disease and the target outcome, and the patient's own disease always receives at least a floor weight of 0.05. The adapted representation enters the same linear head as in the joint model. Each adapter is updated only by the supervision of its own source;

when another disease borrows it, its parameters are used but not updated. The sources can be diseases, outcomes or both; the primary local model uses both banks. Its adapters have rank 8, the query is interacting with width 512, and the router is held fixed for the first two epochs. The primary seed is 42. This is R_{AB}, aliased as `R_L` in the analysis inventory (Table S3). The approach relates to mixtures of experts and parameter-efficient adapters [30, 31, 50–52].

The router was configured on the tuning set in three steps, each ordered from the simplest option: which sources (disease only, outcome only, both), the adapter rank (8, 16, 32), and the routing query (additive with width 128, interacting with width 128, interacting with width 512). Each step chose the option with the highest panel AUC: the AUC difference averaged over estimable cells (at least five cases and five controls), each disease weighing equally, each outcome equally within a disease and each of the six horizons equally. Options whose panel AUC lay within the noise floor of the best were treated as tied and the simplest of them was chosen; an option whose panel IPA was worse than the simplest option's by more than the IPA noise floor was not eligible. The noise floor is the difference between two fits of the same model: the router trained with every source masked, which has the joint model's effective architecture but is fitted independently, against the joint model (**[about 0.004]** tuning AUC). This comparison estimates variation between these two fits and does not prove equivalence of the architectures. The 0.01 margin applies only to claims on the test set, not to these choices. These screens describe the original selection procedure, not a new choice made independently for each figure. The later amendment fixed the current primary local model to the two-bank R_{AB} configuration above. The original router remains a sensitivity fit. Population-pretrained arms are reported separately while their matched downstream comparisons remain pending; the current local OPERA identity is not changed by their availability.

Three controls separate borrowing from simpler explanations. The joint model told the patient's disease adds a learned, zero-initialised shift per disease to the representation, so it can express disease-specific baseline risks but borrows nothing. The own-disease router and the own-outcome router are routed models whose router may use only the patient's own disease adapter, or only the target outcome's own adapter, so they add the adapters' capacity without borrowing. All controls are trained with the same protocol as the routed model. A fourth control keeps the selected source banks but lets each patient use only their own disease's and own outcome's adapter, so the difference from the routed model is the borrowing of the routed model itself. A sensitivity fit, added before stage two, starts the slot that applies no adapter with a larger share of the routing weight (logit 3.0, the code's default, against 0; about 15–18% rather than 1% with 94 sources) and is otherwise identical, to test whether the starting point changes how selectively the router borrows. A rule fixed before either fit finished decided between them (Supplementary Methods, prespecification): the routed model stays primary unless the sensitivity fit is better on the tuning panel by more than the noise floor in AUC, passes the IPA guard and stays better in a second seed of each; the other is reported as the sensitivity analysis. Route entropy, the effective number of sources and the share of weight on a patient's own adapters are reported for both (**[values]**). Route weights describe how the model allocates corrections and are not interpreted as relationships between diseases; what each source contributes is measured by removing it (Supplementary Methods, borrowing map).

Router variants and the primary routed model

On the tuning set, the routed model fixed by the steps above routed poorly. In most patients most of the routing weight sat on one or two sources whose corrections were small (the overall-survival adapter carried 40% of the routed contribution), while the disease adapters, which enter every patient's prediction at the 0.05 floor of their own disease, had grown corrections about 20 times larger. Any weight borrowed from another disease's adapter therefore injected a large correction learned for that disease, and the router learned to park its weight on a source that changed little.

The disease adapters pointed in unrelated directions (median pairwise cosine about 0), and those of related B-cell lymphomas resembled each other, so the adapters had learned disease-specific corrections; it was the routing that had collapsed.

Before any variant was trained, we fixed the variants, their controls and the rule for choosing among them (analysis lock, addenda of 26 October 2026). Two changes address the collapse: a global adapter applied to every prediction outside the routed mix (A), so that general corrections no longer have to pass through a source, and two epochs in which the router is held fixed while the adapters train (B), so that early routing does not lock in before the adapters have learned anything. A third bounds the routing scores (T, three times the cosine of query and key) and was to be used only if A and B did not route healthily; a fourth applies a patient's own disease and outcome adapters at full weight outside the mix, which then holds only borrowed sources (O). Each was trained with controls on the same protocol: the same model with only the patient's own adapters (own-only), the model with the global adapter and no sources (global-only), and a second seed of the candidate. Routing was judged healthy when, measured by contribution rather than weight (route weight times the source's support shrinkage, or its correction size for O), the most frequent top source led in at most half of the rows, at least three different borrowed sources led across diseases, and the median effective number of sources exceeded 1.5.

The routed model with A and B (R_{AB}) routed healthily in both seeds; A alone, and O, did not. Borrowing within R_{AB}, the full model against its own-only control, changed tuning AUC by $+0.014$ (95% interval 0.001 to 0.027), above the seed-to-seed difference of 0.002. Against the first router, R_{AB} differed by -0.004 (-0.011 to 0.002) and O by -0.009 (-0.021 to 0.002). The prespecified rule kept the first router, since R_{AB} was below it by more than the seed-to-seed difference. We amended the rule before any test data were opened and made R_{AB} the primary routed model: the first router's routing is known to have collapsed, so nothing it appears to borrow could be interpreted; R_{AB} routes as intended in both seeds; the selection amendment was informed by tuning diagnostics; and the cost on tuning was small, with an interval including zero. The first router is reported throughout as a sensitivity analysis, and the national routed models use the same readout as R_{AB}. Equation 1 describes this primary R_{AB} fit, including its global adapter; the earlier router omitted G and had no router warm-up. The fitted encoder and all adapters are matched to each variant's own predictions. They are not interchanged at inference.

Hierarchy for effect heterogeneity

Claims rest on the pooled paired-bootstrap differences and their labels (Table 2); the hierarchical model is descriptive. Many cells are small, so their raw estimates are noisy and rankings of individual cells are unreliable. We therefore also fit a simple Bayesian hierarchical model for partial pooling: each cell's effect is the panel mean plus a disease effect, an outcome effect and a cell-specific deviation, so that a noisy cell is shrunk towards what similar cells show, by an amount the data determine. It answers two questions: how large is the typical effect, and how much does it vary between diseases and between outcomes? Because the same patients contribute to every outcome within a disease, the cells' sampling covariance, taken from the shared bootstrap draws, enters the likelihood.

Whether effects depend on the amount of training data for the target is a separate question, and we did not assume a linear relationship: the cell effect varies smoothly with log training events through a cubic spline with few knots and a shrinkage prior towards a flat relationship. As a check that does not depend on smoothing, cells were also grouped into bins of training events fixed before test-set evaluation. Pooling contrasts were fitted as their independent parts (disease pooling, outcome pooling and their interaction), so that every reported combination is an exact sum and all are mutually consistent. Priors are weakly informative on each metric's scale and

checked by prior predictive simulation; convergence and posterior predictive checks are reported, and posterior probabilities of benefit, harm and practical equivalence use a prespecified tolerance and are descriptive, not claim labels. Raw paired estimates are always shown alongside. **[Insert locked priors, tolerances, support-bin edges and sampling diagnostics.]**

Analysis identities and clinical comparison implementation

The primary cohort-pretrained OPERA fit uses the amended two-bank router R_{AB}. In the analysis inventory, `R_L` and `R_AB_L` refer to this same fit; `J_L` identifies the separate joint-model reference. The earlier router, `R_ORIG_L`, and importance-weighted event-sampling fits are sensitivity analyses. Natural patient sampling defines the primary fits. Population-pretrained and cohort-adapted models are separate transfer arms.

Table S3. Models and contrasts in the local analysis. All local sequence models use the cohort-pretrained encoder. The primary local model is OPERA; fixed-arm scientific comparisons retain their own model identities. Population-pretrained arms are separate comparisons and do not change the meaning of the local results.

Model	Analysis identifier	Role
Specialist	`S_L`	One disease and one outcome; reference for supervised pooling.
Disease-pooled	`D_L`	One outcome across diseases.
Outcome-pooled	`O_L`	All outcomes within one disease.
Joint	`J_L`	All diseases and outcomes, shared readout; the fixed pooling and routing reference.
OPERA, primary local router	`R_L` = `R_AB_L`	Both disease and outcome adapter banks, global adapter outside the routed mixture, and two epochs of router warm-up.
Global-only control	`N_A_L`	Global adapter with no source adapters; reference for own and borrowed corrections.
Own-only control	`RBH_AB_L`	Matched routed architecture restricted to the patient's own disease and target outcome sources.
Original router	`R_ORIG_L`	Earlier router; sensitivity analysis, not the primary model.
Sampling sensitivities	`J_IPW_L`, `R_AB_IPW_L`	Importance-weighted event sampling, compared with the corresponding naturally sampled fit.

The local routed fit uses rank-8 adapters, an interacting query of width 512, a residual scale of 0.7, source-support shrinkage with $\tau = 10$, and an own-source floor of 0.05. Seed 42 is the primary local analysis; repeat seeds are diagnostics or sensitivities. The cohort encoder has hidden size 256. The primary run is `seed42/L/fix_R_AB`; the joint reference is `seed42/L/J_all_diseases_all_outcomes`.

In the current tuning figures, clinical indices are compared with the primary local OPERA model on patients with the corresponding index recorded, without an additional disease-specific inclusion list. The shared benchmark is one-year overall survival; native-endpoint analyses are identified separately by endpoint and horizon. Comparisons with fewer than 50 evaluated patients are reported as insufficient support. Model risk-group thresholds are estimated from training predictions by matching the index's low-risk observed risk and highest-category specificity, then frozen for tuning and test evaluation. Tuning comparisons remain exploratory because model development used that split. Native categories and numeric totals are retained where available; already-coarsened registry categories cannot be reconstructed without the original score or labels.

Statistical inference and descriptive analyses

Time-dependent AUC was the prespecified primary performance metric. Brier score, IPA and calibration provided complementary assessments of probability accuracy. Each pooled difference was labelled by a rule fixed before testing: better (or worse) if its 95% interval lay above (below) zero, noted as beyond the practical margin when the difference exceeded 0.01 AUC; otherwise equivalent if the interval lay within ± 0.01; otherwise inconclusive; each with a paired bootstrap P value. The headline comparisons form one family, with P values adjusted by Holm's method: sharing across diseases, across outcomes and both (each against the specialists), the joint model and OPERA against per-task XGBoost, borrowing by the routed model, the national encoder against the cohort encoder, and OPERA against the clinical indices. Everything else, including results for individual diseases, outcomes, cells and disease pairs, is reported as estimates with intervals and is descriptive. Across disease–outcome cells, effects were also summarised, descriptively, by hierarchical models that estimate the typical effect, its variation between diseases and outcomes, and its relation to the number of training events; estimates for individual cells are shown in the Supplementary figures. Calibration regression estimated the intercept and slope jointly from the unchanged model probabilities. These coefficients describe calibration and were not applied to update risks. Fits without a finite solution, including separation of events and non-events, were reported as not estimable. Medians and interquartile ranges describe variation between cells, rather than uncertainty in a pooled mean. Compute, training time and storage were compared for one shared model and the separate models it replaces. Integrated gradients and their event-removal and randomisation checks are described in the interpretability supplement [53, 54].

The generated tuning recipe required at least 50 evaluated patients per clinical-score illustration. This is an operational display threshold, not a sample-size justification or a guarantee of adequate events within score strata. Group counts and uncertainty remain necessary. Crossed or coincident matched thresholds, or thresholds leaving empty training strata, were reported as not estimable rather than artificially separated.

Temporal differences between study periods

The three periods differed in ways that a temporal evaluation has to absorb. National laboratory capture is complete only from 2017, so patients treated earlier had far fewer recorded results (median 119 in the training period against 524 in the test period), and later patients had longer recorded histories (median 11.6 against 22.0 years). Performance status and the disease-specific index category, recorded in the quality registries, were missing for 47% and 58% of test patients against 24% and 36% in training. The reason for this difference in registry completeness was not established. The disease mix shifted too: DLBCL made up 27.5% of training and 22.7% of test patients, while indolent B-cell lymphomas rose from 13.4% to 18.8%, partly through more unspecified B-cell and splenic marginal-zone lymphoma, and test patients were older (median 72 against 69 years). The test set therefore measures performance under clear data shift.

Reference variability in the original router screen

The joint model and the router trained with every source masked implement the same shared prediction architecture but were fitted separately. Their AUC difference was -0.002 (95% CI -0.005 to 0.001). This difference supplied the operational noise floor for the original screening comparisons. It is a comparison of two fits, rather than a general estimate of training variability. The primary amended router used matched global-only and own-only controls, as specified above.

Classification at every ordered risk-group boundary was reported for the index and model in

`classification_at_thresholds.csv`. Higher groups were positive, with model risk greater than or equal to the stored cutoff. Metrics included TP, FP, FN, TN, recall, precision, specificity, NPV, MCC, accuracy and F1. Known-status metrics excluded early censoring and treated competing death as negative. A separate IPCW version used disease-specific censoring weights. Undefined quantities were missing, not zero. The source split, cutoff and input prediction hashes were retained with the report.

Validation of integrated-gradient explanations

Removing the five most credited events lowered the predicted risk more than removing five random events for **[[n]]** of **[[n]]** check outcomes, attributions of a model with randomised weights agreed little with those of the trained model (median rank agreement **[[x]]**), and attributions were stable to the choice of reference (**[[x]]**).

Recent-record removal measures reliance on recent information. It does not validate the attribution method, which was assessed by event-removal, randomisation and reference-sensitivity checks separately.

Supplementary Methods: evaluation and reporting

Prespecification

Model development and analysis amendments used the training and tuning splits. The held-out test split comprises patients treated from 2023. The current figure previews use tuning data and are exploratory. Final reporting must identify the dated analysis lock, the fitted checkpoints and any departures from that plan. **[Attach the final plan, amendment history and held-out evaluation manifest before describing analyses as prespecified or completed.]** Clinical risk-group thresholds are estimated from training data. The separate panel-wide operating-point analysis uses tuning-derived thresholds.

An initial router-rule amendment was made on 21 October 2026, after rough source-screen estimates had been seen and before the test set was opened; all four source banks are compared on the test set, and the rank, query and OPERA results had not been seen. The router screens were then rerun from scratch after an audit showed that the optimiser's numerical constant (Adam's epsilon, 10^{-6}) damped the updates of the routing parameters, whose gradients were mostly smaller; every routing parameter was retrained with 10^{-10} before the router was locked and before the test set was opened, and the earlier screens are not reported as results. The rule for the sensitivity fit's starting point (Supplementary Methods, model details) was added on 25 October 2026, from training-time diagnostics of unfinished fits, before their tuning-panel results or any test result had been seen.

Weighting of summaries

We report two mean paired summaries of AUC, IPA and Brier differences on the same supported disease–outcome–horizon cells. The equal-cell summary gives each cell–horizon combination equal weight. The disease-balanced summary gives each disease equal weight, each outcome equal weight within a disease, and each evaluated horizon equal weight within an outcome. Point-estimate weights and the included cells are held fixed across the paired patient bootstrap draws; unavailable cell estimates are not treated as zero. The Brier difference is oriented as reference minus candidate, so positive values favour the candidate for all three measures. These means describe different task distributions; neither is a patient-population utility measure. Endpoint-family or clinical-domain weighting is distinct from this disease-balanced definition and is labelled separately when used.

Headline differences and Table 2 give each supported cell–horizon combination equal weight. Disease-balanced summaries use the nested disease, outcome and horizon weighting described above. The 52 endpoint families define training-loss weights, not evaluation weights. Clinical-domain weighting is a separate sensitivity analysis. Eligibility and finite-bootstrap support are recorded for each contrast rather than silently replacing missing estimates by zero. AUC requires at least five cases and five controls for the supported-cell summaries. This restriction does not define whether a model can produce a prediction or whether a Brier score can be calculated.

Borrowing map

The routed readout adds to the shared patient representation one routed correction per source disease (routing weight times that disease's adapter output), and the risk head is linear, so each source's contribution to every hazard can be switched off exactly while all other weights stay as trained; no weight is renormalised. A switched-off source's term is replaced by its mean over tuning patients for the same target disease and outcome (mean ablation); diseases without tuning patients use the mean over all tuning patients. Mean replacement retains the average

source contribution while removing its patient-specific variation. Zero replacement changes both components. Zero ablation is reported as a sensitivity analysis. One evaluation of the locked model scored, on the same patients, the full model, the model with other-disease corrections mean-replaced, each source alone (own disease plus that source) and each source left out (all sources but that one). Differences in AUC used one paired patient bootstrap shared by all variants: the AUC lost without borrowing (full minus none), the source alone (alone minus none; primary) and the source left out (full minus left out). The outcome-adapter bank, global correction and shared encoder remain active in these disease-bank ablations. The source-alone contrast measures what a disease adds beyond that baseline, the source left out what it adds with all the others present. The loss without borrowing measures what the fitted model uses; it can exceed the gain over a model trained without borrowing (the own-disease router), because the encoder was trained alongside the borrowed corrections. Per target disease, source and clinical domain, differences were averaged over estimable cells (at least five events and five non-events), each outcome weighing equally and each horizon equally within an outcome. The map shows the source alone with adaptive shrinkage [55]: within each clinical domain all pairs share one prior, a mixture of normal distributions centred on no borrowing with weights fitted to the data, so that the many near-null pairs are pulled towards zero while large, precisely estimated ones are kept. Disease groups are not used in the pooling; they order the map for reading only. Dots mark pairs whose pooled 95% interval lies above zero. The AUC lost without borrowing (full against no borrowing in the fitted router) was estimated per target disease, per outcome domain and over all diseases, each disease weighing equally (Fig. S37B, D), with the same bootstrap, and set beside routing (routed against joint model) and borrowing as trained (routed model against the router trained with its own disease adapter only), and against disease size (Fig. S18A–B). Candidate pairs (a source disease, a target disease and a clinical domain) were listed from clinical reasoning and the literature and kept only if a single-pair estimate could be informative: at least 200 training patients in the source, and at least 50 test patients in the target. These support criteria do not guarantee that an individual pair can be estimated precisely. Smaller diseases remain in the descriptive map. A confirmatory test of these pairs was planned, with the expected pairs and negative controls to be rated by study haematologists without our rationale. Before any rating or test-set evaluation, the routed model was found on the tuning set to borrow broadly, with similar mixtures of sources across diseases (correlation of borrowing profiles 0.83 to 0.89), so a test of a few selected pairs would not describe how it borrows. The pairs are therefore reported descriptively, each with its unshrunk source-alone gain and 95% interval (Table S4, with the excluded candidates and the reason for each), and whether borrowing favours related diseases is described over all diseases by lineage concentration (main text, Methods). IPA differences were computed in the same way as a secondary metric.

Table S4. Prespecified borrowing pairs and negative controls. Fixed before the test set was opened and reviewed by the study haematologists. Source: the disease switched on alone; target: the disease predicted; outcomes: the clinical domain tested. Rating: the study haematologists' expected strength, given blind to the rationale and to whether the pair was proposed as a hypothesis or a control. Pairs were labelled against a practical margin of 0.25 AUC points.

	Source	Target	Outcomes	Rating	Rationale (evidence)
Expected to share signal					
H1	Chronic lymphocytic leukaemia (CLL)	Small lymphocytic lymphoma (SLL)	all outcomes	–	One disease in two presentations (blood vs lymph nodes), managed by the same criteria and treatments. (WHO 5th edition classification (Alaggio et al., Leukemia 2022))

	Source	Target	Outcomes	Rating	Rationale (evidence)
H2	Extranodal marginal-zone lymphoma (EMZL)	Splenic marginal-zone lymphoma (SMZL)	disease course and later malignancy	–	The marginal-zone lymphomas share prognostic factors: the MZL-IPI (LDH, haemoglobin, lymphocyte and platelet counts, subtype) was built for and validated across all subtypes, and advanced disease is treated with the same rituximab-based regimens. First-line treatment often differs (radiotherapy for localised EMZL; rituximab alone or splenectomy for SMZL). (Arcaini et al., EClinicalMedicine 2024, doi:10.1016/j.eclinm.2024.102592)
H3	Multiple myeloma (MM)	Chronic lymphocytic leukaemia (CLL)	infection and antimicrobial use	–	Both cause secondary humoral immune deficiency (hypogammaglobulinaemia in up to 85% of CLL during the disease course, and intrinsic or treatment-related in myeloma), with the same immunoglobulin-replacement indications. (Frontiers in Immunology 2014, doi:10.3389/fimmu.2014.00626; Frontiers in Immunology 2016, doi:10.3389/fimmu.2016.00317)
H4	Chronic lymphocytic leukaemia (CLL)	Multiple myeloma (MM)	infection and antimicrobial use	–	The same, in the other direction. (Frontiers in Immunology 2014, doi:10.3389/fimmu.2014.00626; Frontiers in Immunology 2016, doi:10.3389/fimmu.2016.00317)
H5	Burkitt lymphoma (BL)	Lymphoblastic lymphoma (LBL)	neutrophil, white-cell and platelet findings	–	Both receive short, highly intensive regimens (CODOX-M/IVAC, hyper-CVAD, ALL-type protocols) with marked myelosuppression despite growth-factor support. (REFERENCE NEEDED: toxicity of intensive regimens in adult Burkitt lymphoma)
H6	Burkitt lymphoma (BL)	Lymphoblastic lymphoma (LBL)	metabolic and cell-turnover markers	–	Rapid tumour turnover and intensive induction put both at risk of tumour lysis (potassium, phosphate, urate, creatinine). (REFERENCE NEEDED: tumour lysis in Burkitt and lymphoblastic lymphoma)
H7	Peripheral T-cell lymphoma, NOS (PTCL)	T-cell lymphoma, unspecified (TCL)	disease course and later malignancy	–	Unspecified T-cell lymphoma is managed like peripheral T-cell lymphoma, not otherwise specified (CHOP-based therapy), with a similarly poor course. (WHO 5th edition classification (Alaggio et al., Leukemia 2022))
H8	Peripheral T-cell lymphoma, NOS (PTCL)	Angioimmunoblastic T-cell lymphoma (AITL)	disease course and later malignancy	–	Angioimmunoblastic T-cell lymphoma (nodal T-follicular-helper lymphoma in WHO 5) is treated with the same CHOP-based first-line regimens as PTCL, not otherwise specified, with a similarly poor course; the two are often hard to tell apart histologically. (WHO 5th edition classification (Alaggio et al., Leukemia 2022))
H9	Chronic lymphocytic leukaemia (CLL)	Lymphoplasmacytic lymphoma (LPL)	cardiovascular and thrombotic events	–	BTK inhibitors, used in CLL and in Waldenstrom macroglobulinaemia, cause atrial fibrillation and bleeding. But first-line Waldenstrom treatment in Denmark has mostly been chemoimmunotherapy (e.g. bendamustine-rituximab), so a null result here would be unsurprising. (Brown et al., Haematologica 2017 (pooled AF in ibrutinib trials), PMC5622864)
Negative controls					
C1	Multiple myeloma (MM)	Hodgkin lymphoma (HL)	all outcomes	–	No shared biology, treatment or complication profile (young patients on ABVD-type therapy vs older patients on myeloma regimens); MM is the second-largest source, so this also tests whether a large source helps regardless of relevance. (–)
C2	Hairy-cell leukaemia (HCL)	Diffuse large B-cell lymphoma (DLBCL)	disease course and later malignancy	–	No shared biology or course (indolent, cladribine-treated vs aggressive, R-CHOP-treated). (–)
C3	Anaplastic large-cell lymphoma (ALCL)	Chronic lymphocytic leukaemia (CLL)	all outcomes	–	No shared biology, treatment or complication profile (a T-cell lymphoma treated with CHOP-type therapy vs an indolent B-cell leukaemia). (–)
Candidates excluded before rating (estimate not informative)					
X1	Diffuse large B-cell lymphoma (DLBCL)	Transformed follicular lymphoma (tFL)	disease course and later malignancy	–	target has 44 test patients
X2	Diffuse large B-cell lymphoma (DLBCL)	Richter transformation (RT)	disease course and later malignancy	–	target has 25 test patients
X3	Chronic lymphocytic leukaemia (CLL)	Richter transformation (RT)	infection and antimicrobial use	–	target has 25 test patients

	Source	Target	Outcomes	Rating	Rationale (evidence)
X4	Multiple myeloma (MM)	Plasma-cell leukaemia (PCL)	disease course and later malignancy	–	target has 38 test patients
X5	Multiple myeloma (MM)	Amyloidosis	renal injury and support	–	target has 17 test patients
X6	Anaplastic large-cell lymphoma (ALCL)	Solitary plasmacytoma	all outcomes	–	target has 32 test patients
X7	Amyloidosis	Hodgkin lymphoma (HL)	all outcomes	–	source has 22 training patients
X8	Transformed follicular lymphoma (tFL)	Hairy-cell leukaemia (HCL)	disease course and later malignancy	–	HCL has no second-line outcomes and almost no deaths, so no estimable cells

Sample size and precision

The cohort was not sized for this study. It comprises every eligible patient in Denmark over the study period, so the relevant question is what precision this fixed sample allows, not how many patients would be needed. Formal criteria for the sample size of a new prediction model [5] assume one outcome and a fixed set of predictors, and were not used to justify the size of this shared-parameter model. Shared training does not remove the need for adequately supported validation. We therefore report, for every disease–outcome cell, the number of training events and of test events and non-events (Fig. S5), evaluate a cell only when the test period has at least five events and five non-events, and judge each estimate by the width of its interval. For validation, about 100 events and 100 non-events are commonly recommended for precise estimates of discrimination and calibration [56]. [N] cells reached this at one year, and cells below this threshold are interpreted with particular caution. Pooling evidence across cells can support a panel-level conclusion but does not validate a sparsely observed cell for individual clinical use.

Risk sets and paired comparisons

The unit of evaluation was one disease–outcome–horizon combination. Its risk set was determined by eligibility and ascertainment of the outcome, not by the models compared. For each split we report eligible patients, outcome events, competing deaths and patients whose status was unknown at the horizon. Within a cell, every model was evaluated on the same patients and labels. Contrasts can cover different sets of evaluable cells, so their cell and patient counts are reported. Cells in which a model could not be fitted or a metric could not be estimated are shown with the reason (Fig. S4) and not treated as zero effects.

A pooled model could predict cells in which no specialist could be fitted. These cells have absolute performance estimates but no paired specialist comparison, so they are reported as added coverage and kept out of the paired averages. Outcome definitions and the reconstruction of treatment lines are in the definitions supplement.

Clinical prognostic indices

Which index each subtype has. Table S5 lists, for every subtype, the prognostic index derived for it, how far that index is established, and what is used in practice. A subtype was classed as having no dedicated prognostic index if no multivariable index had been derived for that entity and externally validated in an independent cohort; indices developed for other entities (such as the IPI, built for aggressive lymphoma in 1993 and applied to Richter transformation, transformed follicular lymphoma and anaplastic large-cell lymphoma) or for broader families (the MZL-IPI for all marginal-zone lymphomas) did not count. This literature criterion was

preferred to guideline adoption, which differs between guidelines. Two sensitivity analyses vary it: counting family-level indices removes nodal marginal-zone lymphoma from the stratum, and requiring routine, guideline-recommended use adds angioimmunoblastic T-cell lymphoma, whose dedicated scores have limited external validation, and Burkitt lymphoma, whose validated index does not yet assign treatment.

NCCN-IPI is the main clinical comparator in DLBCL [2]. IPI comparisons are reported separately, including eligible lymphoma subtypes with a recorded score. How the indices were compared, and how model risk groups were anchored to them, is described under "Comparison with clinical prognostic indices". All methods in an index comparison used the same patients, those with the index recorded at diagnosis, and how often it was recorded is reported by disease and calendar period. Because the models read the wider history, a gain over an index is a gain of the whole pipeline, not of the model alone. Index categories are not probabilities, so they were used for ranking and stratification only, not for Brier scores or calibration. **[Index, version, population and availability inventory from the secure analysis.]**

Risk groups, age and operating points

Continuous model risks were converted to groups fixed at baseline with cutoffs estimated on training predictions from the fitted model, while index groups kept their native definitions. These are apparent training predictions, not out-of-fold predictions. The low-risk cutoff targets the index's training low-risk event rate and the high-risk cutoff its highest-category specificity. Intermediate boundaries preserve the index's intermediate-group proportions. Unsupported or crossed boundaries are reported rather than repaired silently. Kaplan–Meier estimates describe endpoints that include death, and Aalen–Johansen estimates non-fatal endpoints with death as a competing event. Group sizes, events and 95% bootstrap intervals accompany every estimate. Patients without a recorded index were not included in index comparisons.

In the age figures, patient counts, mean predicted risk and observed risk are shown as separate rows on the same age bands and groups. The predicted row is the sequence model's mean probability, also within index categories. The index itself supplies no probability. Estimates are suppressed for cells with fewer than ten patients, while counts remain visible.

Clinical operating points use these frozen training-derived thresholds. Specificity is not forced to match again in tuning or test patients. We report the achieved sensitivity, specificity, positive and negative predictive values and patient counts. Confusion counts use patients with known status at the horizon, and the number excluded by censoring is reported. Censoring-weighted quantities are labelled as such. At every non-low-risk boundary, positive means risk greater than or equal to the cutoff. Exports include recall, precision, specificity, negative predictive value, MCC, accuracy, F1 and confusion counts, separately for known-status and IPCW analyses. These clinical threshold exports currently contain point estimates, not bootstrap confidence intervals.

For the panel as a whole, every model was also evaluated at six decision points, each fixed per model, cell and horizon on that model's tuning predictions. The first flagged as many tuning patients as had the event, so that every model raised the same number of alerts. The second and third maximised Youden's index and the MCC. The remaining three were the lowest thresholds with a tuning specificity of at least 80%, 90% and 95%. The test period then shows how far the achieved specificity drifts from its tuning target. For each decision point we report the threshold, the confusion matrix, sensitivity, specificity, positive and negative predictive values, MCC and the share of patients flagged, with 95% intervals from the shared patient bootstrap and paired differences between OPERA and XGBoost, the joint model and the specialist, and the routed and joint models. Disclosure checks are required before exporting confusion counts. The new clinical threshold CSVs do not yet enforce small-count suppression automatically. Cells with fewer than five tuning events or non-events have no decision point.

Table S5. Prognostic indices by lymphoid subtype. Dedicated index: a multivariable index derived for that entity. Status: *routine*, externally validated and recommended in international guidelines; *validated*, externally validated but not the basis of routine decisions; *proposed*, derived without independent validation; –, none. In practice: what is applied when there is no dedicated index (borrowed indices in italics). †Prespecified stratum without a dedicated index. **[Check the guideline column against the current Danish (DLG, DMSG, DCLLSG) guidelines.]**

Subtype	Dedicated index	Status	In practice
DLBCL and related aggressive B-cell lymphomas			
Diffuse large B-cell lymphoma	IPI, NCCN-IPI, CNS-IPI [1, 2, 57]	Routine	IPI (Danish guideline)
Richter transformation†	Richter score [58]	Proposed	*IPI*; clonal relatedness, TP53
Transformed follicular lymphoma†	–	–	*IPI*; managed as DLBCL
Burkitt and lymphoblastic lymphoma			
Burkitt lymphoma	BL-IPI [59]	Validated	Low/high-risk criteria
Lymphoblastic lymphoma†	–	–	ALL-type risk and response
Mantle-cell lymphoma			
Mantle-cell lymphoma	MIPI [47]	Routine	MIPI
Indolent B-cell non-Hodgkin lymphoma			
Follicular lymphoma	FLIPI [46]	Routine	FLIPI
Lymphoplasmacytic lymphoma	IPSSWM [60]	Routine	IPSSWM
Extranodal marginal-zone lymphoma	MALT-IPI [61]	Routine	MALT-IPI
Splenic marginal-zone lymphoma	HPLL/ABC score [62]	Validated	HPLL/ABC score
Nodal marginal-zone lymphoma†	– (MZL-IPI covers all MZL [63])	–	*FLIPI* or *IPI*
B-cell lymphoma, unspecified†	– (category)	–	*IPI*
CLL and small lymphocytic lymphoma			
Chronic lymphocytic leukaemia	CLL-IPI [3]; Rai/Binet stage	Routine	CLL-IPI, stage
Small lymphocytic lymphoma	CLL-IPI (same disease)	Routine	CLL-IPI, stage
Hairy-cell leukaemia, Hodgkin lymphoma			
Hairy-cell leukaemia†	–	–	Treatment criteria [64]
Hodgkin lymphoma	IPS [49]; early-stage risk groups	Routine	IPS, risk groups
T-cell non-Hodgkin lymphoma			
Peripheral T-cell lymphoma, NOS	PIT [65]	Validated	*IPI* most often; PIT
Angioimmunoblastic T-cell lymphoma	AITL score [66]	Proposed	*IPI* or PIT
Anaplastic large-cell lymphoma†	– [67]	–	*IPI*; ALK status
T-cell lymphoma, unspecified†	– (category)	–	*IPI* or *PIT*
Myeloma and other plasma-cell neoplasms; amyloidosis			
Multiple myeloma	ISS, R-ISS [4, 48]	Routine	ISS, R-ISS
Plasma-cell leukaemia†	PCL score [68]	Proposed	*ISS, R-ISS*; treated as high-risk myeloma
Solitary plasmacytoma†	Risk factors only [69]	–	Light-chain ratio, marrow involvement
AL amyloidosis	Mayo stage [70]	Routine	Mayo stage

Censoring, calibration and robustness

The primary metrics weight patients by the inverse probability of remaining uncensored, estimated separately by disease, outcome and split; an analysis on patients with known status, without weights, is a prespecified sensitivity analysis. We report the censoring survival at each horizon and the distribution of weights, and flag long horizons where little follow-up makes estimates unstable. The weights are held fixed across bootstrap resamples, so intervals do not include their uncertainty, and require independent censoring within the estimation strata, which may not hold where monitoring follows the patient's condition.

Calibration is reported as calibration-in-the-large, the calibration slope where estimable, and reliability curves for outcomes chosen by a fixed rule. No model was recalibrated on test data. Sensitivity analyses by horizon, calendar year, amount of recorded history and timing of the subtype diagnosis are reported on their own risk sets. Raw paired effects are reported with the hierarchical summaries, and the support curve, unsmoothed and binned analyses are treated as views of the same data rather than independent replications.

Regional performance groups each patient by the region with the most pre-treatment contacts. The models were trained nationally, so this describes variation within the development health system rather than validation in regions excluded from training.

Patient representations

Representation figures identify their split and available model stages. Each stage has its own projection, fitted on tuning patients and applied to the evaluation split. Coordinates are not comparable across independently fitted stages. Updated galleries colour patient points by predicted risk. Observed-risk and calibration surfaces, when shown, use local IPCW smoothing and are labelled separately. Router arrows require a shared projection of the base and corrected representations. Projection geometry and arrows are descriptive and do not establish predictive gains or biological mechanisms.

Numerical tables

The printed supplementary tables summarise the comparisons that the figures show: the panel-level comparisons behind Table 2, paired differences by comparison and clinical domain, calibration by model and domain, the evaluated cells, events and non-events by domain and horizon, threshold metrics at the six decision points, and, where available, the comparisons with each index on its original endpoint and the computational resources of each model. Intervals condition on the fitted checkpoints and describe patient-sampling uncertainty. Limited router seed controls do not provide systematic estimates of training-run uncertainty. The complete cell-level results, including every confusion matrix, accompany the paper as machine-readable files, after disclosure review, including suppression of small counts where required.

[Reviewed aggregate summary tables have not yet been exported. Complete cell-level numerical tables will accompany the figures.]

What the model relies on: attributions and input restriction

This secondary analysis asks what OPERA reads in a patient's history, so that clinicians can judge whether its predictions rest on sensible information. It is computed once on the model selected on tuning data, before any attribution is inspected, and every figure and table is a summary of that one computation. Outcomes, sampling, the reference, the expectations and the checks were specified beforehand.

Attribution. Let $f_k(E; C)$ be the prediction for outcome k at one year, where E holds the content embeddings of the recorded events (which diagnosis, drug or laboratory result and value bin each event was) and C the context held fixed (each event's time, the age at it, its position and the patient's subtype). The reference E^0 replaces every event's content by the mean content embedding of training events, an average event, while keeping the timeline. The encoder was pretrained to predict the next event, so the mask token was never trained and is not a suitable reference. Integrated gradients [53] move the record step by step from this average record to the patient's own, and add up how much each event's content changes the prediction along the way; the integral below is that running sum, and α runs from the average record ($\alpha = 0$) to the patient's record ($\alpha = 1$). Event j receives the credit

$$a_{jk} = \sum_d (E_{jd} - E^0_{jd}) \int_0^1 \frac{\partial f_k(E^0 + \alpha(E - E^0); C)}{\partial E_{jd}} \, d\alpha, \tag{2}$$

and the credits add up to $f_k(E; C) - f_k(E^0; C)$. For every outcome f_k is the log-odds of the predicted cumulative incidence, so that contributions are comparable across rare and common outcomes and across low- and high-risk patients. For death, serious infection and severe neutropenia the absolute risk is also attributed, for the removal checks and the explanations of single patients. All targets share one forward pass along the integration path, and the attribution passes through the router, so it explains the prediction the model actually makes. For death the attributed quantity is the risk of death, so a positive contribution raises predicted mortality.

The path integral is evaluated by Gauss–Legendre quadrature with 8 nodes and repeated with 32 when the absolute gap between the summed credits and $f_k(E; C) - f_k(E^0; C)$, divided by $\max(|f_k(E; C) - f_k(E^0; C)|, 0.001)$, exceeds 0.05. Patient–outcome pairs that still fail are reported by outcome and excluded from the summaries. The attribution manifest records the split, patients, outcomes and numerical settings actually processed. Current tuning exports must not be described as completed held-out explanations. Patient-level outputs stay in the secure environment.

Episodes and concepts. The unit of interpretation is the episode of care: events within one day of each other, such as one admission or one diagnostic work-up, because the diagnoses, results and drugs of one admission are recorded at different times. Concepts group codes by structure only: ICD-10 to three characters, ATC to the pharmacological subgroup, procedures to four characters, pathology SNOMED codes to three, and laboratory results to the test and the third of its value range (low, middle or high) of the latest result. Where reviewed mappings are available, codes are named from the official classifications (ICD-10 in its Danish SKS form, the WHO ATC index, the NPU terminology for laboratory tests, SKS for procedures and ICD-O morphology for pathology), at the finest level of the code's hierarchy that had a name, and grouped into the chapters of each classification and into clinical groups whose diagnosis codes are those of the outcome definitions (main-text Outcomes section), so a finding and the outcome it resembles are defined by the same codes. **[The names of the concepts shown in the figures were reviewed by [reviewer].]**

What makes patients differ. Because the credits add up, each patient's prediction is $f_k(E^0; C) + \sum_g c_g + r$, with c_g the summed credit of source (or time window) g and r the numerical residual. The between-patient variance of the predictions therefore divides exactly into $\mathrm{cov}(f_k(E^0; C), f_k) + \sum_g \mathrm{cov}(c_g, f_k) + \mathrm{cov}(r, f_k)$. Each share, a covariance divided by the variance, is signed and the shares sum to one. The share of the reference is the part of the differences between patients carried by the timing of care, age and subtype rather than by what the events were. Intervals come from 1,000 weighted patient-bootstrap resamples.

Outcome-specific use of the history. For each patient and pair of outcomes, the episode contributions were ranked within each outcome and correlated (Spearman). Patient correlations were averaged on the Fisher-z scale with the sampling weights, for patients with at least five episodes and pairs with at least 20 such patients. Agreement describes similarity in credited episodes. It does not identify whether the model has learned frailty or an outcome-specific mechanism.

Explanations for one patient. For death, serious infection and severe neutropenia, explained in absolute risk, each patient's episodes of care are summed and labelled with their timing and main concepts. Two records are extracted. For every patient and outcome, the three episodes that most raise and the three that most lower the predicted risk, each with its contribution in percentage points, is the format a clinical tool could show beside each prediction. For one patient, the five episodes that move any of the three risks most are shown with what each adds for every outcome, so that the reference, the episodes and the rest of the history add up to the prediction. Concepts require a reviewed code-to-name mapping before publication, with laboratory results named by the value third of the result. Both records stay in the secure environment and pairs that failed the completeness check are left out. The manuscript shows a synthetic patient.

Findings and expected effects. Thirteen findings (seven laboratory, three diagnostic and three medication findings) are summarised for every outcome as the mean change in predicted log-odds among the patients who have them, with 95% patient-bootstrap intervals, suppressed below 20 patients. For 21 finding–outcome cells a direction was fixed before the analysis (for example, high LDH raising predicted death and treatment failure, and low neutrophils raising predicted severe neutropenia and serious infection). Each is seen or opposite when the interval excludes zero on the expected or other side, and unclear otherwise. The other cells are descriptive. Agreement with expectations shows plausibility, not faithfulness.

Checks. For faithfulness, the k events with the largest attributions were replaced by the reference (and, separately, deleted) and the drop in predicted risk was compared with removing k random events and the k least attributed events, in 200 patients, with $k = 5$ prespecified. Deletion can create unfamiliar inputs, which is why replacement is primary. In 100 patients the attributions were recomputed with the readout, and with the readout and encoder layers, redrawn at their original scale [54] (agreement should be low), and with a per-source reference (agreement should be high). Intervals describe the explanation sample given the fitted model and the reference, not training-run uncertainty.

Input restriction. Separately, the same model was evaluated on the whole test set with restricted input and without retraining: the timing of care only (every event replaced by the average event, as in the reference), without laboratory results, without medication, without diagnoses, without the region and specialty of care (two regions contribute richer hospital-record data, so region partly marks which data a patient has), with the last year of history only, and with the final 7, 14, 21 or 28 days of recorded events removed. These recent-record gaps retain the same fitted model and evaluate sensitivity to information immediately before treatment. For each outcome the held-out AUC at one year was compared with the full input, using paired patient-bootstrap resamples shared across outcomes and restrictions, and averaged over outcomes and clinical domains. This measures what the fitted model relies on, not what a model trained on less information could achieve.

Supplementary figures

Supplementary figures group study design, performance, clinical comparisons, representations and explanation diagnostics; all are cross-referenced in the text.

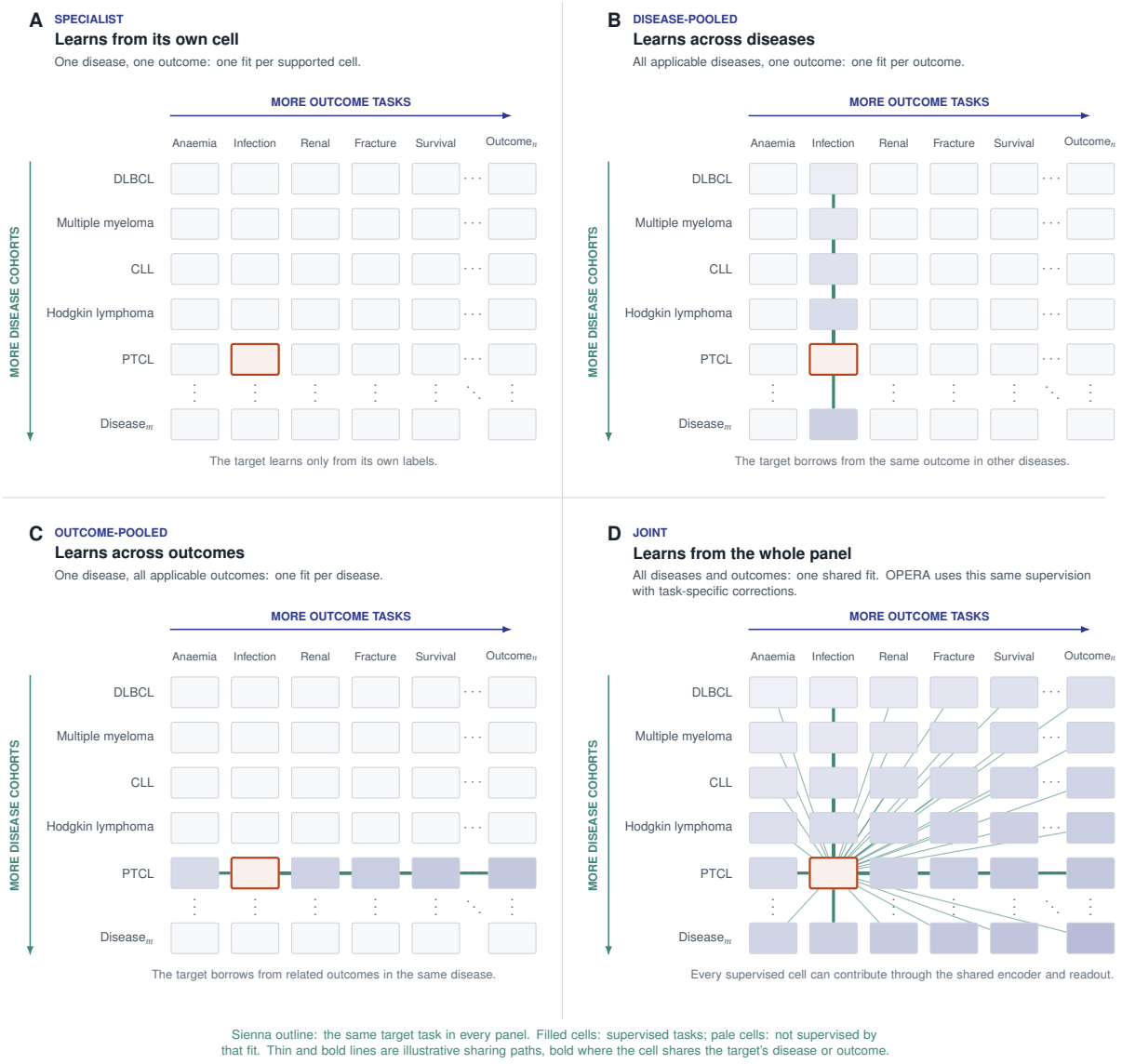


Figure S1. Supervision scopes on the disease–outcome matrix. The matrix of Fig. 1E, with one target task (outlined) evaluated identically in every panel. Diseases are rows and outcomes are columns. **A**, A specialist learns one outcome in one disease. **B**, A disease-pooled model learns one outcome across all diseases. **C**, An outcome-pooled model learns all outcomes within one disease. **D**, The joint model learns all outcomes in all diseases. Filled cells indicate supervision, not performance. The routed model uses the same panel as the joint model and changes only how predictions are read out. Every model predicts all six horizons.

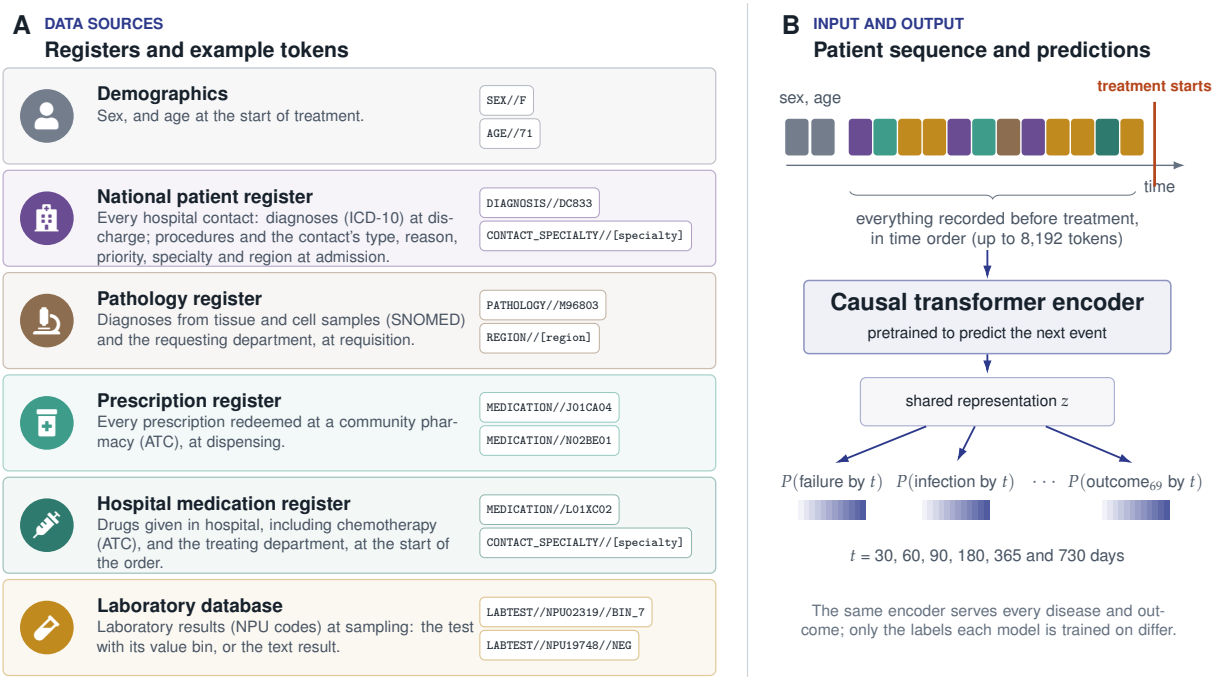


Figure S2. Data sources and input representation of the sequence model. A, the six sources linked through the civil registration number, what each contributes and when its events are dated, with two example tokens per source. Codes are illustrative, and laboratory values enter as quantile bins fitted on events before 2022. **B**, each patient's history becomes one token sequence, coloured by source, in time order and ending at the start of first-line treatment. The pretrained encoder reads it once into a shared representation, from which the model gives the probability of every outcome by each of six horizons.

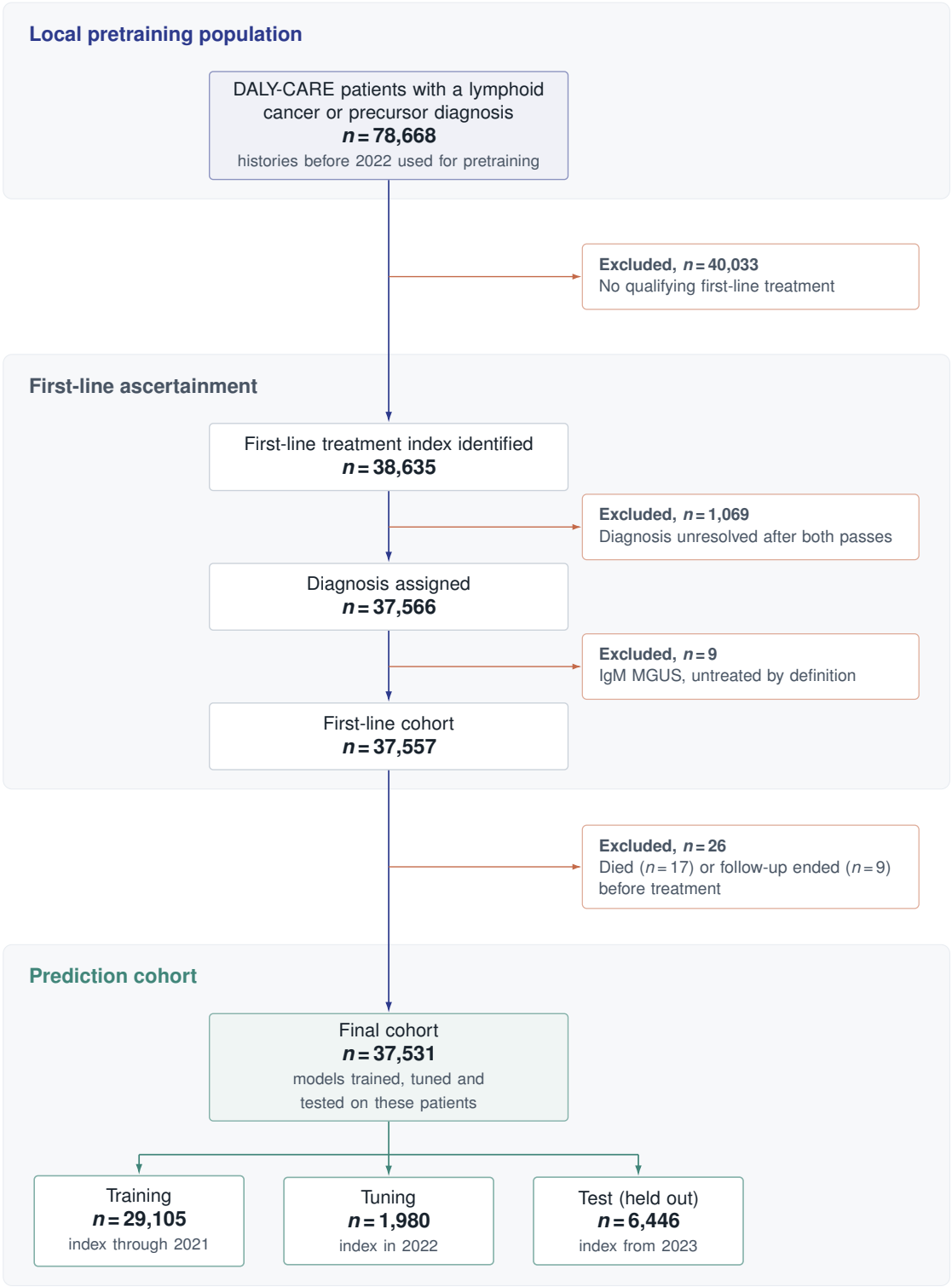


Figure S3. Cohort flow diagram. Of the local pretraining population of 78,668 adults in DALY-CARE with a lymphoid cancer or precursor (MGUS, MBL) diagnosis, whose pre-2022 histories were used for pretraining, 38,635 started first-line treatment. A diagnosis was assigned to 37,566 of them: to 37,357 from diagnoses recorded up to 14 days after treatment start, and to a further 209 on reassessment up to day 90 for late-coded diagnoses. The 37,531 with an assigned lymphoid disease who were alive and followed at treatment start form the final cohort, the patients whose outcomes were predicted: every prediction model was trained, tuned and tested on them, by treatment start in the training (to 2021), tuning (2022) or test (from 2023) period. Counts are unique patients. Not every patient contributes to every endpoint: outcome-specific eligibility, competing events and follow-up are applied within each split, and the resulting risk sets are given in the Supplementary Methods.

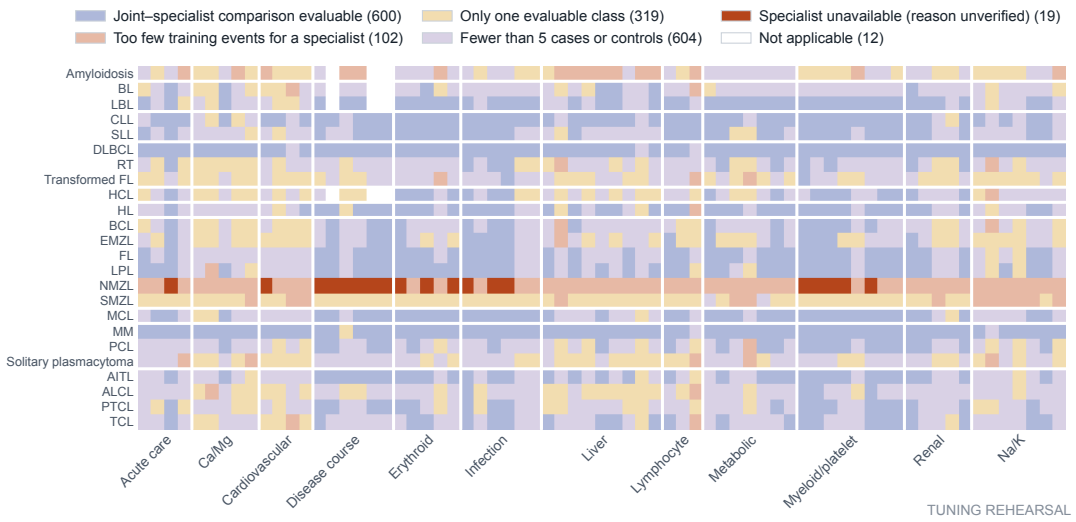


Figure S4. Estimability of disease–outcome cells. Status of every disease–outcome cell at one year: all models evaluated, too few training events to train a specialist, insufficient evaluation cases or controls, unavailable training support or specialist predictions, or not applicable. Missing specialist predictions are not labelled failed fits without a confirming run log. Counts in the legend.



Figure S5. Training events per disease–outcome cell. A, Training events by one year in every disease–outcome cell, including cells excluded from performance estimation (log scale, grey for zero, hatching for unavailable counts and white for not applicable). Crosses mark fewer than five evaluation cases or controls when those counts are exported. This criterion is separate from training support. **B**, Cumulative distribution of known counts, including zeros. Unavailable counts are reported separately. The descriptive training-support band edges are marked.

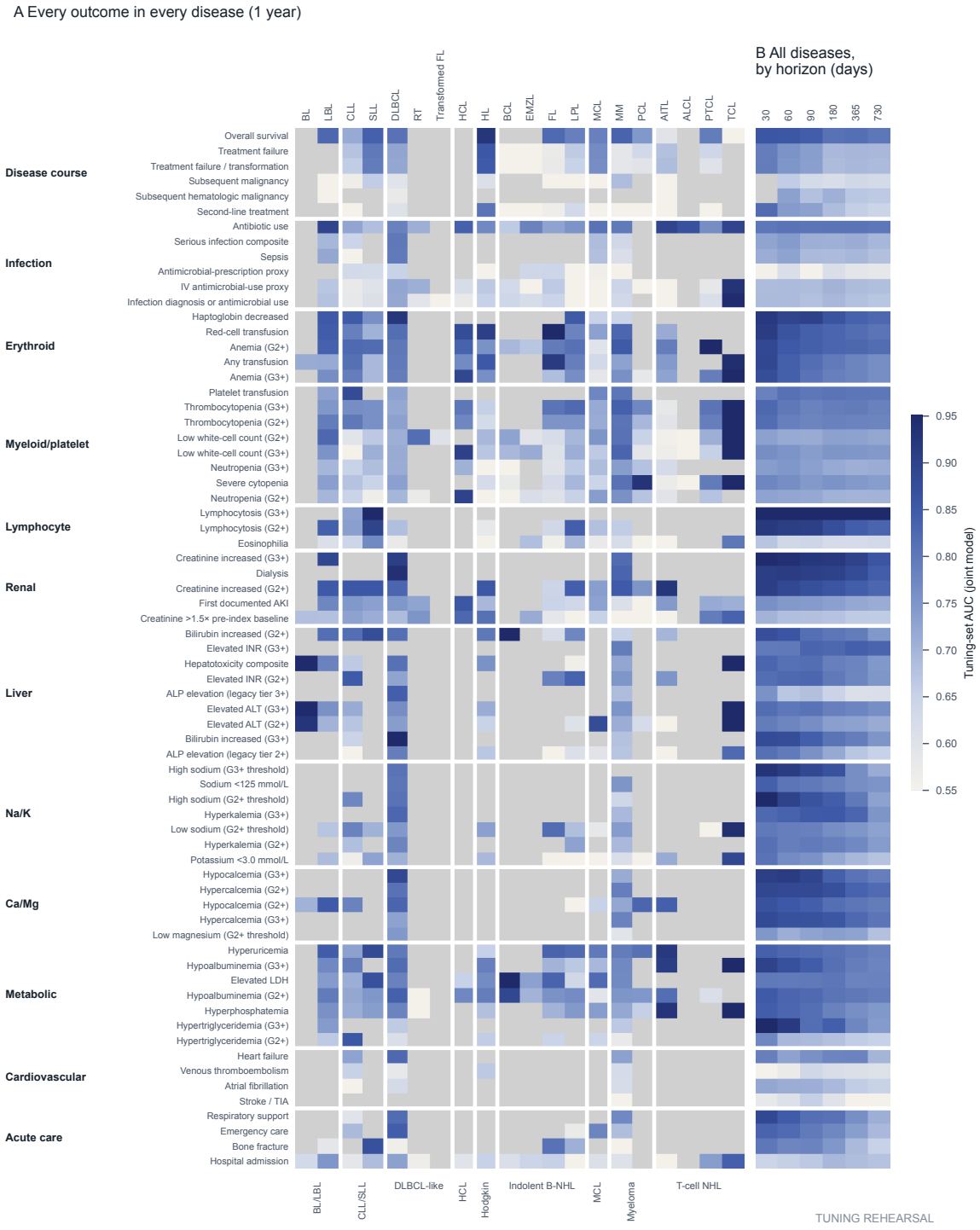


Figure S6. Discrimination of the joint model for each outcome in each disease. A, Test-period one-year AUC of the joint model for each outcome (rows, by clinical domain) in each disease (columns, by disease group). Grey cells had too few events to estimate. **B**, The same outcomes over all diseases at each horizon.

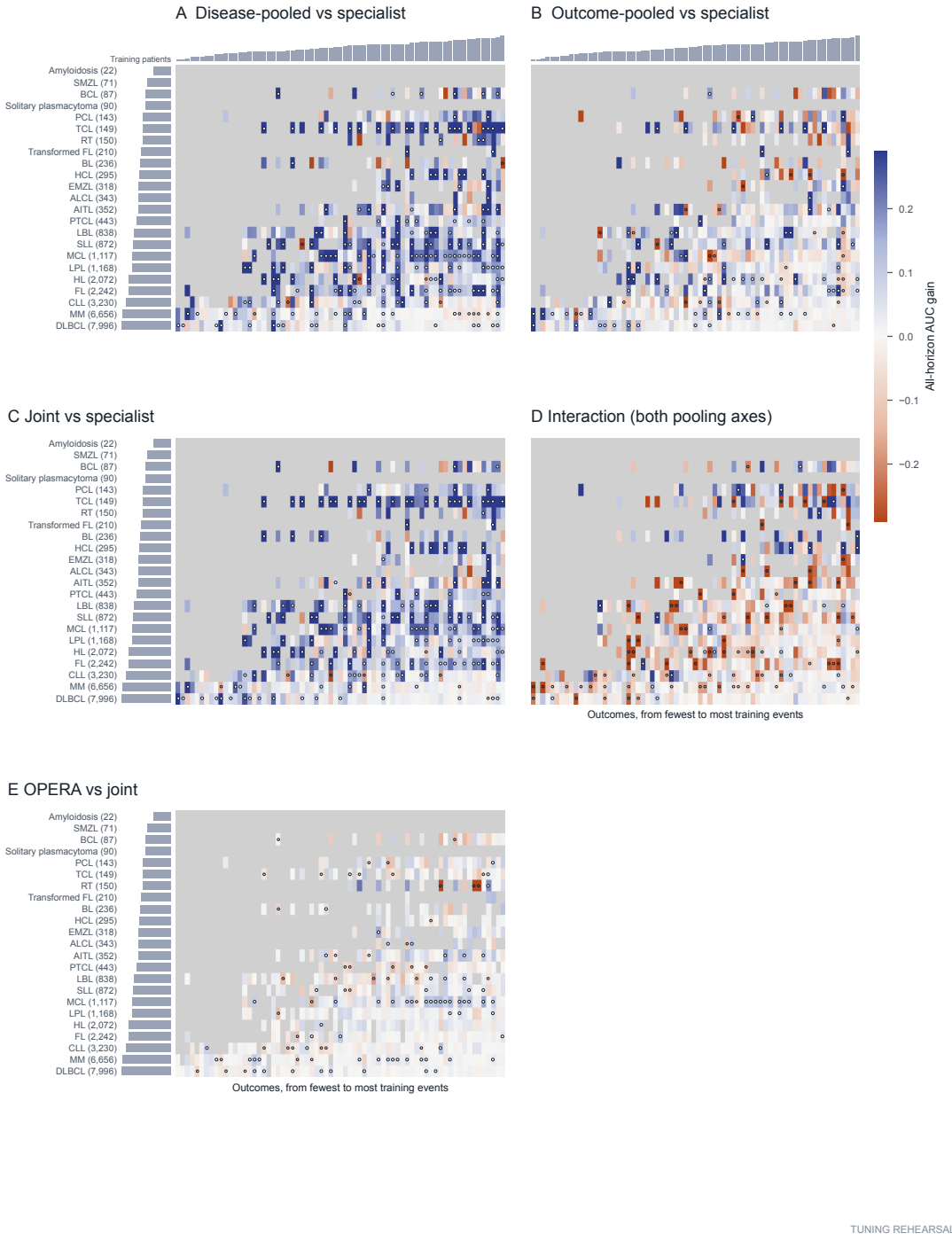


Figure S7. Pooling gains for each disease–outcome cell. Disease-pooled, outcome-pooled and joint gains over the specialist, the factorial interaction, and OPERA versus joint are shown separately. The last comparison adds routed corrections and is not a pooling-axis control. Diseases (rows) run from fewest to most training patients and outcomes (columns) from fewest to most training events, with log-scale bars showing both. Dots mark cells whose 95% interval excludes zero: white for a gain, an open ring for a loss. Most cells are small, so colours alone do not establish benefit or harm. Dots mark unadjusted pointwise intervals, not multiplicity-corrected discoveries. Grey cells could not be estimated.

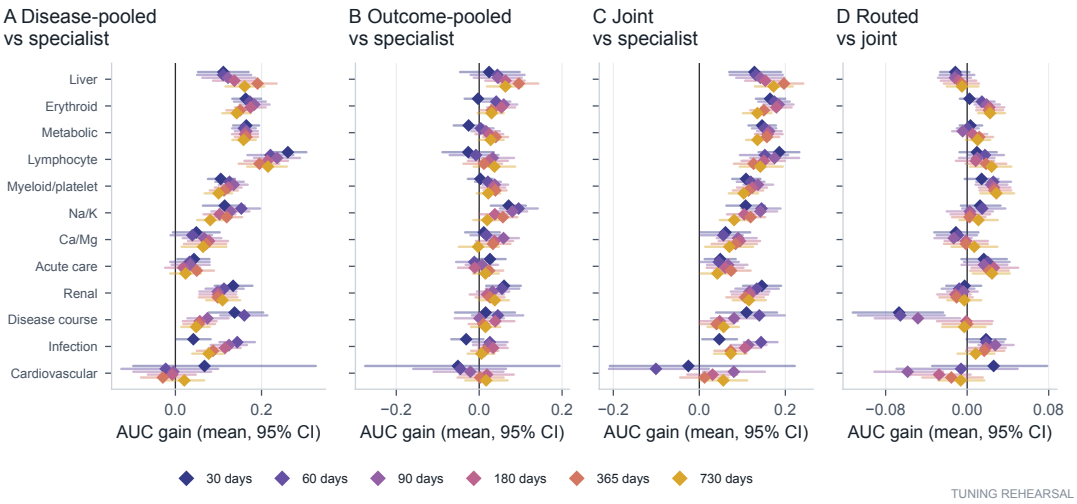


Figure S8. Pooling gains across horizons. Mean paired AUC gain over the specialist by clinical domain at each horizon for the disease-pooled, outcome-pooled and joint models (A–C). Panel D shows the routed model against the joint model. Bars show paired 95% confidence intervals from the saved domain summaries. Descriptive median/IQR fallbacks in design previews are labelled explicitly.

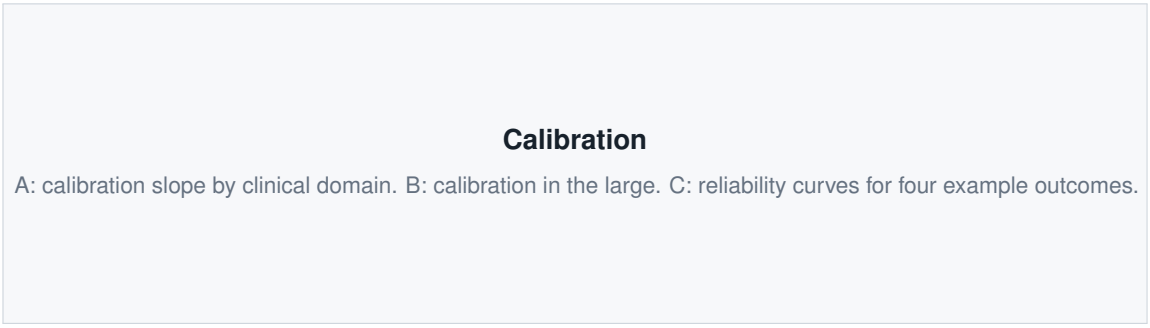


Figure S9. Calibration of the joint, routed, specialist and XGBoost models. A, Calibration slope at one year for the joint, routed, specialist and XGBoost models by clinical domain (median and interquartile range across cells, ideal 1; where an interval runs past the axis it is cut). **B**, Calibration in the large, observed minus mean predicted risk (ideal 0). **C**, Reliability curves for four example outcomes. Observed risks are weighted by the inverse probability of censoring and point size reflects the number of patients per bin; bins with fewer than 20 patients are not drawn. No model was recalibrated on test data.

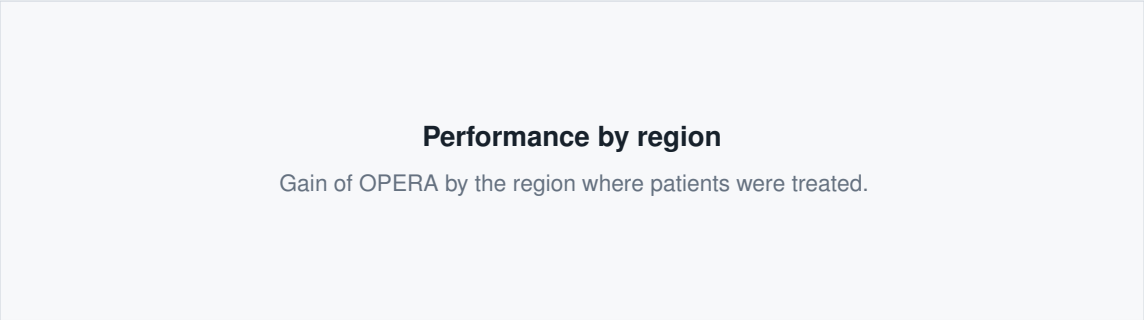


Figure S10. Performance by region. The models were trained on patients from all five regions, and test patients are grouped by the region of most of their pre-treatment contacts and every comparison is repeated within each region on the same fits. Maps show **A**, the one-year AUC gain of OPERA over per-task XGBoost, **B**, its one-year AUC, and **C**, its calibration in the large (observed minus predicted risk). Below, the same with 95% paired bootstrap intervals over equally weighted cells: **D**, gains over the specialist and XGBoost, **E**, AUC of OPERA and XGBoost, **F**, calibration in the large, and **G**, recorded events and laboratory results before treatment (median and interquartile range), alongside regional differences. Region boundaries from geoBoundaries (CC BY 4.0) [71], with Bornholm as an inset.

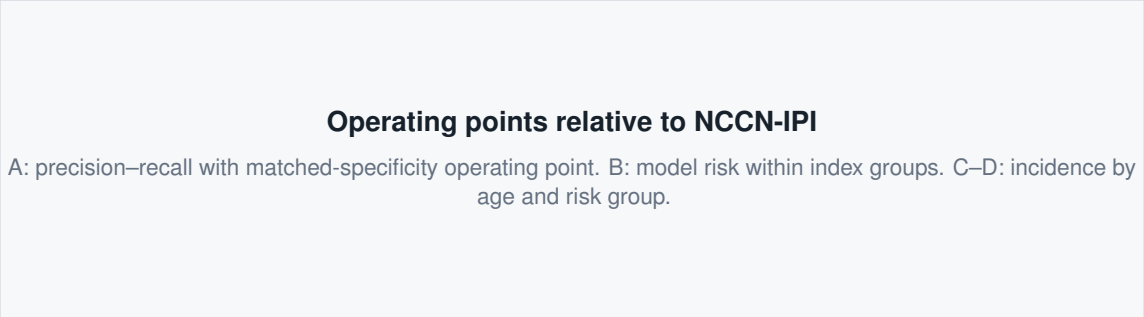


Figure S11. Operating points and reclassification relative to NCCN-IPI in DLBCL. A, Precision–recall curves for death within one year (patients with known status at one year) for the model and for each NCCN-IPI category threshold. The square marks the NCCN-IPI high-risk category. Dotted lines project it onto the model's curve: straight up gives the model's precision at the NCCN-IPI's recall, and straight right its recall at the NCCN-IPI's precision. The diamond marks the model at a threshold fixed on training data to target the index's highest-category specificity. AP, average precision. **B**, Predicted one-year risk of each patient within each NCCN-IPI category, coloured by whether the patient died, with the model risk groups shaded. **C–D**, One-year overall survival (95% bootstrap interval) and number of patients by age band and risk group for the model (**C**) and the NCCN-IPI (**D**). Counts below five are shown as <5, and estimates are suppressed for cells with fewer than ten patients.

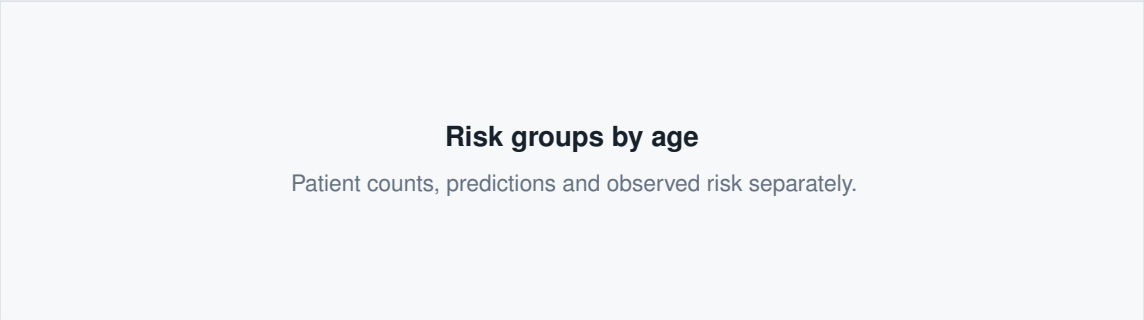


Figure S12. Model and NCCN-IPI risk groups by age in DLBCL. Patients with DLBCL and a recorded NCCN-IPI, grouped by model risk group (left) and NCCN-IPI category (right). Rows show, by age band, the number of patients, the mean predicted one-year overall survival and the observed Kaplan–Meier estimate, each with its 95% bootstrap interval. Estimates are suppressed for cells with fewer than ten patients, but counts remain visible.

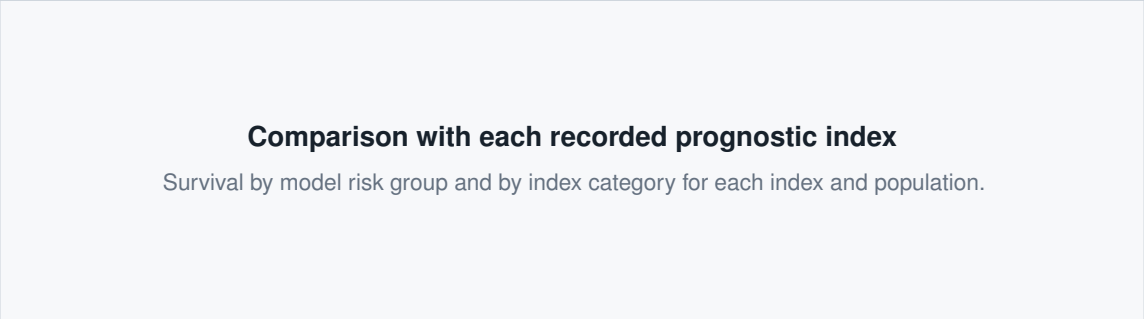


Figure S13. Model risk groups compared with each recorded prognostic index. One row per index and the population it applies to, on patients with that index recorded at diagnosis. Left, survival (or freedom from the endpoint) by model risk group. Middle, the same by index category, on the same patients. Right, estimates at one year with 95% CI (M, model, and I, index). Shaded bands are 95% bootstrap intervals and the dashed line marks one year. Model cutoffs were fixed on training data for each index, matching its low-risk observed risk and high-risk specificity. Every index recorded for at least 50 patients of a subtype is shown, and the NCCN-IPI in DLBCL. Group sizes, observed risk, sensitivity and specificity of both systems are in Supplementary Table [S]. **[Endpoint per index.]**

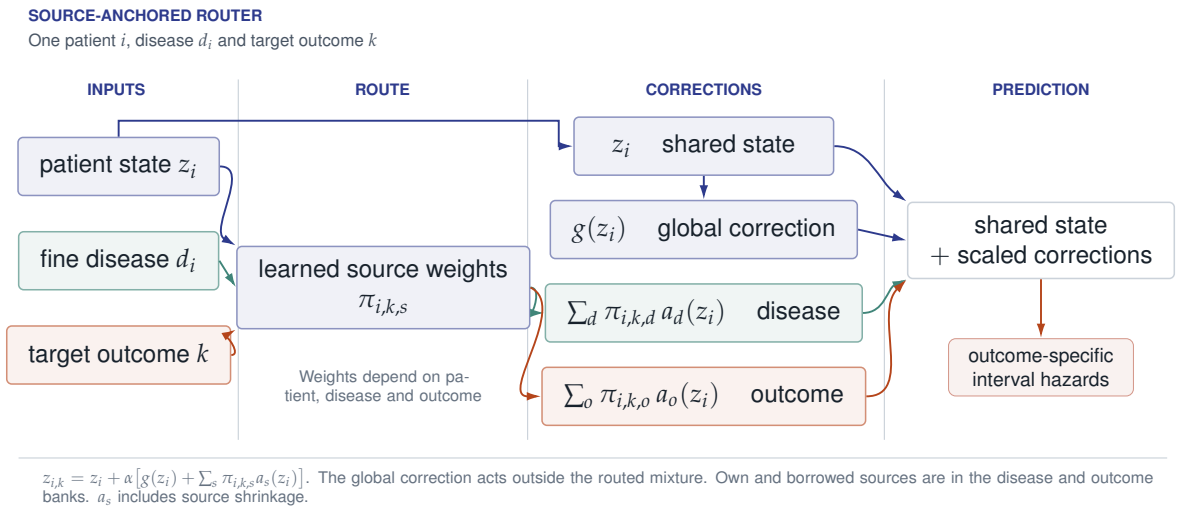


Figure S14. Source-anchored routing. For each patient and target outcome, the router mixes a shared representation with residual adapters, one per disease and, in the design screened on tuning data, one per outcome; the locked routed model uses disease adapters only. Its weights depend on the patient, disease and target. Sources with no training support are masked and low-event sources are shrunk. An adapter is updated by its own eligible supervision, while borrowed adapter parameters are detached. The mixed representation enters a shared linear head for interval hazards and cumulative incidence. This is a model schematic, not an empirical route-weight or performance plot.

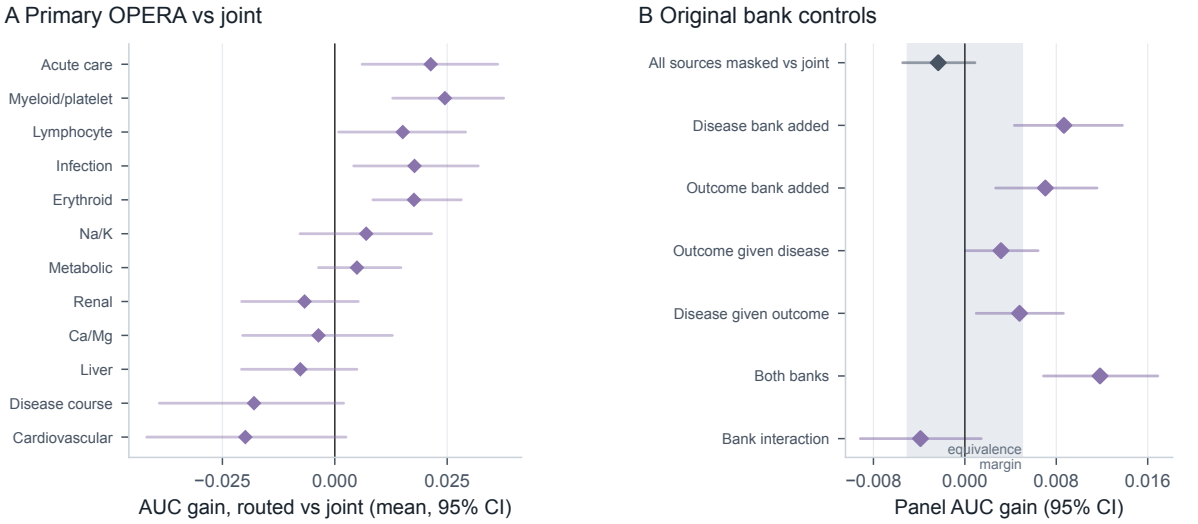


Figure S15. Routed versus joint model, and adapter-bank comparisons. A, Paired one-year AUC gain of the routed over the joint model by clinical domain, with the prespecified margin of practical equivalence shaded. **B**, Test-set panel gains of the four source banks, trained with the locked rank and query under the same protocol: the router trained with every source masked against the joint model (the noise floor of refitting, and a check that the architecture alone changes nothing), disease adapters, outcome adapters, each added to the other, both, and their interaction; and the controls (the joint model told the disease, the own-disease and own-outcome routers). The primary amended router uses both disease and outcome adapter banks. The earlier router is a sensitivity analysis.

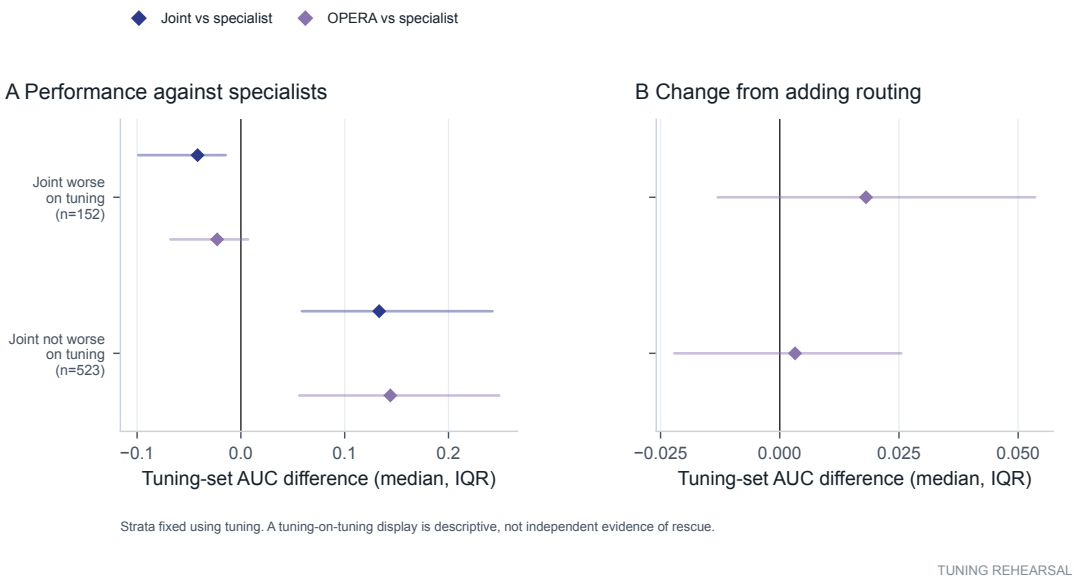


Figure S16. Routing in cells where joint learning reduced performance on the tuning set. Cells were classified on the tuning set by whether the joint model was worse than the specialist. **A**, Evaluation-period gains of the joint and routed models over the specialist in each stratum. **B**, Routed-versus-joint differences within those same strata. Diamonds and bars show descriptive medians and interquartile ranges. Tuning-on-tuning previews do not provide independent evidence of recovering harmed tasks.

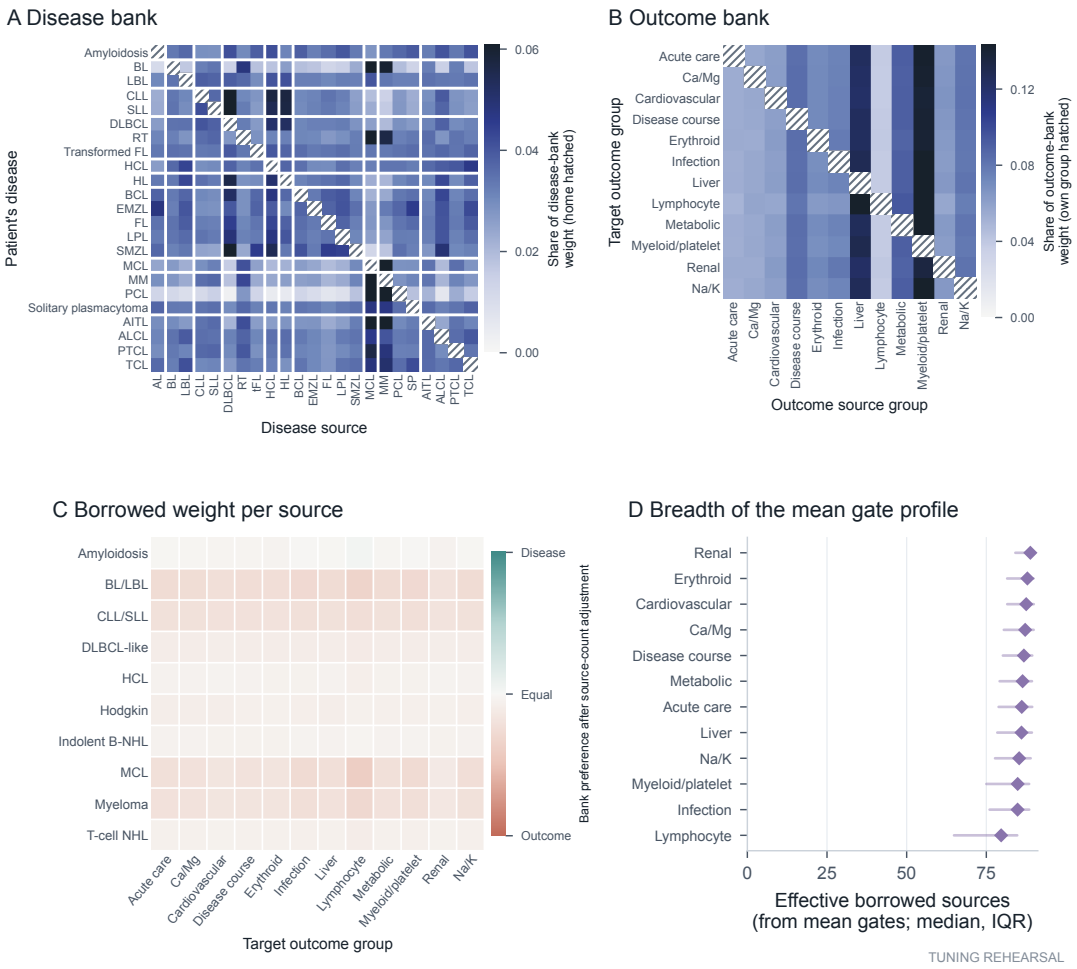


Figure S17. Router weights by disease and clinical domain. **A**, Disease-bank shares, normalized within each disease–outcome task before averaging (own disease hatched). **B**, Outcome-bank shares aggregated by clinical domain (own domain hatched). Sources without an estimable target row remain in the source bank. **C**, Mean gate weight assigned to other diseases, other outcomes and the remaining own/shared sources, by patient disease group. These shares sum to one and describe allocation, not correction magnitude. **D**, Effective borrowed-source count (exponential entropy) computed from mean gates, excluding the shared slot and own-disease/own-outcome slots; points and bars show the median and interquartile range across disease summaries. This is not the mean of patient-level effective source counts. The base encoder path remains present independently of the gate. Weights describe allocation, not predictive benefit (Fig. S37).

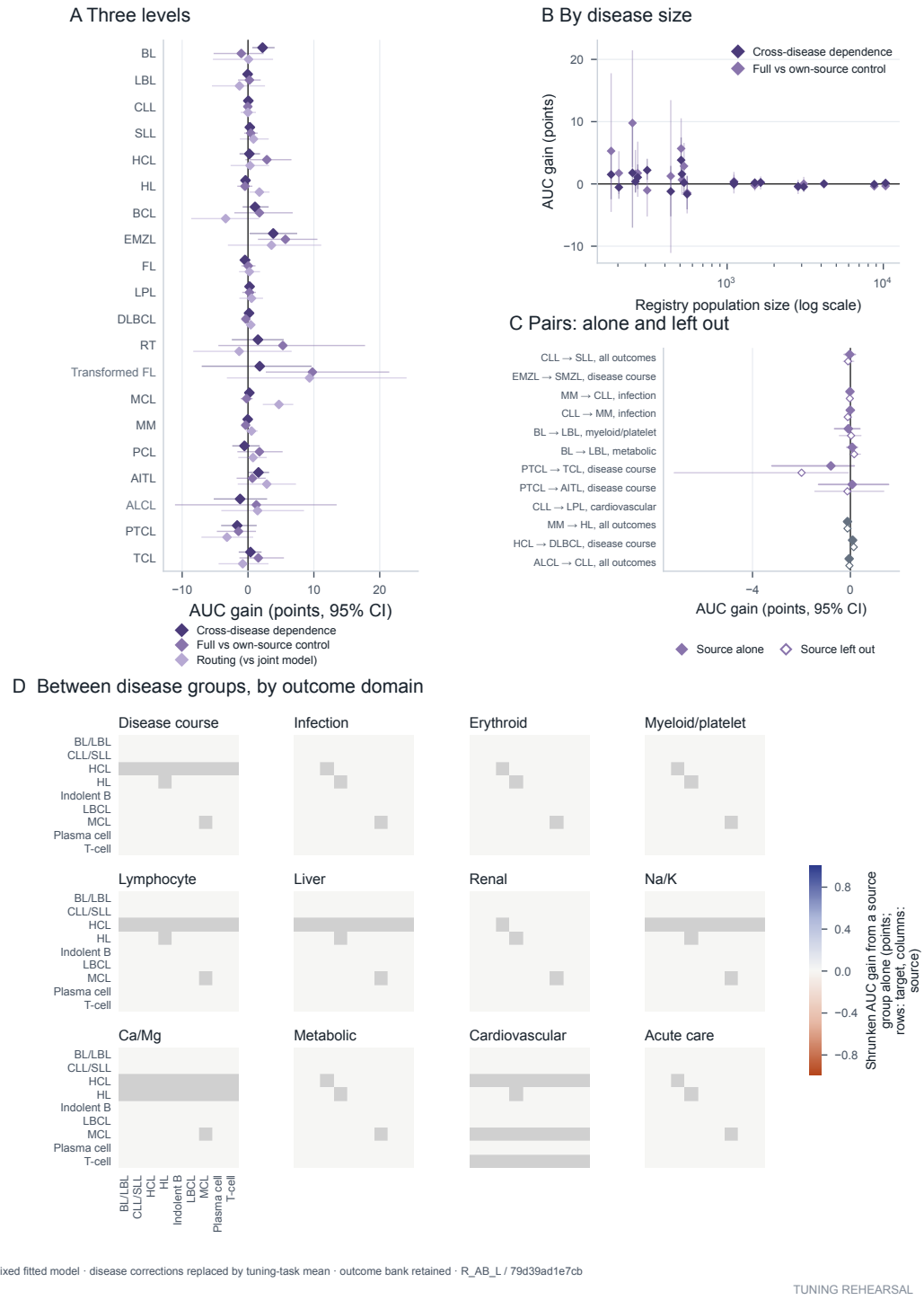


Figure S18. Borrowing in detail. A, For each target disease, over all outcomes and horizons: cross-disease source dependence (other-disease terms retained versus replaced by their tuning means, with the outcome bank retained, as in Fig. S37B), the gain over the own-source control (a separately trained router) and routing (against the joint model). The last two compare separately trained models and carry the noise of refitting; cross-disease dependence can exceed the trained gain because the encoder was trained alongside the borrowed corrections. **B**, Cross-disease dependence and the gain over the own-source control against the number of patients with the disease. **C**, The pairs of the draft candidate-pair plan, shown descriptively, with the source alone (filled) and the loss when only that source is left out of the full model (open). **D**, Mean unshrunk gain from a source disease alone between disease groups (rows: target, columns: source) in each of the 12 clinical domains, on one scale; diseases with few estimable outcomes are omitted, and the groups summarise for reading only (they do not enter the pooling). AUC points with paired patient-bootstrap 95% intervals; IPA differences are in **[Supplementary Table]**.

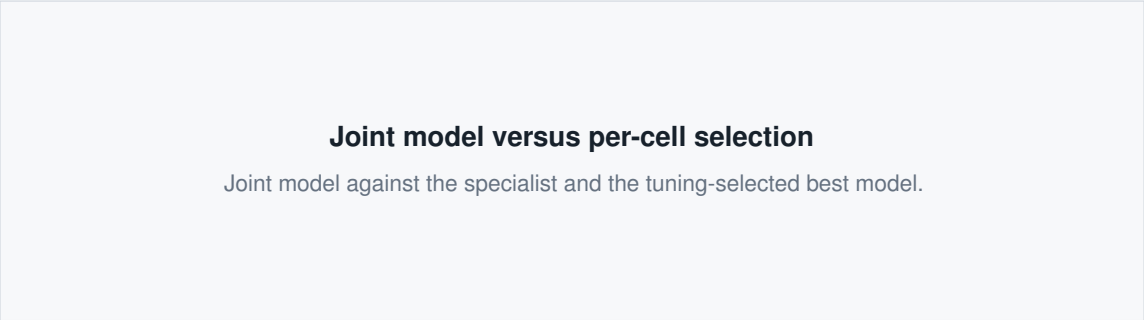


Figure S19. Joint model versus tuning-based selection for each cell. A, Test AUC of the joint model against the specialist in each cell, with point size reflecting training events. **B**, Against the best of the specialist, disease-pooled and outcome-pooled models chosen per cell on the tuning set using mean horizon log loss. **C**, Which model was chosen by clinical domain in an expanded candidate set that also includes the joint model and OPERA. This expanded selection does not change the original specialist/disease-pooled/outcome-pooled policy compared in B.

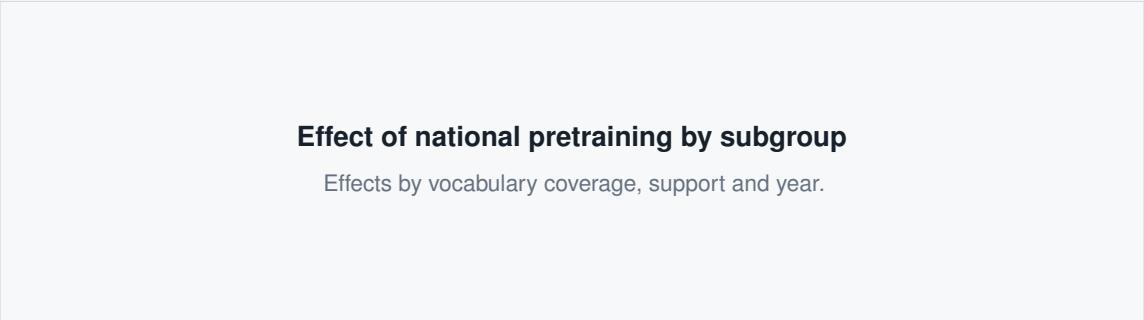


Figure S20. Effect of national pretraining by vocabulary coverage, training support and year of treatment. Patient-weighted improvement in one-year Brier score ($\times 100$, 95% CI) of the joint model initialised from the national encoder, used directly or after local adaptation, over the locally pretrained joint model, by the share of a patient's history covered by the national vocabulary, by the number of training events of the predicted combination, and by year of treatment.

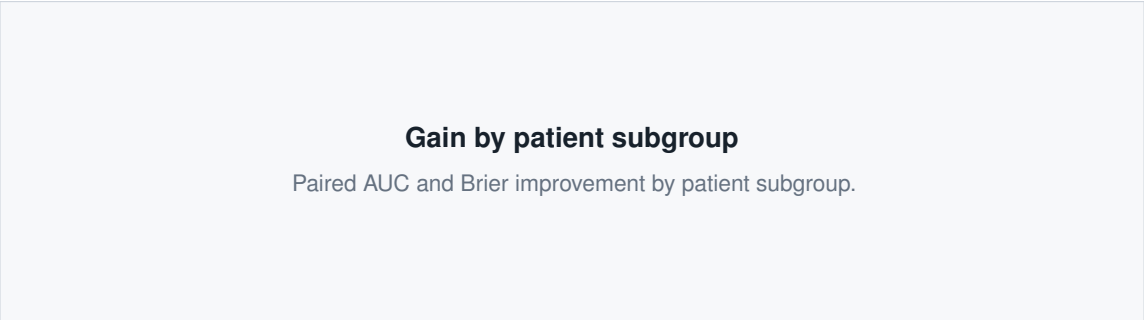


Figure S21. Gain by patient subgroup. (A) AUC improvement recomputed within each subgroup and averaged equally over estimable disease–outcome–horizon cells with finite estimates in every bootstrap replicate. (B) Patient-weighted Brier improvement pooled across cells and horizons. Differences and 95% paired patient-bootstrap intervals are multiplied by 100; positive values favour the first model in each comparison. Groups are defined from pre-treatment characteristics; the two panels use different weighting rules and can have different estimable cell inventories. Missing estimates are left blank.

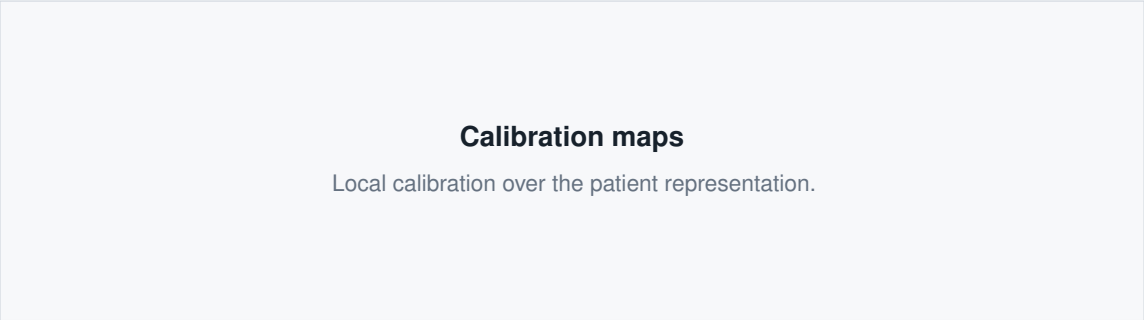


Figure S22. Local calibration across the patient representation. Two-dimensional projection of test patients' representations for example outcomes. Rows show event status, projected routed corrections, IPCW kernel estimates of observed risk, patient-level final-model predicted risk and local calibration residuals (observed minus predicted). Kernel residual estimates are sampled at patient positions for the residual colours, rather than displaying each patient's binary event minus predicted risk. Arrow magnification is shared across outcomes and stated in the key.

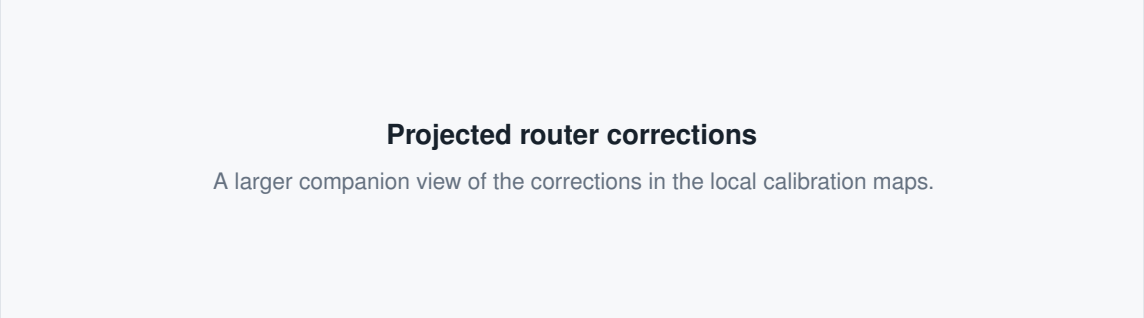


Figure S23. Projected router corrections by outcome. The same patient coordinates and routed corrections as in Fig. S22, with one shared arrow magnification. Arrows show mean local displacement in supported bins that meet the stated coherence rule. They describe changes in the model representation, not causal changes in patients. An absent vector export is labelled explicitly.

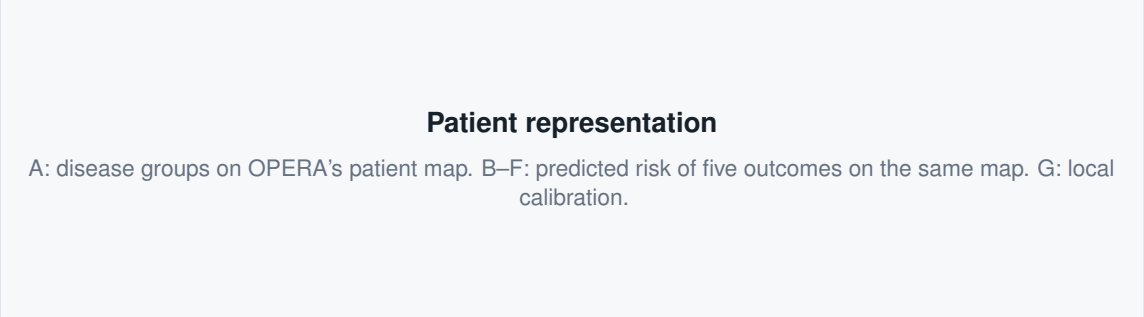


Figure S24. Predicted outcome risk across OPERA's patient representation. A, Test patients on a two-dimensional projection of OPERA's patient representation (fitted on tuning patients), coloured by disease group, with numbered group medians linked to the legend. **B–F**, Patient-level predicted one-year risk of five outcomes from different clinical domains, on a common probability scale. **G**, IPCW kernel estimates of observed minus predicted risk of OPERA for overall survival when available, sampled at patient positions. This is a smoothed local calibration estimate, not an individual residual. All 69 outcomes are shown in Fig. S26–Fig. S31. The map is descriptive and carries no performance claim.

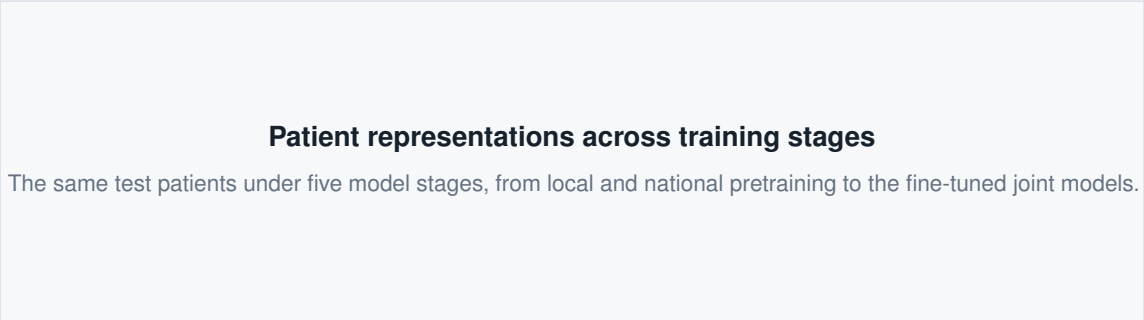


Figure S25. Patient representations at each training stage. The same test patients under five model stages: pretraining on the cohort, the joint model from that encoder, national pretraining, national pretraining adapted to the cohort, and the nationally initialised joint model. The first row colours patients by disease group, with density contours. Other rows colour individual patients by age, recorded events before treatment (log colour scale) and native IPI category. Missing categories remain grey. Each column's projection was fitted on tuning patients and applied unchanged to test patients, so coordinates are comparable within a column but not between columns. The maps are descriptive and carry no performance claim.

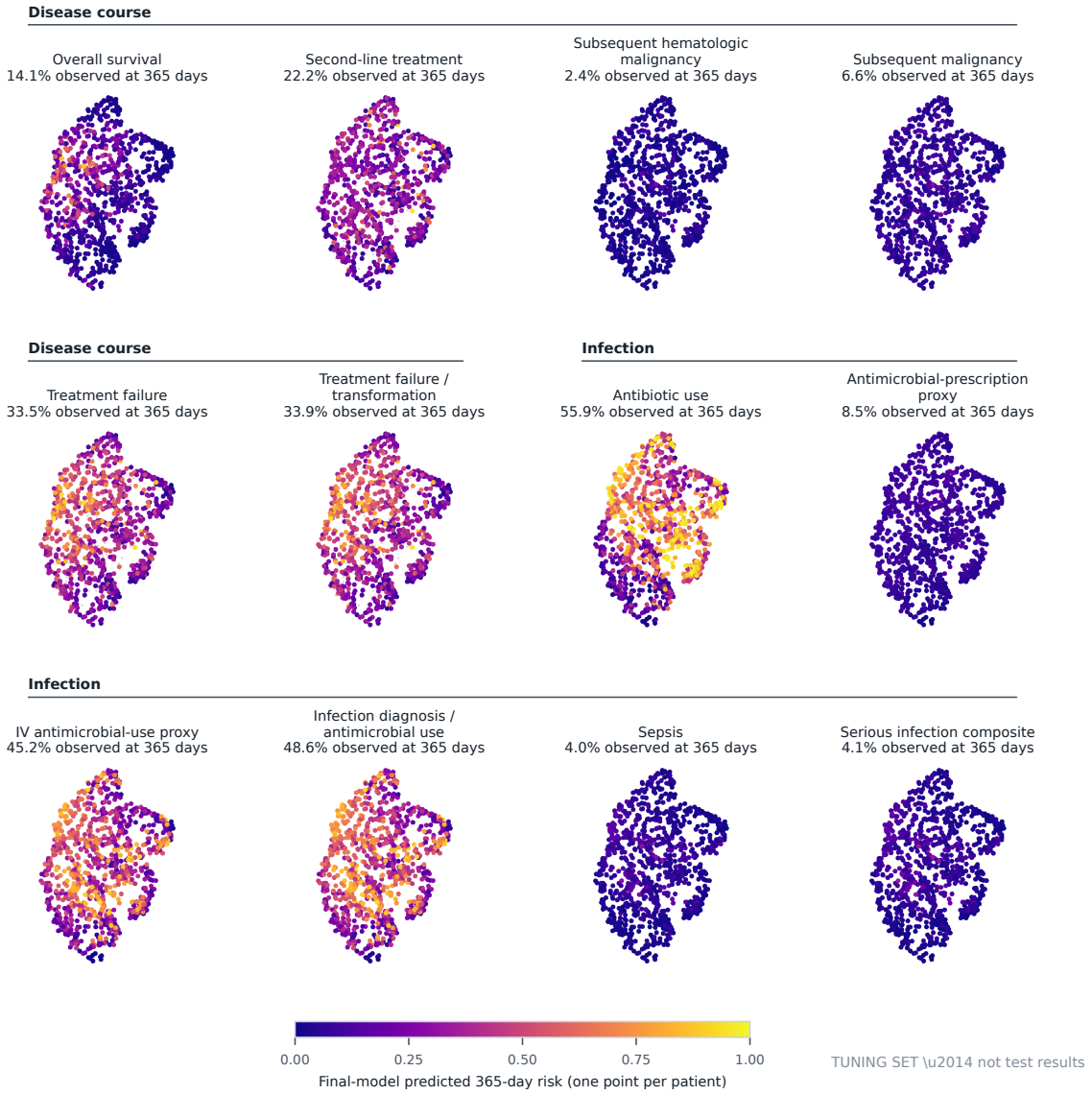


Figure S26. Predicted one-year risk of each outcome across the patient representation (1 of 6). Every panel shows the same two-dimensional projection of OPERA's patient embedding (fitted on tuning patients) for test patients. Colour is the predicted one-year risk for each patient. Observed-risk surfaces, when supplied as a separate sensitivity view, are labelled as IPCW kernel estimates. Outcomes are grouped by clinical domain, named above each group.

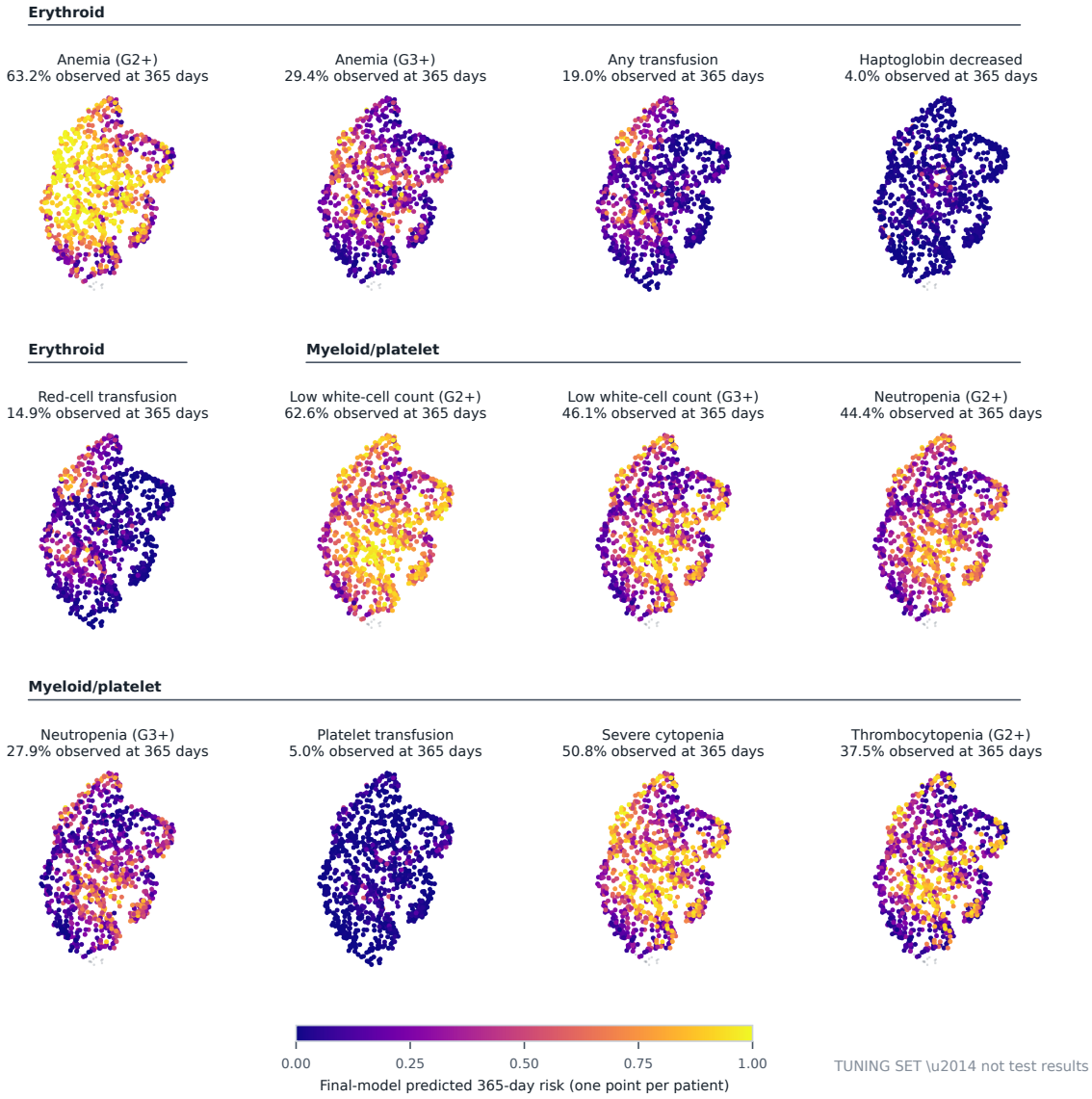


Figure S27. Predicted one-year risk of each outcome across the patient representation (2 of 6). As Fig. S26, continued.

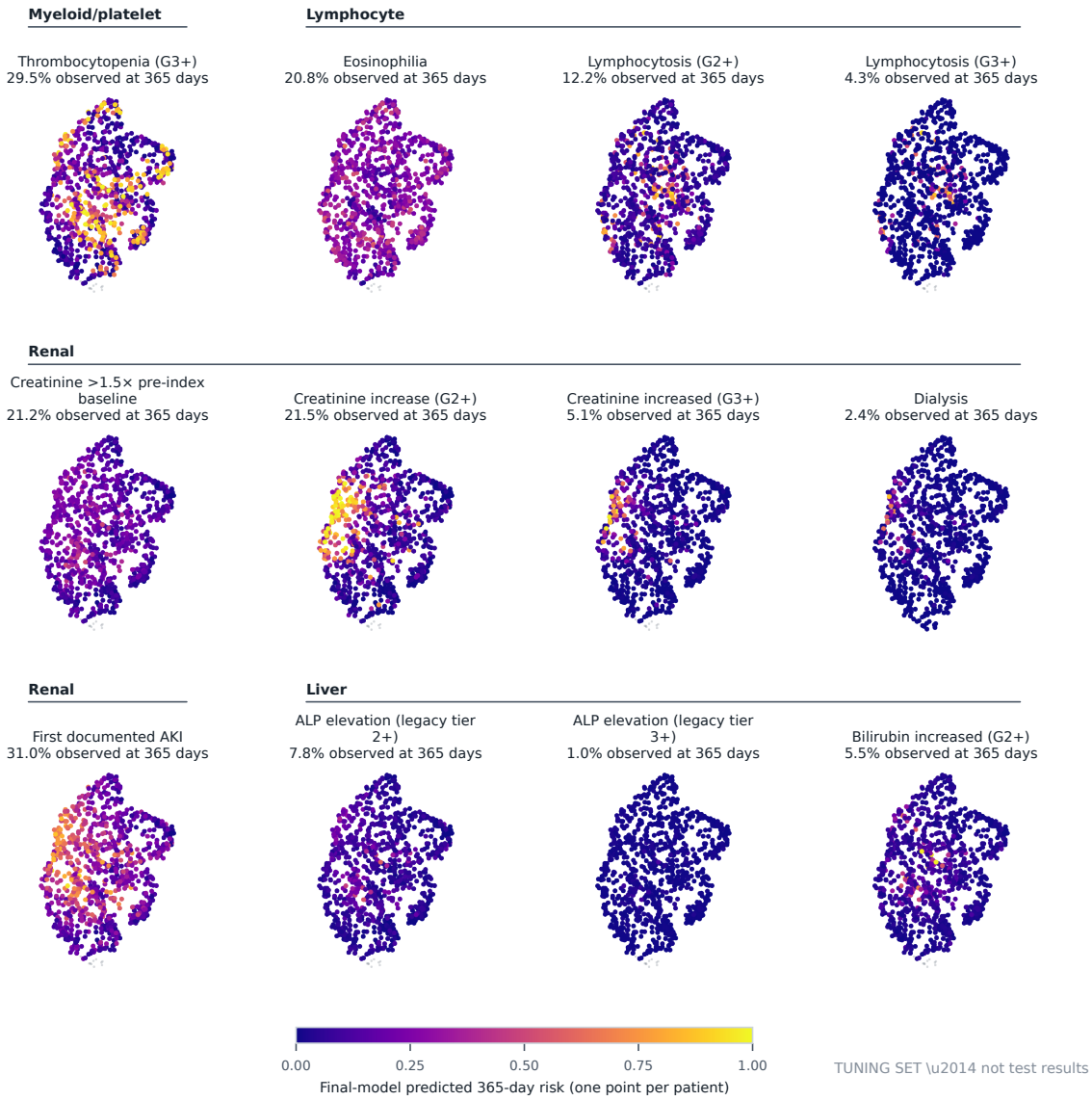


Figure S28. Predicted one-year risk of each outcome across the patient representation (3 of 6). As Fig. S26, continued.

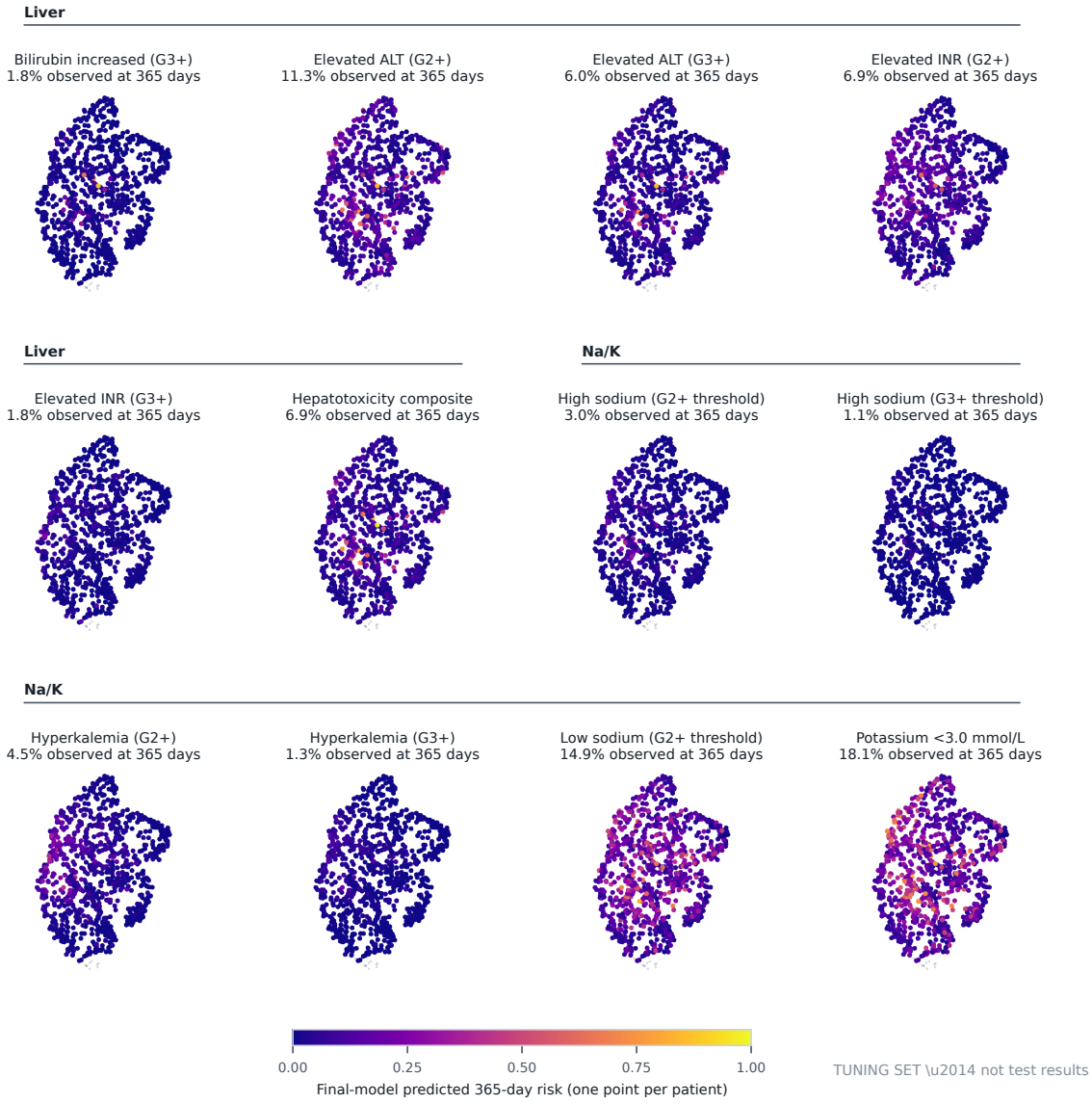


Figure S29. Predicted one-year risk of each outcome across the patient representation (4 of 6). As Fig. S26, continued.

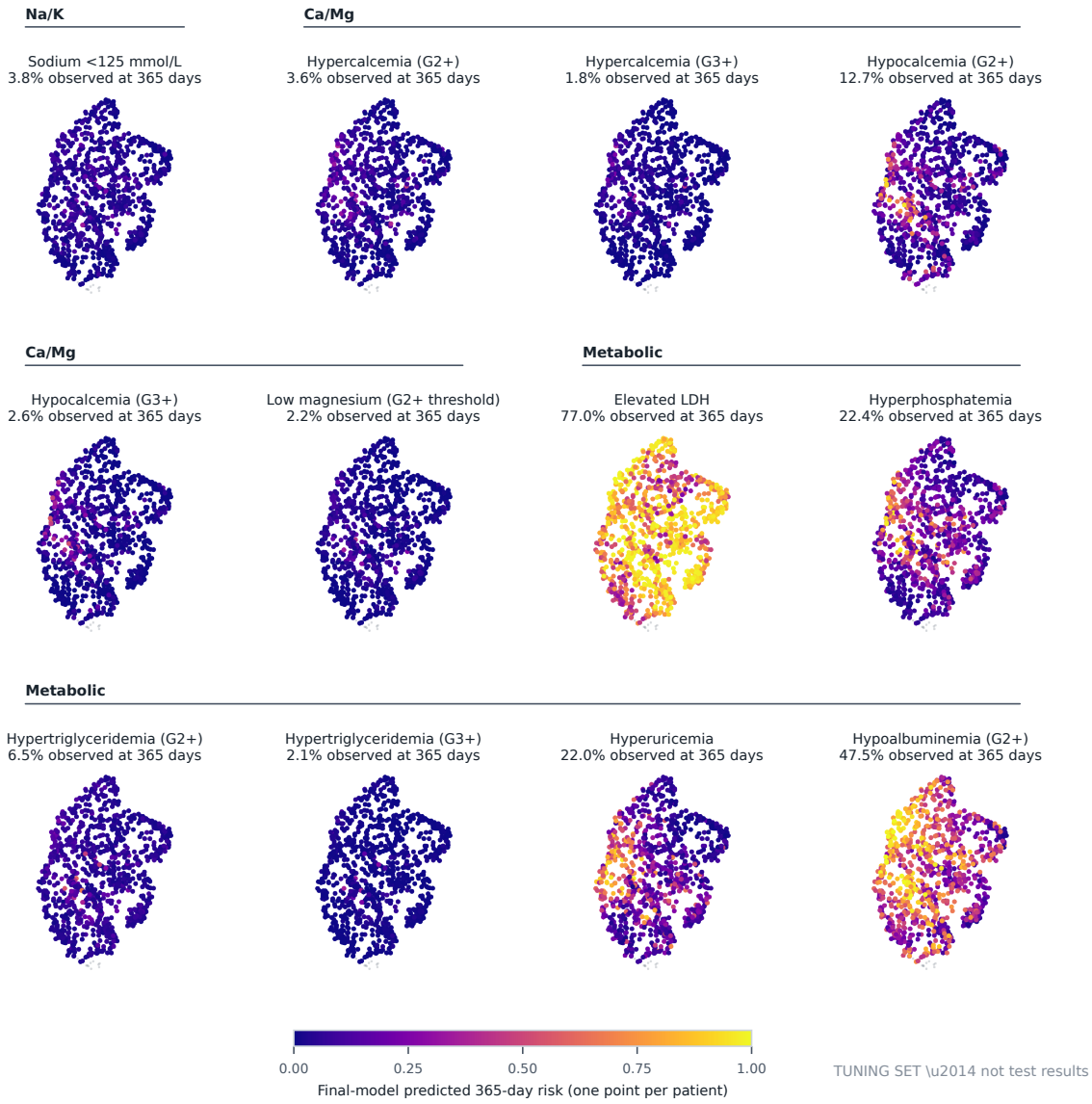


Figure S30. Predicted one-year risk of each outcome across the patient representation (5 of 6). As Fig. S26, continued.



Figure S31. Predicted one-year risk of each outcome across the patient representation (6 of 6). As Fig. S26, continued.

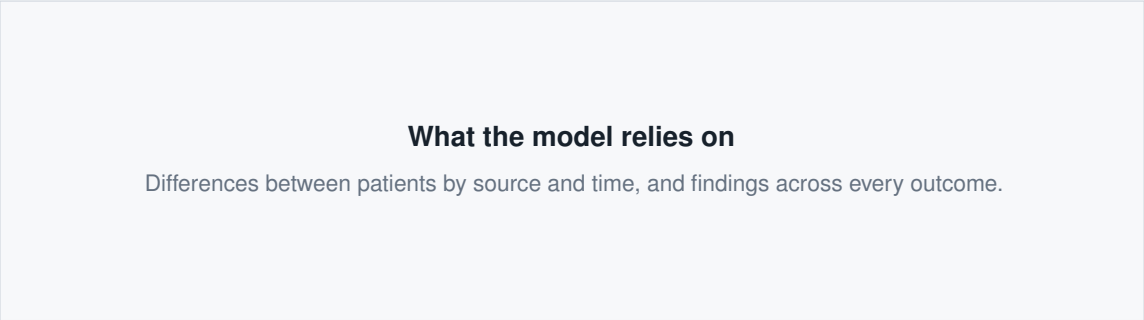


Figure S32. What the model relies on, for every outcome. Rows are outcomes grouped by clinical domain. Integrated gradients split each held-out patient's predicted log-odds at one year into a reference, which keeps the timing of care, age and subtype but replaces every recorded event by an average event, and the contribution of each event. **A**, The variance of the predictions between patients, divided into the part carried by the timing of care alone (grey) and the parts carried by what was recorded, by source. Shares are signed and add up to 100% per outcome. **B**, The same division by time before treatment. **C**, Mean change in predicted log-odds among the patients who have each finding, shown as the corresponding odds ratio. Boxed cells are the expected directions documented in the analysis plan, dots mark 95% patient-bootstrap intervals that exclude zero, and grey cells had fewer than 20 patients with the finding. Proposed mappings and concepts added during tuning are exploratory and separately flagged. No unmatched concept is counted as a contradicted expectation.

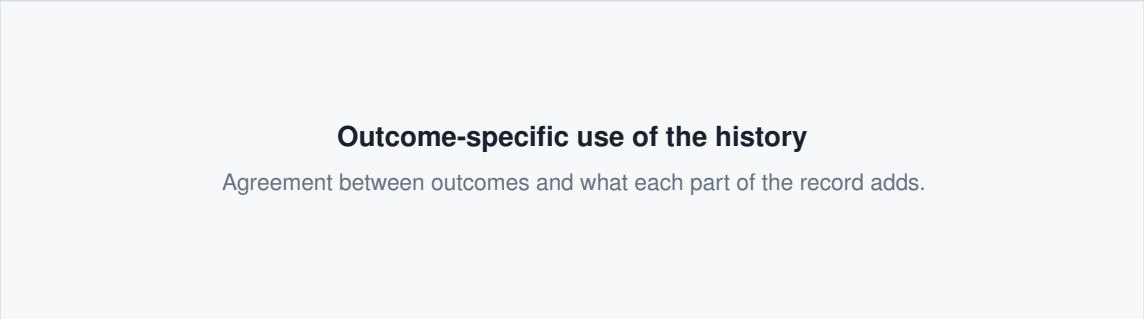


Figure S33. The model uses the same history differently for different outcomes. A, For every pair of outcomes, how much they agree on which episodes of a patient's history raise or lower risk (Spearman correlation of episode contributions within patients, averaged over patients). Episodes are events within one day of each other, and each clinical domain is outlined. **B**, Change in held-out AUC ($\times 100$) when the same fitted model reads only part of the record, averaged over all outcomes and over the outcomes of each domain. The model is not retrained, so this measures what it relies on, not what a model trained without that input could achieve.



Figure S34. One synthetic patient's history read for six complications. Each track is one outcome from a different clinical domain. Circles are the episodes of care (events within one day of each other) that move its predicted risk, at their time before treatment, sized by the change in predicted log-odds and coloured by direction (sienna raises, indigo lowers). The episodes that matter most for any outcome are labelled with their main findings and timing. The predicted one-year risk is given at the right, with the risk for the same timing of care and average findings. For every patient, the episodes that most raise and lower each risk can be extracted in the same way and shown beside the prediction. The patient and its contributions are invented, and records of real patients stay in the secure environment.

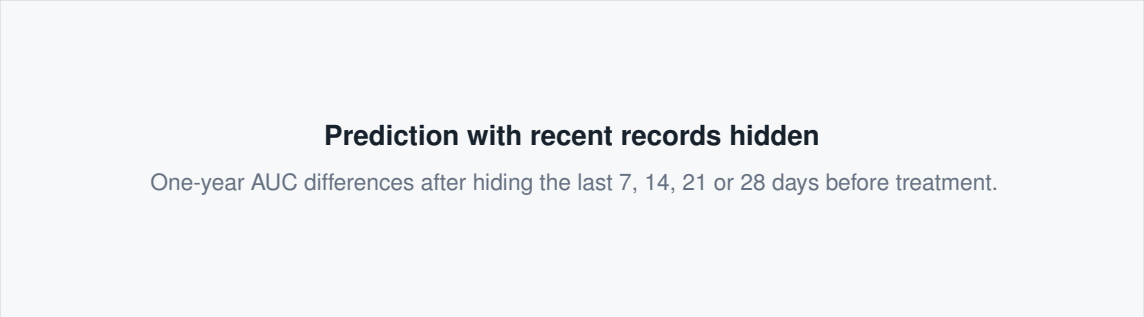


Figure S35. Dependence on recent records. The selected checkpoint was evaluated with the last 7, 14, 21 or 28 days of clinical events removed. Earlier history and prediction/background tokens were retained. Differences are restricted-input minus full-input AUC at one year, averaged across evaluable outcomes overall and within clinical domains, with paired patient-bootstrap 95% intervals. The model was not retrained.

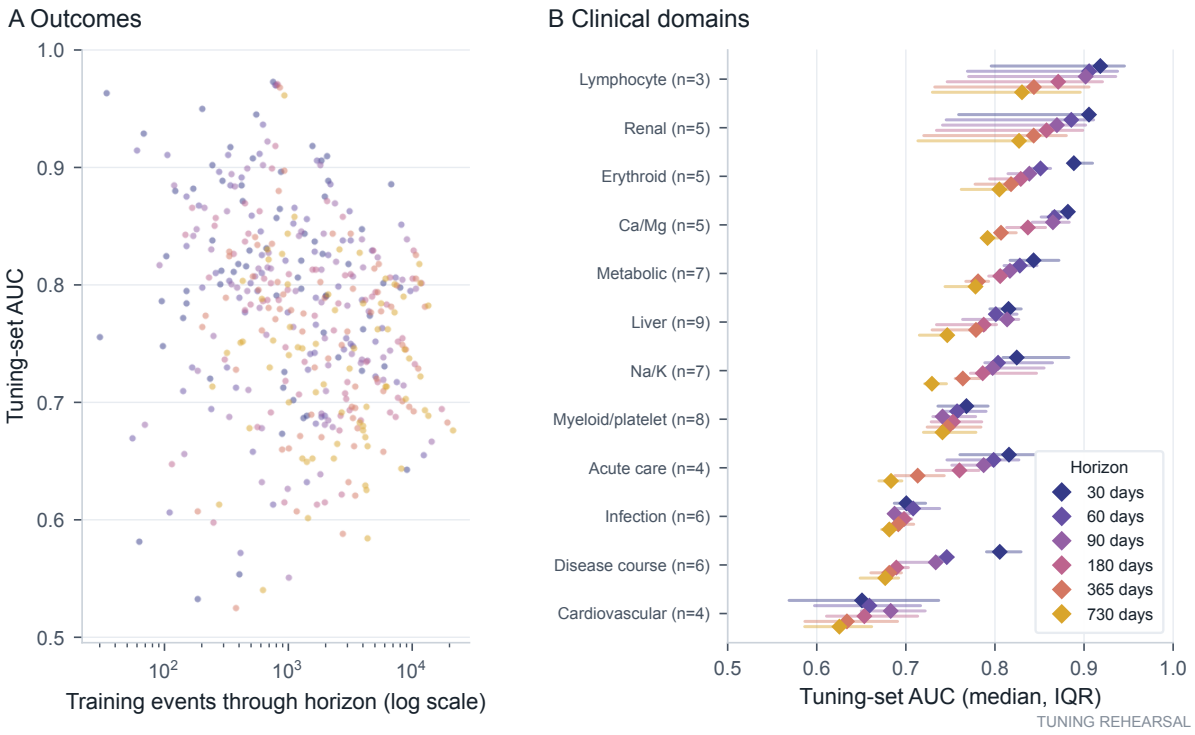


Figure S36. Discrimination of the joint model across outcomes and horizons. **A**, Test-period AUC for each of the 69 outcomes at each horizon (colour), against its training events. **B**, The same results summarised by clinical domain. Outcomes or horizons without enough test events are not shown, and every outcome in every disease is in Fig. S6.

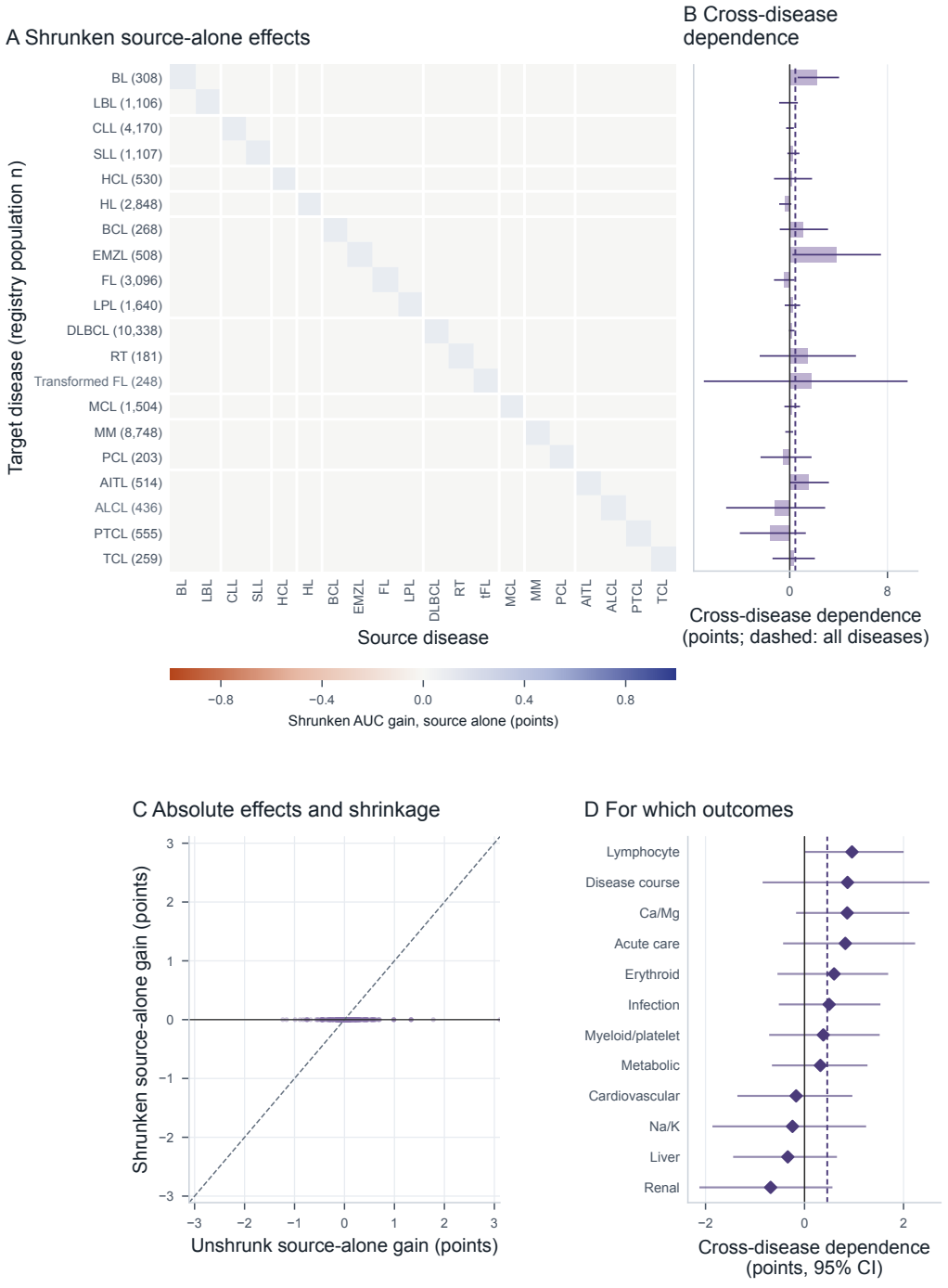


Figure S37. What the fitted model borrows from other diseases. Other-disease adapter terms are replaced by their tuning means without retraining or renormalizing the remaining gates; own-disease terms, the outcome bank, global corrections and the base encoder path are retained. Comparisons use the same evaluated patients (AUC points). **A**, Unshrunk gains when one source disease is retained alongside the target's own disease. Own-source cells are grey. Colour alone does not establish an effect for an individual pair. Lineage groups organise the display. **B**, Unshrunk cross-disease dependence: full-model AUC minus AUC with other-disease terms replaced; paired patient-bootstrap intervals and a dashed all-disease mean. **C**, Unshrunk versus shrunken source-alone estimates on an absolute AUC-point scale. The diagonal indicates no change under shrinkage. Near-zero shrunken effects are not normalised into lineage shares. **D**, Cross-disease dependence by clinical domain, with all diseases weighing equally; dashed line, all outcomes. Source-alone gains are not additive in AUC and need not sum to the total dependence. Further contrasts appear in Fig. S18.

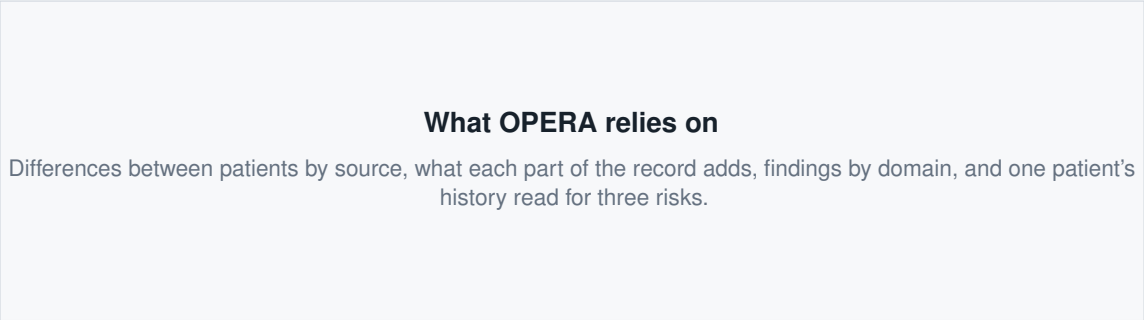


Figure S38. What OPERA relies on. Rows in A–C are clinical domains. Integrated gradients split each held-out patient's prediction into a reference, which keeps the timing of care, age and subtype but replaces every recorded event by an average one, and the contribution of each event. **A**, The signed covariance decomposition of predicted one-year log-odds between patients, including the share carried by the reference retaining timing, age and subtype (grey) and by each source of records, averaged over the outcomes of each domain. **B**, Change in held-out AUC ($\times 100$) when the fitted model reads only part of the record, without retraining: reference event content, no laboratory results, no medication, no diagnoses, no region or specialty of care, or only the last year of history. **C**, Mean change in predicted log-odds among the patients with each finding, averaged over the outcomes of each domain and shown as the corresponding odds ratio, with findings grouped by source. Boxed cells contain a prespecified expected effect, drawn solid where it was seen and dashed where it was not; grey cells had too few patients with the finding to report. **D**, One synthetic patient's history read for three risks, with the episodes that move them most labelled above the tracks. Each track is one outcome, and each circle is an episode of care at its time before treatment, sized by how much it changes the predicted one-year risk (percentage points) and coloured by direction. The predicted risk is given at the right, with the risk for the same timing of care and average findings. Sienna and indigo mark effects throughout (raising risk or losing AUC, and lowering risk), and the other colours identify sources of records. Every outcome is shown in Figs. S32 and S33, and the patient's history read for six complications in Fig. S34.

Supplementary Methods: cohort, prediction time and outcome definitions

This supplement describes how the study cohort, the index date (start of first-line treatment), the temporal split and the 69 outcomes, from disease-course endpoints to treatment-related and acute-care outcomes, were defined in the Danish Lymphoid Cancer Research (DALY-CARE) data resource [1]. Reported cohort counts describe the analysis release (Supplementary Table D2). The R and Python code that implements every step is released with the paper.

Conventions. Dates are calendar dates without time of day unless stated otherwise. The *index date* is the start of the reconstructed first-line treatment episode (Section D2). *End of follow-up* is the earlier of death and last follow-up as recorded in the DALY-CARE patient table. The *administrative censoring date* is the earlier of end of follow-up and the data cutoff of 19 March 2026. Unless stated otherwise, an outcome event is the *first* qualifying date d with index date $< d \leq$ censoring date; events on the index date itself were never counted. For each outcome, patients for whom the outcome could not be ascertained were excluded from that outcome's risk set (masked supervision); exclusion was never interpreted as absence of the event.

How to read this supplement. Sections D1–D4 describe the data and who enters the study, including the reconstructed treatment index and disease assignment. Section D5 explains disease-course outcomes, particularly second-line treatment. Section D6 gives the clinical outcome dictionary; Section D8 explains who can contribute a label and when observation ends. The final sections distinguish panel selection and reporting groups from the clinical definitions and state their limitations.

Three questions for every outcome
What counts? The qualifying event and its threshold or code rule.
Who is eligible? Required baseline information and source coverage.
How long is it observable? Ascertainment, censoring and competing death.
An unavailable label is not a negative outcome.

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D1 Data sources and linkage

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All sources were accessed within the DALY-CARE secure environment and linked at patient level by the DALY-CARE pseudonymised patient identifier. Supplementary Table D1 lists the sources and their use.

Supplementary Table D1: Data sources.

Source	Content used	Used for	Coverage and date handling
DALY-CARE patient table	Vital status, date of death or end of follow-up, sex, date of birth	Death, end of follow-up, demographics	End of follow-up is the earlier of death and last follow-up.
DALY-CARE haematological diagnoses	Harmonised haematological diagnoses with date, reliability rank and data source	Diagnosis anchor, disease classification, new haematological malignancy	Diagnoses from the Danish clinical quality registries (RKKP) were assigned the highest reliability rank.
Danish National Patient Registry: diagnoses	All ICD-10 diagnoses (SKS diagnosis codes)	New malignancy, cardiovascular, infection and fracture outcomes	LPR2 until 2019 and LPR3 thereafter; no era restriction.
DALY-CARE ICD-10 disease dictionaries	ICD-10 code patterns per haematological disease; SKS radiotherapy codes	Disease classification; radiotherapy as first-line treatment	Supplemented locally (Section D3).
Harmonised antineoplastic treatment data (version of 10 September 2026)	Drug code (ATC or SKS), drug text, indication code and text, source, start date	First- and second-line treatment episodes	Treatment-plan records were excluded; records from the LYFO, DaMyDa and DCLLR quality registries were regarded as reliable.
Curated CLL and MM treatment-line data (CLL version 4, MM version 5)	Dated treatment lines	First-line anchoring in CLL/MM; second-line events in CLL/SLL and MM	Assumed complete for CLL/SLL and MM patients.
Danish Lymphoma Registry (LYFO)	Date of second-line treatment	Second-line treatment in lymphoma	Data from the CLL (DCLLR) and myeloma (DaMyDa) registries were not used for second-line dating.
Laboratory results	Analytes identified by NPU codes, cleaned with the DALY-CARE standard procedure	Laboratory outcomes and laboratory ascertainment	Complete capture assumed from 1 January 2017 (Section D8).
DALY-CARE acute kidney injury data	Start dates of acute kidney injury episodes derived from creatinine by a separate DALY-CARE algorithm	Acute kidney injury outcome	Laboratory ascertainment rule.
Danish National Patient Registry: admissions, contacts and procedures (LPR2, LPR3)	Admissions and outpatient/emergency contacts (LPR2); physical contacts (LPR3); surgical and other procedures (SKS)	Procedure outcomes; acute hospitalisation	Procedures dated by the procedure start date, not the contact start.
Hospital Medicines Register (SMR)	In-hospital medicine administrations with ATC code and route	Intravenous antimicrobial-use outcome	Reporting starts 15 May 2018; outcome restricted to index dates on or after this date.
Danish National Prescription Registry	Dispensed prescriptions with ATC code and dispensing date	Prescription antimicrobial-use outcome	No era restriction.
International Prognostic Index (IPI)	IPI at diagnosis	Population description and prognostic benchmark	Not an outcome or sequence-model input.

Linkage of procedures. Procedure records were linked to their admission or contact through the full record identifier and dated by the start of the procedure rather than of the admission.

Data cutoff. The administrative data cutoff was 19 March 2026 for all endpoints. A death recorded after the cutoff was treated as censoring at the cutoff, not as an event.

D2 Study population and index date (first-line treatment)

First-line treatment is not recorded as such in any Danish register, so its start, the index date of every prediction, was reconstructed from the treatment records. The reconstruction has to separate lymphoma-directed treatment from three kinds of record that look alike in the data: drugs given for another cancer, drugs given as prophylaxis, maintenance or supportive care, and isolated administrations that do not amount to a course of therapy. Each of the choices below addresses one of these.

Where to look. Each patient's earliest DALY-CARE haematological diagnosis of any kind, including MGUS, MBL and unspecified codes, served as the diagnosis anchor, and treatment records from 30 days before to 3,650 days after it were eligible. Starting before the anchor allows for treatment that begins while the diagnosis is still being registered.

Which records count. A record needed a patient, a date and a drug code, or a drug text where no code was given. Records were then judged by their indication, in order of trust. Records from the clinical quality registries (LYFO, DaMyDa, DCLLR) were kept, as were records whose indication carried a haematological ICD-10 code (DC81–DC89, DC90–DC96, DD47). Records were discarded when the indication was a listed non-haematological cancer (DC18, DC19, DC20, DC25, DC34, DC43, DC50, DC56, DC61, DC71, DC78), when the indication text matched a list of non-haematological Danish terms, or when the drug is not used for lymphoid malignancy (e.g. fluorouracil, capecitabine, imatinib, azacitidine, hydroxycarbamide, cisplatin, carboplatin). Records without an informative indication were kept as uncertain rather than discarded, because much routine treatment is recorded without one.

From records to courses of therapy. Records belonged to one episode as long as consecutive treatment dates were at most 21 days apart, a gap rule that follows the cycle length of most regimens rather than a limit on duration; the episode started on its earliest record. An episode then had to look like therapy, and the evidence required depended on how specific its drugs were. The strongest evidence is a drug used almost only for lymphoid malignancy, and an episode containing one of these *anchor* agents qualified outright. The anchors were the CLL agents (fludarabine, cyclophosphamide, bendamustine, chlorambucil, obinutuzumab, ibrutinib, zanubrutinib, acalabrutinib, idelalisib, venetoclax, alemtuzumab; ATC and SKS BWHA/BOHJ codes); aggressive B-cell lymphoma agents and regimens (e.g. SKS codes for CHOP, COP and CHOEP; mitoxantrone, ifosfamide, carboplatin, gemcitabine, vincristine, melphalan, cytarabine, bortezomib, lenalidomide, thiotepa, carmustine); myeloma agents and regimens identified by drug text (e.g. daratumumab, carfilzomib, pomalidomide, autologous stem-cell transplantation), including the generic myeloma treatment categories recorded as drug text (complex or other cytostatic treatment, other antineoplastic agents, monoclonal antibodies and antibody–drug conjugates); SKS BWHA135 and the drug text "vcr"; and radiotherapy (SKS codes described as "strålebehandling", except listed prefixes, plus AFB04F3 and AFB03F3). Drugs that are also given for prophylaxis, maintenance or other diseases were kept out of the anchors on purpose, so that they could not confirm a course on their own: rituximab (ATC L01FA01, L01XC02), unspecified CD20 antibody (drug text and SKS BOHJ11), methotrexate (drug text and ATC L01BA01, L04AX03), gemcitabine and cisplatin identified by text, doxorubicin, etoposide, oxaliplatin, selinexor, basic cytostatic treatment, the generic cytostatic codes BWHA1, BWHA2 and BWHA4, and bevacizumab.

Without an anchor, more evidence was required the more easily the episode could be something else. A quality-registry record sufficed, unless the episode also contained one of the suspect agents above and fewer than two distinct codes. Anti-CD20 treatment (the codes above, or the drug texts rituximab, ofatumumab or CD20 antibody) on two or more dates sufficed, so that rituximab monotherapy, a genuine first-line treatment in indolent lymphoma, is not lost. An episode with any other suspect agent needed at least three treatment dates and two distinct codes, and any remaining episode at least two dates or two distinct codes.

Curated treatment lines. For patients in DALY-CARE's curated treatment-line data for CLL (version 4) and multiple myeloma (version 5), the curated first-line date was trusted over the reconstruction. An episode that contained this date, or lay within 30 days of it, qualified without the therapy-like criterion, and the episode closest to the curated date was chosen. The index date remained the start of that episode, not the curated date, so that all diseases share one definition of the index.

The first line. All other patients received their earliest qualifying episode, and patients without one had no index date and were not included.

Cohort flow. Supplementary Table D2 shows the cohort flow. Patients enter the outcome population only if they had a follow-up record and both end of follow-up and the data cutoff fell after the index date; a patient with a missing follow-up record, end of follow-up on or before the index date (including death on the index date) or an index date on or after the cutoff was therefore excluded from all outcomes. Twenty-six patients of the first-line cohort were excluded in this way: for all of them, death (17) or the end of follow-up (9) was recorded on or before the reconstructed index date.

Supplementary Table D2: Cohort flow.

Stage	Patients
Patients in DALY-CARE: lymphoid cancer or precursor (MGUS, MBL) diagnosis (pretraining population)	78,668
No qualifying first-line treatment	40,033
First-line index date identified (pass 1)	38,635
Diagnosis assigned in pass 1 (diagnoses from 20 years before to 14 days after index)	37,357
Unassigned after pass 1	1,278
Additionally assigned in pass 2 (coding-lag reassessment, up to 90 days after index)	209
Unassigned after pass 2 (excluded)	1,069
Diagnosis assigned after both passes	37,566
IgM MGUS excluded	9
First-line cohort	37,557
No follow-up after the index date or index on/after the data cutoff	26
Supervised population	37,531
Training (index date up to 31 December 2021)	29,105
Tuning (index date in 2022)	1,980
Held-out test (index date from 1 January 2023)	6,446

D3 Disease classification

Diagnosis codes were mapped to disease groups using the DALY-CARE ICD-10 patterns with local refinements: SLL only by the exact code DC830; lymphoplasmacytic lymphoma by DC830B or DC880; nodal marginal-zone lymphoma by DC830C; splenic marginal-zone lymphoma by DC830D; the generic marginal-zone group was not used; and DD472A (IgM MGUS) was mapped to an exclusion group.

Pass 1 (contemporaneous diagnoses). Disease identity was taken from the diagnoses recorded around the start of treatment, from 20 years (7,300 days) before to 14 days after the index date. The long look-back captures an indolent disease diagnosed years before it needed treatment; the short forward window allows for codes registered during the first weeks of treatment. A single code can be a working diagnosis, so a disease group counted as confirmed only when supported by a diagnosis of reliability rank 3 or better (quality-registry diagnoses rank 1) or by at least two diagnoses at least 14 days apart. Catch-all labels (generic DALY-CARE, unspecified, unspecified B-cell lymphoma, unspecified lymphoma and unspecified leukaemia) could not be assigned, nor could MGUS and MBL, which by definition are not treated.

When several groups were confirmed, three principles decided between them. First, a specific label beat a broad one: the catch-all labels and the broad groups B-cell lymphoma, T-cell lymphoma, plasma-cell disorder and Hodgkin lymphoma gave way to any other confirmed specific label in the window, so that Hodgkin lymphoma was assigned only when no other specific lymphoma was confirmed. Second, an aggressive disease beat an indolent one, because first-line treatment in a patient with both is usually given for the aggressive disease: Richter transformation, DLBCL, BL, PTCL, ALCL, AITL, LBL and PCL took precedence over all other labels, including MCL. Third, remaining ties were broken by a ranking of aggressiveness (Richter transformation; then DLBCL, BL, PTCL, ALCL, AITL, LBL, PCL and MCL; then MM and solitary plasmacytoma; then CLL, SLL and LPL; then FL, MZL, EMZL and HCL; then others), and then by the most reliable, most recent and most specific diagnosis.

Transformation was recognised from the order of diagnoses. A patient with DLBCL confirmed in the window was labelled Richter transformation if CLL or SLL had been recorded anywhere in their history, confirmed or not, at least 90 days before their first DLBCL code, and transformed follicular lymphoma under the same rule for FL. The 90-day gap keeps a CLL or FL code registered during the same work-up as the DLBCL from being read as a transformation.

Pass 2 (coding-lag reassessment). A specific lymphoma diagnosis is often registered only after the diagnostic work-up is complete, which can be after treatment has started. Patients with an index date but no assignable disease after pass 1 therefore entered a second pass. Their earliest confirmed, assignable diagnosis anywhere in their history became a new anchor; first-line treatment was re-identified with the same algorithm in a window from 90 days before this anchor to 20 years after it (curated CLL/MM anchors still applied), and disease was re-assigned with a 10-year look-back and a 90-day forward window. Pass-2 patients therefore received a new index date, replacing their pass-1 index. This allowed a specific diagnosis registered up to 90 days after treatment start to define disease identity.

Exclusions. Patients still unassigned after pass 2, and patients assigned IgM MGUS, were excluded. Characteristics of the first regimen (for example radiotherapy alone) were recorded descriptively and did not lead to exclusion.

Disease taxonomy. The analysis used 24 disease subtypes nested in 10 disease groups (Supplementary Table D3).

Supplementary Table D3: Disease groups and subtypes.

Disease group	Subtypes
CLL/SLL	CLL, SLL
Indolent B-NHL	FL, extranodal MZL, nodal MZL, splenic MZL, LPL, B-cell lymphoma without further specification
DLBCL-like	DLBCL, Richter transformation (coded or derived), transformed FL
MCL	MCL
HL	HL
MM	MM, plasma-cell leukaemia, solitary plasmacytoma
T-NHL	AITL, ALCL, PTCL, T-cell lymphoma without further specification
HCL	HCL
BL/LBL	BL, LBL
Amyloidosis	Amyloidosis

D4 Temporal split and prediction time

Temporal split. Patients were assigned by index date to **training** (index date before 1 January 2022), **tuning** (1 January to 31 December 2022) or **held-out test** (from 1 January 2023). Each patient had the same index date and split in every outcome; composite outcomes required agreement of index date and split across components.

Prediction time. The model input for each patient comprised all records timestamped before 00:00 on the index date, followed by a prediction token. Treatment and laboratory records on the index date itself were therefore not model inputs, although an index-date laboratory draw counted towards laboratory ascertainment (Section D8). Because most records carry a date but no time, events on the index date cannot be ordered relative to treatment start; such events were not counted as outcomes.

Pretraining. The cohort encoder was pretrained with an exclusive calendar cutoff of 1 January 2022, applied in the same way, but this cutoff alone does not establish the provenance of vocabulary counts or laboratory-bin boundaries. The local and national preprocessing artifacts must separately document their fitting populations and dates. National vocabulary counts may include a wider period than encoder training, as noted in the model-details supplement. Pre-2022 histories of patients later assigned to tuning or held-out test were included in pretraining; the evaluation therefore concerns future treatment episodes, not unseen individuals.

Disease identity at prediction time. Disease labels could rely on codes registered up to 14 days (pass 1) or 90 days (pass 2) after the index date. Models conditioned on disease therefore used information that was not necessarily available at the index date for these patients; a prespecified sensitivity analysis excludes the 209 patients whose disease was assigned after day 14 (Supplementary Methods, Prespecification).

D5 Disease-course endpoints

D5.1 Second-line treatment

Second-line treatment was identified from the same filtered treatment records as first-line treatment, with disease-specific rules.

Lymphoma groups (DLBCL-like, HL, BL/LBL, MCL, T-NHL, HCL, indolent B-NHL). (1) If the first-line episode spanned a single day, its end was imputed as start plus an expected regimen duration. (2) Treatment

records dated more than a minimum gap after the first-line end were grouped into episodes (21-day gap rule). (3) Episodes were excluded as autologous stem-cell transplant consolidation (transplant or conditioning codes within the consolidation window), consolidation (a share of first-line codes reappearing at or above an overlap threshold within the consolidation window), maintenance (only anti-CD20 or lenalidomide within the maintenance window), or single-date anti-CD20. (4) Remaining episodes had to be therapy-like (as for first line, except that quality-registry episodes needed $\geq$2 codes or $\geq$2 dates) *and* have at least moderate evidence of treatment for lymphoid malignancy: quality-registry source; an anchor agent with haematological indication; an anchor agent without suspect agents; or haematological indication with $\geq$2 dates *and* $\geq$2 distinct codes. Anti-CD20-only episodes on $\geq$2 dates were therefore counted as retreatment and qualified as second line when they met these evidence criteria (e.g. obinutuzumab, an anchor agent, or haematological indication with two distinct codes), except within the maintenance window in MCL and indolent B-NHL. The event date was the start of the earliest qualifying episode. (5) A second-line date recorded in LYFO after the index date replaced the derived date if it was earlier or if no derived date existed. The disease-specific parameters are listed in Supplementary Table D4.

Supplementary Table D4: Disease-specific parameters for second-line treatment in lymphoma groups. Durations and windows in days; – not applied.

Group	Imputed first-line duration	Minimum gap	Consolidation window	Maintenance window	Overlap threshold	Reliability
DLBCL-like	126	60	90	–	0.45	high
HL	120	60	90	–	0.45	high
BL/LBL	28	45	90	–	0.50	low
MCL	140	60	180	548	0.40	moderate
T-NHL	126	60	90	–	0.40	moderate
HCL	7	90	–	–	0.30	low
Indolent B-NHL	168	90	120	730	0.35	moderate

Ascertainment in lymphoma groups. Patients whose only post-first-line retreatment evidence was weak were excluded from the risk set rather than counted as event-free. Because second-line identification was unreliable in BL/LBL and HCL, these patients were excluded unless a LYFO second-line date created an event; BL/LBL, HCL and amyloidosis were consequently excluded from all second-line-dependent outcomes.

CLL/SLL and MM. The event was the earliest second-line date in the curated CLL or MM treatment-line data (matched on disease group), if later than the index date. CLL/SLL and MM patients without such a date were event-free, assuming completeness of the curated data. Amyloidosis was not included.

Censoring. Second-line treatment was censored at the earlier of the date of death and 19 March 2026; an event counted only if on or before this date, and patients censored on or before the index date were excluded. Death was not part of the endpoint and was handled as a competing event (Section D8). The endpoint is time to next qualifying treatment, not directly observed progression.

D5.2 Treatment failure and transformation

Treatment failure was the earlier of second-line treatment and death. **Treatment failure or transformation** was the earliest of second-line treatment, death and a new haematological malignancy. Transformation was defined operationally as a new three-character ICD-10 block among C81–C96 and D45–D47 first recorded more than 31 days after the index date and not recorded on or before index +31 days; it is not histologically confirmed transformation and also captures unrelated second haematological neoplasms. Both composites inherited the second-line ascertainment rules and the exclusion of BL/LBL, HCL and amyloidosis (Section D10).

D5.3 Overall survival and new malignancy

Overall survival was the time to death on or before 19 March 2026, censored at the administrative censoring date. A new haematological malignancy and a new malignancy of any type were defined by the first post-index diagnosis in a three-character ICD-10 block never recorded on or before index +31 days; codes registered around treatment start, which typically reflect the presenting disease, were thereby not counted. Relapse coded in the same block therefore never counted, whereas a new block (for example, FL, C82, to DLBCL, C83) did.

Supplementary Table D5: Disease-course endpoints (6 outcomes). OP, outcome population (Section D8).

Outcome	Definition	Source	Ascertainment and censoring	Competing event
Overall survival	Death on or before 19 March 2026.	Patient table	OP; administrative.	None
Second-line treatment	As above.	Treatment data, curated CLL/MM lines, LYFO	Disease-specific; censored at death or cutoff; excludes BL/LBL, HCL, amyloidosis.	Death
Treatment failure	Earliest of second-line treatment and death.	Composite	Intersection of components; earliest component censoring.	None
Treatment failure or transformation	Earliest of second-line treatment, death and new haematological malignancy.	Composite	As above.	None
New haematological malignancy	New ICD-10 block among C81–C96, D45, D46, D47 after index +31 days.	DALY-CARE haematological diagnoses	OP; administrative.	Death
New malignancy (any)	New ICD-10 block among C00–C97, except C44 and C80, after index +31 days.	DNPR diagnoses	OP; administrative.	Death

D6 Treatment-related and acute-care outcomes

D6.1 Outcome classes

Each outcome belongs to one of the following classes.

Graded laboratory (DALY-CARE grading).
Laboratory values were graded by the DALY-CARE standardised adverse-event grading algorithms, whose upstream graders cite CTCAE v5 (correspondence with CTCAE in Section D7). The event was the first post-index measurement at or above the stated grade.

Local laboratory threshold.
As above, with the numeric threshold stated in the table.

Baseline-relative laboratory.
The event was defined relative to the patient's own pre-index value. The baseline was the median value on the latest day with a measurement in the 365 days strictly before the index date; patients without a gradable baseline were excluded. Many patients with lymphoid malignancy have abnormal values before treatment (for example, through hepatic or renal involvement); a fixed threshold would then record a pre-existing abnormality at the first post-index draw as an event, whereas a baseline-relative threshold requires worsening.

Registry diagnosis or procedure.
The first post-index ICD-10 diagnosis or SKS procedure code. For *new-diagnosis* outcomes, a post-index code counted only if its three-character block had never been recorded on or before the index date (or index + grace, where stated).

Medication-use proxy.
Repeated antimicrobial administration or dispensing (Section D6.2).

Composite.
Earliest of its components (Section D10).

Except for the baseline-relative definitions and the alkaline phosphatase tiers, laboratory events are the first post-index documentation of an abnormal value, not new onset or worsening: abnormal baseline values were not excluded. No outcome is a verified CTCAE v6 grade; grade labels denote the output of the grading algorithm ("CTCAE-aligned"). Prediction horizons were 30, 60, 90, 180, 365 and 730 days, all prespecified; pooled summaries weigh every horizon equally, and the main figures show 365 days. For every outcome in Supplementary Tables D6–D12, death was a competing event.

Abbreviations in Supplementary Tables D6–D12. *Sources*: DNPR-Dx, National Patient Registry diagnoses; DNPR-Proc, National Patient Registry procedures (LPR2 and LPR3); Lab, laboratory results; SMR, Hospital Medicines Register; Rx, National Prescription Registry. *Baseline*: Lab rule, shared laboratory ascertainment rule (Section D8.1); OP, all patients in the outcome population. *Ascertainment*: Admin, censored at the administrative censoring date; Lab, laboratory censoring (Section D8.1). *Class*: Grd, DALY-CARE grading; Loc, local threshold; Rel, baseline-relative; Reg, registry; Prx, medication-use proxy; Cmp, composite.

D6.2 Infection and antimicrobial use

Diagnosis-based infection outcomes used the first post-index code without exclusion of prior episodes, because infections are acute events and a new post-index episode is clinically relevant even after an earlier one. The medication-use proxies grouped administrations (SMR, intravenous route only) or dispensings (prescription registry) of antimicrobials (DALY-CARE ATC selection) into episodes separated by gaps of more than 7 days; an episode qualified if it spanned at least 1 day, that is, $\geq$2 administrations or dispensings on different days. The event was the first date of the first qualifying post-index episode. No prophylaxis exclusion was applied to intravenous administrations; for prescriptions, ATC J01EE01 (sulfamethoxazole–trimethoprim), J05AB01 (aciclovir), J05AB11 (valaciclovir) and the exact code P02 were excluded as typical prophylaxis. These outcomes measure repeated antimicrobial use, not confirmed infection, days of supply or treatment intent. Because SMR reporting began on 15 May 2018, absence of intravenous administrations before that date is not evidence that none occurred; the intravenous proxy and the composite that contains it were therefore restricted to patients with index date on or after 15 May 2018.

Supplementary Table D6: Infection and antimicrobial-use outcomes (6 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Sepsis	ICD-10 A40, A41; first post-index code.	DNPR-Dx	OP	Admin	Reg
Serious infection	ICD-10 A40, A41, B44, B45, B46, B59 (sepsis, invasive fungal infection, pneumocystosis) as one code set.	DNPR-Dx	OP	Admin	Reg
Antibiotic treatment (procedure)	SKS procedure codes beginning BPH; dated by procedure start.	DNPR-Proc	OP	Admin	Reg
Intravenous antimicrobial use	Qualifying episode of intravenous antimicrobial administration.	SMR	OP; index date $\geq$ 15 May 2018	Admin	Prx
Prescription antimicrobial use	Qualifying episode of antimicrobial dispensing, excluding prophylaxis codes.	Rx	OP	Admin	Prx
Infection diagnosis or antimicrobial use	Earliest of serious infection, prescription and intravenous antimicrobial use.	Composite	Intersection; index date $\geq$ 15 May 2018	Earliest component	Cmp

D6.3 Blood counts, cytopenia and transfusion

Supplementary Table D7: Blood count, cytopenia and transfusion outcomes (16 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Anaemia, grade $\geq$2; grade $\geq$3	Haemoglobin graded $\geq$2 / $\geq$3.	Lab	Lab rule	Lab	Grd
Neutropenia, grade $\geq$2; grade $\geq$3	Neutrophil count decreased, $\geq$2 / $\geq$3.	Lab	Lab rule	Lab	Grd
Leukopenia, grade $\geq$2; grade $\geq$3	White blood cell count decreased, $\geq$2 / $\geq$3.	Lab	Lab rule	Lab	Grd
Thrombocytopenia, grade $\geq$2; grade $\geq$3	Platelet count decreased, $\geq$2 / $\geq$3 (grade 3/4 boundary 25×10^9/L, CTCAE v5).	Lab	Lab rule	Lab	Grd
Lymphocytosis, grade $\geq$2; grade $\geq$3	Lymphocyte count increased, $\geq$2 / $\geq$3. In CLL largely reflects disease activity.	Lab	Lab rule	Lab	Grd
Eosinophilia	Eosinophil count increased, grade $\geq$1.	Lab	Lab rule	Lab	Grd
Haptoglobin decreased	Haptoglobin decreased (binary grading). Not a diagnosis of haemolysis.	Lab	Lab rule	Lab	Grd
Severe cytopenia	Earliest of neutropenia, thrombocytopenia and anaemia, each grade $\geq$3.	Composite	Intersection	Earliest component	Cmp
Red-cell transfusion	SKS BOQA0, BOQA00–BOQA05; dated by procedure start.	DNPR-Proc	OP	Admin	Reg
Platelet transfusion	SKS BOQC, BOQC3.	DNPR-Proc	OP	Admin	Reg
Any transfusion	Union of the red-cell and platelet code sets (single code-set outcome).	DNPR-Proc	OP	Admin	Reg

D6.4 Renal

Supplementary Table D8: Renal outcomes (5 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Acute kidney injury (first documented)	First post-index start of an acute kidney injury episode derived from creatinine by the DALY-CARE algorithm.	Lab	Lab rule	Lab	Grd
Creatinine increased, grade ≥2; grade ≥3	Creatinine graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Creatinine rise >1.5× pre-index baseline	Creatinine (µmol/L) >1.5 times baseline on any post-index day: the baseline arm of CTCAE v5 grade ≥2 alone, not an acute kidney injury diagnosis.	Lab	Lab rule; baseline value >0 (no reference-limit requirement)	Lab	Rel
Dialysis	SKS procedure codes beginning BJFD.	DNPR-Proc	OP	Admin	Reg

D6.5 Hepatic and coagulation

Supplementary Table D9: Hepatic and coagulation outcomes (9 outcomes). ULN, upper limit of normal.

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
ALT increased, grade ≥2; grade ≥3	ALT (U/L), with reference ULN >0. If baseline > its ULN: value >1.5 / >2.0 times baseline. Otherwise: value >3 / >5 times the ULN of the measurement.	Lab	Lab rule; gradable baseline (value >0, baseline ULN >0)	Lab	Rel
Alkaline phosphatase elevation, tier ≥2; tier ≥3	>2.5 / >5 times the sample ULN if the running mean of prior samples was normal, otherwise >2.5 / >5 times that prior mean; missing ULN set to 105 U/L. Time-varying baseline; CTCAE v6 grades ALP at grade 1 only.	Lab	Lab rule	Lab	Grd
Bilirubin increased, grade ≥2; grade ≥3	Bilirubin graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hepatotoxicity	Earliest of ALT increased grade ≥3 (baseline-relative), AST increased grade ≥3 (a component only, not a panel outcome) and bilirubin increased grade ≥3.	Composite	Intersection, hence gradable ALT baseline	Earliest component	Cmp
INR increased, grade ≥2; grade ≥3	INR graded ≥2 / ≥3; anticoagulant therapy not considered.	Lab	Lab rule	Lab	Grd

D6.6 Electrolytes and metabolic

Supplementary Table D10: Electrolyte and metabolic outcomes (19 outcomes). LLN/ULN, lower/upper limit of normal of the measurement.

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Hyponatraemia, grade ≥2	Sodium <130 mmol/L and < LLN (LLN 136 mmol/L if missing). Symptoms not observed.	Lab	Lab rule	Lab	Loc
Sodium <125 mmol/L	Sodium <125 mmol/L and < LLN. Does not capture symptomatic hyponatraemia at 125 to <130 mmol/L.	Lab	Lab rule	Lab	Loc
Potassium <3.0 mmol/L	Potassium <3.0 mmol/L on any post-index day. Symptoms and intervention not observed.	Lab	Lab rule	Lab	Loc
Hypernatraemia, grade ≥2; grade ≥3	Sodium graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hyperkalaemia, grade ≥2; grade ≥3	Potassium graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hypercalcaemia, grade ≥2; grade ≥3	Calcium graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hypocalcaemia, grade ≥2; grade ≥3	Calcium graded ≥2 / ≥3 (albumin correction not established).	Lab	Lab rule	Lab	Grd

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Hypomagnesaemia, grade ≥2	Magnesium (NPU02647) <0.5 mmol/L and < LLN (LLN 0.7 mmol/L if missing).	Lab	Lab rule	Lab	Loc
Hypertriglyceridaemia, grade ≥2; grade ≥3	Triglycerides graded ≥2 / ≥3; fasting status unknown.	Lab	Lab rule	Lab	Grd
Hypoalbuminaemia, grade ≥2; grade ≥3	Albumin graded ≥2 / ≥3.	Lab	Lab rule	Lab	Grd
Hyperphosphataemia	Phosphate > ULN (ULN 1.45 mmol/L if missing); NPU03096 used where the standard phosphate code was absent. Laboratory elevation only.	Lab	Lab rule	Lab	Loc
Hyperuricaemia	Urate > ULN; no event if the ULN was missing. Laboratory elevation only.	Lab	Lab rule	Lab	Loc
LDH increased	Lactate dehydrogenase > ULN; no event if the ULN was missing.	Lab	Lab rule	Lab	Loc

D6.7 Cardiovascular and thrombotic

New-diagnosis rule without grace period: a post-index code counted only if its three-character block had not been recorded on or before the index date. Blocks were evaluated separately, so that, for example, a prior I80 did not exclude a new I26.

Supplementary Table D11: Cardiovascular and thrombotic outcomes (4 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Atrial fibrillation	ICD-10 I48; new diagnosis.	DNPR-Dx	OP	Admin	Reg
Heart failure	ICD-10 I50; new diagnosis.	DNPR-Dx	OP	Admin	Reg
Stroke or TIA	ICD-10 I63, I64, G45; new diagnosis per block.	DNPR-Dx	OP	Admin	Reg
Venous thromboembolism	ICD-10 I26, I80, I81, I82; new diagnosis per block.	DNPR-Dx	OP	Admin	Reg

D6.8 Acute care and other events

Acute hospitalisation. The definition followed the Danish database of acute hospital contacts. In LPR2, inpatient admissions (acute or planned) were combined with outpatient and emergency-room contacts with an acute mode of admission (LPR2 patient type: inpatient, outpatient, or emergency room before 2014; from 2014 emergency-room contacts were registered as outpatient contacts with acute mode of admission). Missing admission and discharge hours were set to 09:00 and 21:00; outpatient and emergency contacts were treated as point events. In LPR3, physical attendances were used, acute if registered with acute priority; planned LPR3 contacts were retained only if they lasted ≥12 hours. Psychiatric specialties were excluded in both eras. Contacts separated by at most 4 hours were merged into one episode. An episode was *acute* if any of its contacts was acute, and a *hospitalisation* if it contained an LPR2 inpatient admission, lasted ≥12 hours or crossed midnight. The outcome comprised acute hospitalisations only, so a planned admission counted only if it was merged with an acute contact; planned admissions, which typically deliver treatment, were not outcome events. The event date was the start date of the first post-index acute hospitalisation.

Supplementary Table D12: Acute-care and other outcomes (4 outcomes).

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Acute hospitalisation	As described above.	DNPR admissions and contacts	OP	Admin	Reg
Emergency department stay	SKS procedure codes beginning NAB, excluding codes containing NAB1.	DNPR-Proc	OP	Admin	Reg
Respiratory support	SKS BGDA0, BGDA1, BGDA2, BGDA7.	DNPR-Proc	OP	Admin	Reg

Outcome	Definition and threshold	Source	Baseline requirement	Ascertainment and censoring	Class
Bone fracture	ICD-10 S02, S12, S22, S32, S42, S52, S62, S72, S82, S92, T02, M80, M84.3, M84.4 (including subcodes); first post-index code, no exclusion of prior fractures.	DNPR-Dx	OP	Admin	Reg

D7 Correspondence with CTCAE

Laboratory outcomes were defined with reference to the NCI Common Terminology Criteria for Adverse Events, version 5.0 [2] and, where noted, 6.0 [3], using laboratory criteria only: the symptom, intervention and hospitalisation criteria of some grades cannot be observed in the data, and none changes a laboratory threshold below. An outcome is *derived* from CTCAE when its thresholds equal the laboratory thresholds of the stated grade, *analogous* when it follows the same grade but differs in the stated way, and has *no counterpart* otherwise. Graded outcomes come from the DALY-CARE graders (package commit a0742b0, every grader citing CTCAE v5); the version installed on the analysis server is to be confirmed. Table D13 lists every laboratory outcome; outcomes were not redefined. Where a DALY-CARE grader uses R's default interval rule, a value exactly at a cut-off falls in the more severe band ($\leq$ where CTCAE uses $<$), which matters for haemoglobin, reported to 0.1 mmol/L, and albumin, reported in whole g/L.

Supplementary Table D13: Correspondence of laboratory outcomes with CTCAE.

Outcome	CTCAE term and grade	Relation	Difference
Thrombocytopenia $\geq$2; $\geq$3	Platelet count decreased, v5	Derived	None; v6 changed only the grade 3/4 boundary.
Lymphocytosis $\geq$2; $\geq$3	Lymphocyte count increased, v5	Derived	None.
Haptoglobin decreased	Haptoglobin decreased, v5	Derived	None.
INR increased $\geq$2; $\geq$3	INR increased, v5	Derived	The arm for patients on anticoagulation is not applied.
Hyponatraemia $\geq$2	Hyponatremia, v5	Derived	None for grade $\geq$2.
Potassium $<$3.0 mmol/L	Hypokalemia grade $\geq$3, v5	Derived	Named by its threshold.
Hypernatraemia $\geq$2; $\geq$3	Hypernatremia, v5	Derived	None.
Hyperkalaemia $\geq$2; $\geq$3	Hyperkalemia, v5	Derived	None.
Hypomagnesaemia $\geq$2	Hypomagnesemia, v5	Derived	None.
Hypertriglyceridaemia $\geq$2; $\geq$3	Hypertriglyceridemia, v5	Derived	None (fasting not specified by CTCAE).
Hyperuricaemia	Hyperuricemia grade $\geq$1, v5	Derived	Unit conversion of urate to be confirmed.
LDH increased	Blood lactate dehydrogenase increased, v5	Derived	None.
ALT increased $\geq$2; $\geq$3	Alanine aminotransferase increased, v6	Derived	Equals v6 in both baseline arms (v5 uses $\times$3–5 of an abnormal baseline).
Anaemia $\geq$2; $\geq$3	Anemia, v5	Analogous	Values exactly at 6.2 or 4.9 mmol/L fall in the more severe grade.
Neutropenia $\geq$2; $\geq$3	Neutrophil count decreased, v5	Analogous	Values exactly at a cut-off fall in the more severe grade; v6 moved every grade down one step, so these are v6 grades $\geq$1 and $\geq$2.
Leukopenia $\geq$2; $\geq$3	White blood cell decreased, v5	Analogous	Values exactly at a cut-off fall in the more severe grade.
Hypoalbuminaemia $\geq$2; $\geq$3	Hypoalbuminemia, v5	Analogous	Values exactly at 30 or 20 g/L fall in the more severe grade.
Creatinine increased $\geq$2; $\geq$3	Creatinine increased, v5	Analogous	Arm relative to the upper limit of normal only; the arm relative to baseline is not applied.
Creatinine $>$1.5$\times$ baseline	Creatinine increased grade $\geq$2, v5	Analogous	Arm relative to baseline only; v6 restricts that arm to a baseline below the lower limit of normal.
Bilirubin increased $\geq$2; $\geq$3	Blood bilirubin increased, v5	Analogous	Arm relative to the upper limit of normal only.
Eosinophilia	Eosinophilia grade $\geq$1, v5	Analogous	The condition "above baseline" is not applied.
Hypercalcaemia $\geq$2; $\geq$3	Hypercalcemia, v5	Analogous	CTCAE grades corrected or ionised calcium; whether the total-calcium codes are albumin-corrected is to be confirmed (derived if so).
Hypocalcaemia $\geq$2; $\geq$3	Hypocalcemia, v5	Analogous	As for hypercalcaemia.
Sodium $<$125 mmol/L	Hyponatremia grade $\geq$3, v5	Analogous	Laboratory arm only; symptomatic values of 125–129 are not included.

Outcome	CTCAE term and grade	Relation	Difference
Hyperphosphataemia	Hyperphosphatemia grade 1, v5	Analogous	CTCAE gives no number for "laboratory finding only"; the upper limit of normal is used.
Alkaline phosphatase tier ≥2; ≥3	Alkaline phosphatase increased, v5	Analogous	v5 multiples applied to a running mean of earlier samples; v6 has grade 1 only, so these are named tiers, not grades.
Acute kidney injury	Acute kidney injury (clinical), v5	No counterpart	Episode algorithm from creatinine, close to KDIGO.
Severe cytopenia	–	No counterpart	Composite of grade ≥3 anaemia, neutropenia and thrombocytopenia.
Hepatotoxicity	–	No counterpart	Composite of ALT (v6), AST (v5) and bilirubin (v5) grade ≥3.

D8 Outcome-specific eligibility, ascertainment and censoring

Outcome population. All outcomes started from the first-line cohort restricted to patients with a follow-up record whose end of follow-up and data cutoff both fell after the index date (37,531 patients; Section D2).

D8.1 Laboratory ascertainment

Laboratory outcomes are only observable in patients whose blood is tested. Absence of an abnormal value in a patient without measurements is not evidence that the abnormality did not occur. All graded, local-threshold and baseline-relative laboratory outcomes and acute kidney injury therefore used a shared ascertainment rule based on the dates of *any* laboratory measurement (not only the outcome analyte). A patient was eligible if all three conditions held:

- *source coverage*: index date on or after 1 January 2017 (the date from which laboratory capture was regarded as complete), or any measurement between 365 days before the index date and the administrative censoring date;
- *baseline*: any measurement between 365 days before the index date and the index date inclusive;
- *entry*: any measurement between the index date and 30 days after it, inclusive, *or* death within 30 days of the index date with index date on or after 1 January 2017 (so that early death did not remove patients from the risk set).

A single measurement on the index date satisfied both the baseline and the entry condition. Observing an event did not override these criteria.

Laboratory censoring. Patients with an event were censored at the administrative censoring date. Event-free patients were censored at the earlier of their last post-index measurement of any analyte plus 180 days and the administrative censoring date, so that time without laboratory monitoring was not credited as event-free. The ascertainment status of each patient was recorded (before laboratory coverage, missing baseline, no post-index measurement, insufficient monitoring, eligible with event, eligible with competing death, eligible event-free).

Baseline-relative and supplementary threshold outcomes. The risk set was that of the corresponding graded outcome (creatinine increased grade ≥2; ALT increased grade ≥2; for potassium <3.0 mmol/L, hypokalaemia grade ≥3), further restricted to patients with a gradable pre-index baseline where required, and censored by the laboratory rule.

D8.2 Non-laboratory outcomes

Diagnosis, procedure, hospitalisation, medication-use and survival outcomes were administratively ascertained: all patients in the outcome population were eligible and censored at the administrative censoring date. The only calendar restriction was the SMR start date (index date on or after 15 May 2018) for the intravenous antimicrobial-use proxy and the infection composite that contains it. Diagnosis, procedure and hospitalisation outcomes were not restricted to one register era, so calendar trends in registry capture are part of these labels.

D8.3 Competing death and horizon labels

For a fixed horizon h, a patient was a case if the event occurred by h and before any competing death; a known non-case if follow-up reached h or a competing death occurred by h; and otherwise of unknown status (right-censored) and excluded from the fixed-horizon loss and from binary metrics. Death precludes the subsequent

occurrence of every non-fatal outcome; treating it as ordinary censoring would overstate the risk of these outcomes, whereas treating it as a known non-case keeps patients who die early in the denominator. A death counted as a competing event only if it occurred on or after the index date, on or before the outcome's censoring date and within the horizon. For laboratory outcomes, an event-free patient who died more than 180 days after the last measurement was therefore censored rather than counted as a competing death. Endpoints that include death (overall survival and both treatment-failure composites) had no competing event.

D9 Selection of the outcome panel

The panel was selected from 87 candidate outcomes using training and tuning data only. Each outcome needed at least 100 events within a year of the index date in training and 20 in tuning, and clinical judgement of the definition took precedence over support (Supplementary Table D14). Test-period counts played no part in the selection.

Supplementary Table D14: Selection of the outcome panel from 87 to 69 outcomes.

Step	n	Outcomes and reason
Candidate outcomes	87	
Definition not reliable	−6	Hypofibrinogenaemia grade ≥2 and grade ≥3 (sparse; assay and grading not verified); leukocytosis grade ≥2 (grading did not match the label, and the term is discouraged for disease-related rises); DIC-like coagulopathy, Hy's-law liver injury and tumour lysis syndrome (laboratory co-occurrence proxies without monitoring-based ascertainment).
Definition concerns, excluded	−5	Lymphopenia grade ≥2 and grade ≥3 (CTCAE v6 has no grade 2/3 tier); hypokalaemia grade ≥2 and grade ≥3 (local thresholds ≤3.0 and ≤2.5 mmol/L, not CTCAE v6); major adverse cardiovascular events (composite included atrial fibrillation, venous thromboembolism and heart failure).
Below support threshold	−9	AST increased grade ≥2 and grade ≥3; herpes zoster; hypermagnesaemia grade ≥3; hypoglycaemia grade ≥2 and grade ≥3; hypomagnesaemia grade ≥3; invasive fungal infection; myocardial infarction.
Retained without renaming	60	
Retained with revised names (part of the 67 retained)	7	Named for what they measure: alkaline phosphatase elevation tier ≥2/≥3; sodium <125 mmol/L (for hyponatraemia grade ≥3); intravenous, prescription and combined infection outcomes as antimicrobial-use proxies; hyponatraemia grade ≥2. ALT increased and hepatotoxicity use the baseline-relative ALT definition.
Added	+2	Creatinine rise >1.5× pre-index baseline; potassium <3.0 mmol/L.
Final panel	69	

D10 Composite outcomes, endpoint groups and display groups

Composite rule. A patient was eligible for a composite only if eligible for every component (intersection of risk sets). The composite was censored at the earliest component censoring date, and the event was the earliest component event, counted only if on or before that censoring date. Components had to agree on index date and split. Death was an event only when overall survival was a component; otherwise death remained censoring in each component and was handled as a competing event. The composites were treatment failure, treatment failure or transformation, severe cytopenia, hepatotoxicity, and infection diagnosis or antimicrobial use. Any transfusion and serious infection were defined as single code-set outcomes rather than composites.

Endpoints. The training loss and the checkpoint criterion (the tuning loss) averaged the horizon-wise binary cross-entropy within each outcome and then equally over endpoints. Endpoints combined the grade variants of one endpoint (grade ≥2 and grade ≥3), the two alkaline phosphatase tiers, and hyponatraemia grade ≥2 with sodium <125 mmol/L, giving 52 endpoints for the 69 outcomes. Related diagnoses, proxies and composites were not merged.

Clinical domains. Each outcome was assigned once to one of 12 clinical domains: disease course and later malignancy; infection and antimicrobial use; erythroid findings and transfusion; neutrophil, white-cell and platelet findings; lymphocyte and eosinophil counts; renal injury and support; liver injury and coagulation; sodium and potassium; calcium and magnesium; metabolic and cell-turnover markers; cardiovascular and thrombotic events; and acute care and other events. Creatinine increase was placed under renal injury and support and hypokalaemia under sodium and potassium. Domains organise the figures, define the sensitivity

analysis that weighs every domain equally, and group outcomes in the borrowing analysis; they do not enter training.

D11 Limitations of the outcome definitions

1. **Proxies rather than adjudicated events.** Antimicrobial-use outcomes measure repeated antimicrobial administration or dispensing, not confirmed infection; the prescription proxy has no calendar restriction and intravenous prophylaxis was not excluded. Transformation is a new haematological ICD-10 block, not histologically confirmed transformation. Second-line treatment is time to next qualifying treatment, not observed progression.
2. **CTCAE-aligned, not CTCAE.** Laboratory grades derive from the DALY-CARE grading algorithms (all citing CTCAE v5) and local thresholds, some using reference limits with fixed fallback values. Symptoms, interventions and clinical context required by CTCAE for several grades are not observed; no outcome is a verified CTCAE v6 grade [2, 3].
3. **Prevalent versus incident abnormalities.** Except for the baseline-relative definitions, laboratory events are the first post-index documentation of an abnormal value, not incident toxicity.
4. **Analyte-agnostic laboratory ascertainment.** Eligibility and censoring use measurements of any analyte, so a patient can be eligible for an outcome whose analyte was rarely measured.
5. **Coding lag and retrospective disease identity.** Disease labels can use codes registered up to 14 days (pass 1) or 90 days (pass 2) after the index date and are used as model input.
6. **First-line identification.** The search is anchored at the first DALY-CARE diagnosis of any kind and limited to 10 years in pass 1; treatment records without indication are retained as uncertain; curated anchors exist for CLL and MM only.
7. **Registry era.** Diagnosis, procedure and hospitalisation outcomes are not restricted across LPR2 and LPR3, so changes in coding practice over calendar time are part of the labels; for LPR3 contacts without recorded clock times, the 12-hour criterion reduces to ending on a later calendar day. Diagnosis-based infection and fracture outcomes do not exclude continued coding of a pre-index episode.
8. **Second-line ascertainment.** Second-line treatment depends on the completeness of the curated CLL and MM data, excludes patients with only weak retreatment evidence (enriching the risk set for events) and is censored at death or the data cutoff rather than at end of follow-up as for other outcomes.
9. **Tumour lysis syndrome** is not part of the panel, because no laboratory definition could be ascertained reliably.

References

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[2] National Cancer Institute. *Common Terminology Criteria for Adverse Events (CTCAE), Version 5.0*, 2017. URL `https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events/ctcae-v5-5x7.pdf`. Published 27 November 2017.

[3] National Cancer Institute. *Common Terminology Criteria for Adverse Events (CTCAE), Version 6.0*, 2025. URL `https://dctd.cancer.gov/research/ctep-trials/for-sites/adverse-events/ctcae-v6.pdf`. Published 22 July 2025.